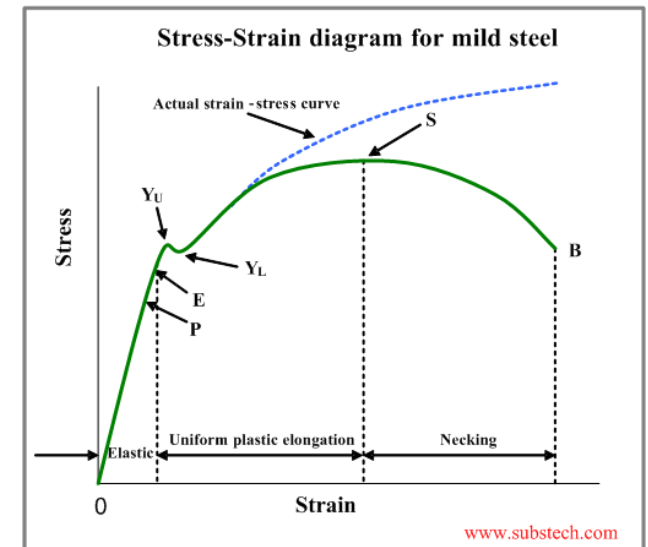
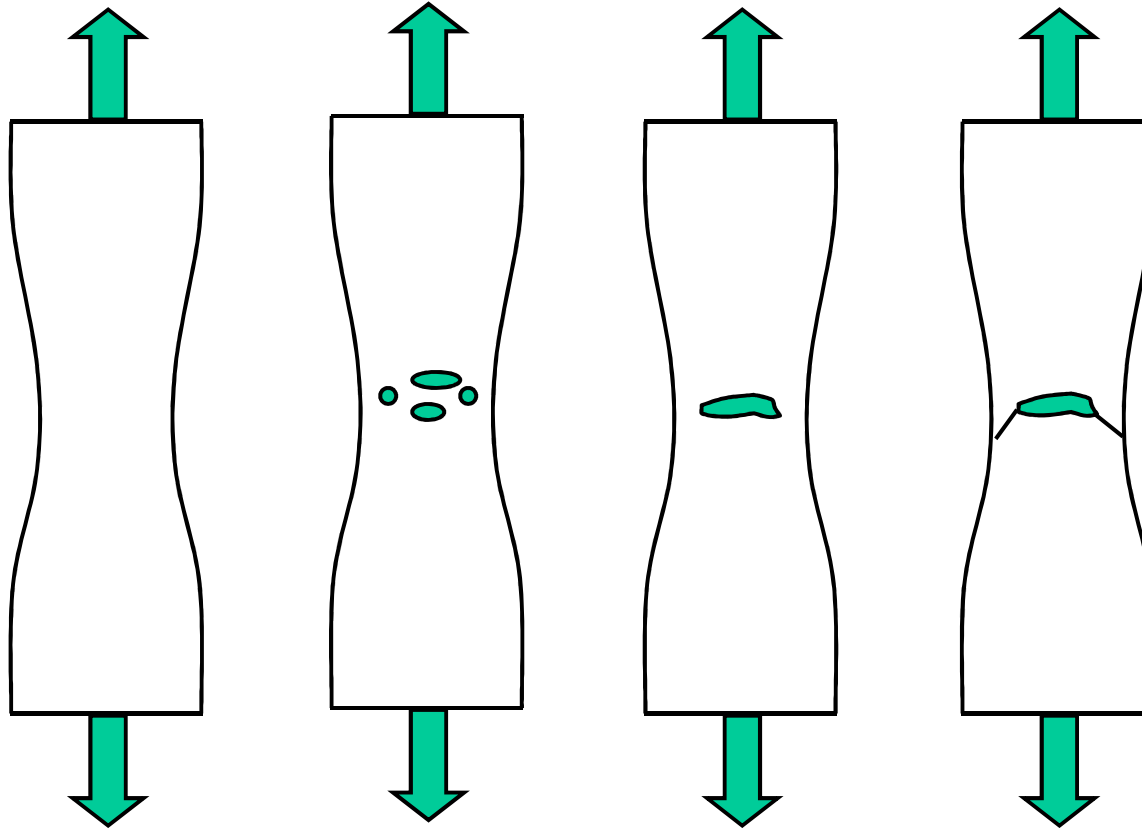


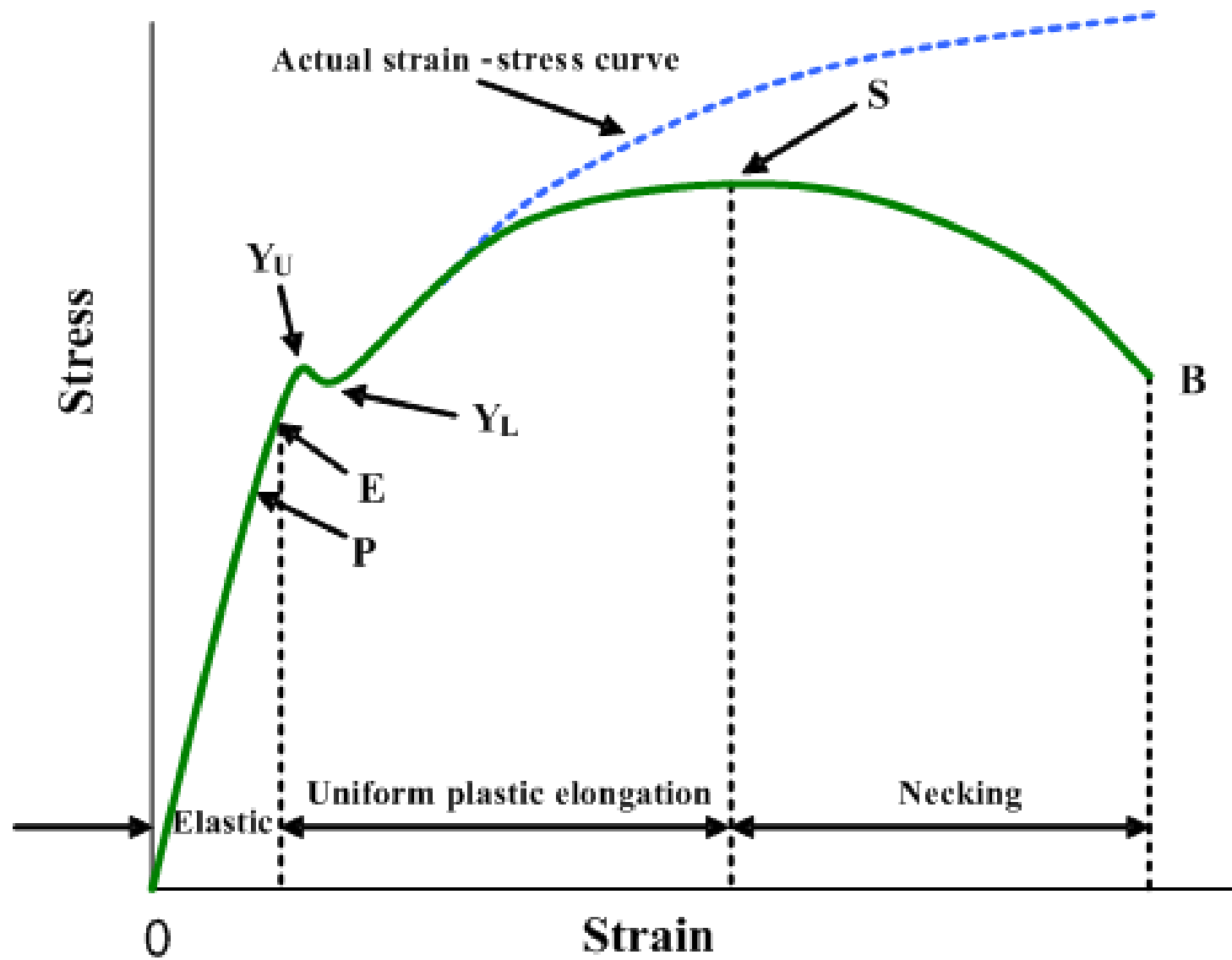
**MECHANICAL PROPERTIES
OF MATERIALS:
Additional details and testing
methods**

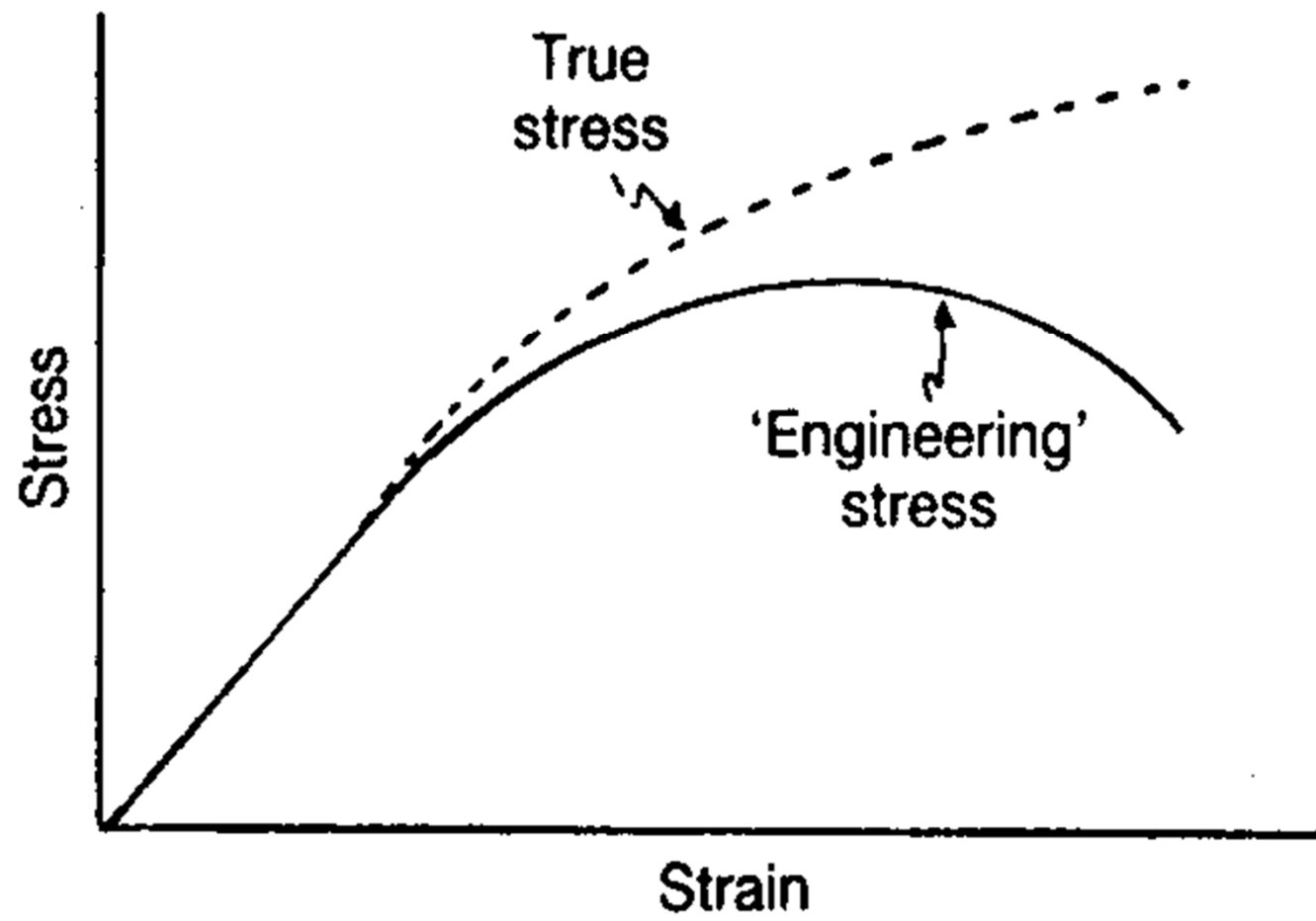
“True” engineering curve

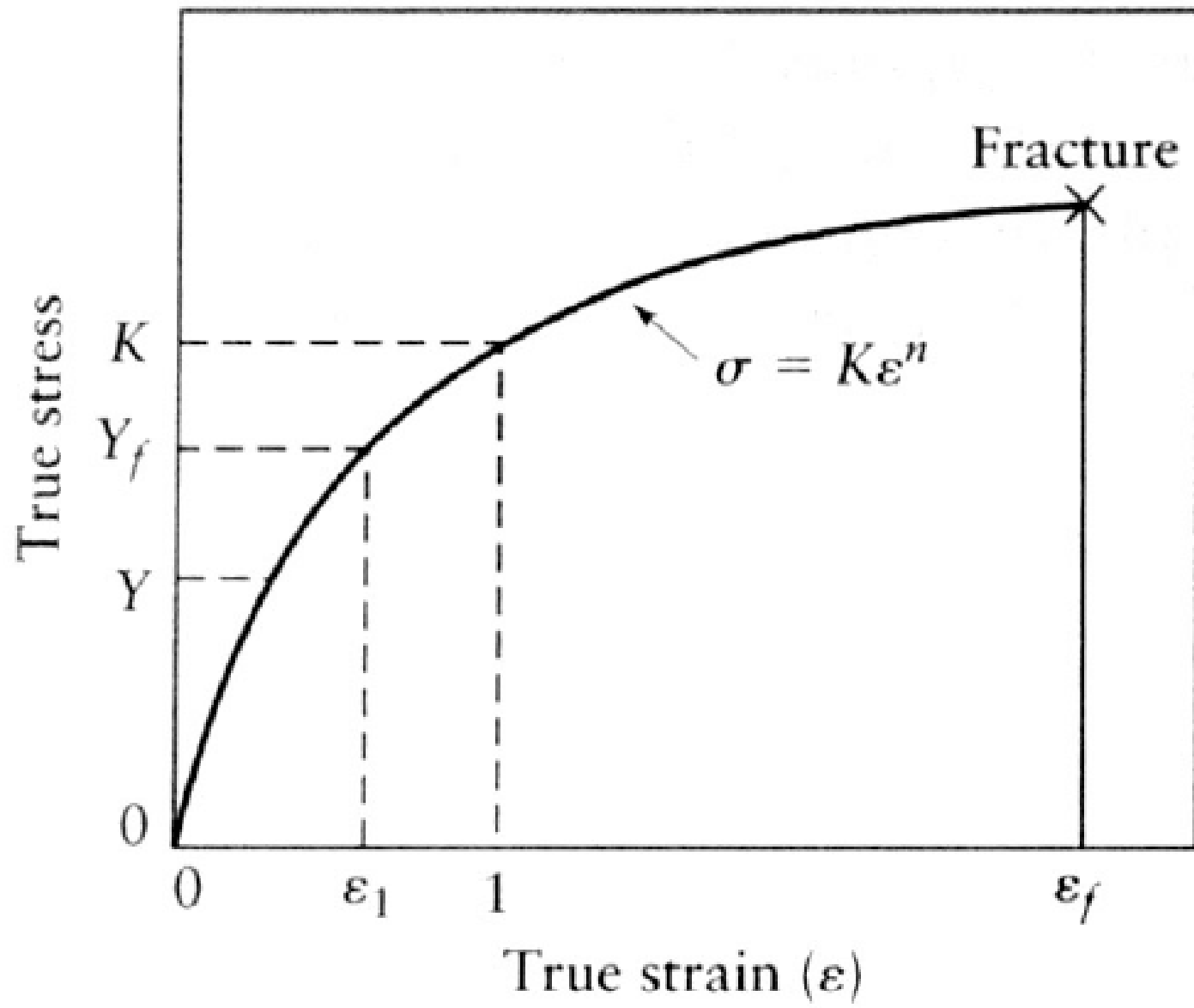


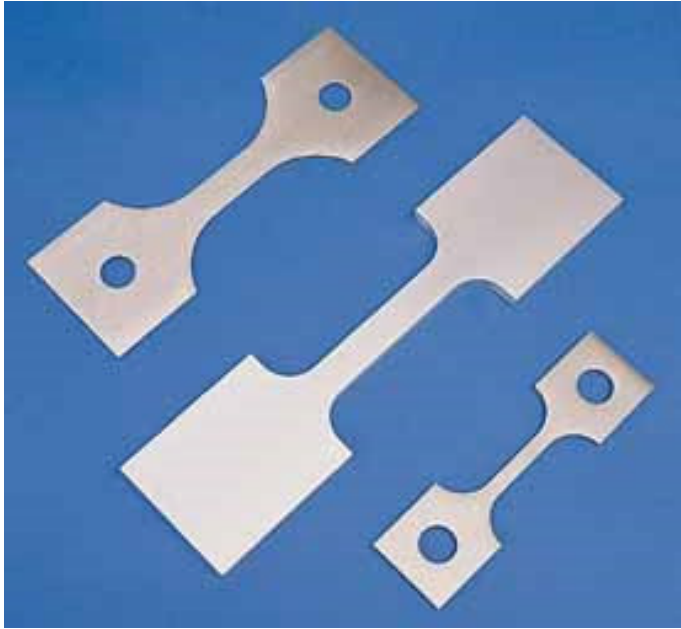
necking Formation of pores
Pores coalescence Crack propagation

Stress-Strain diagram for mild steel







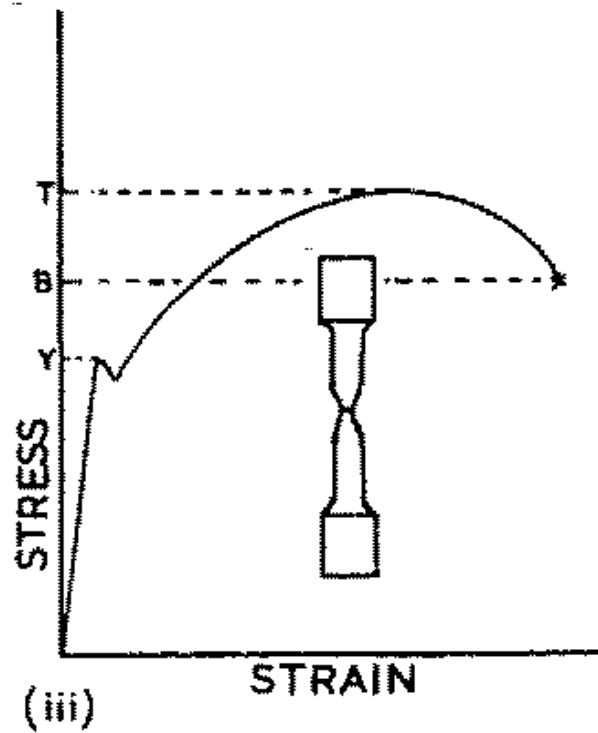


*Excellent video on
tensile test*

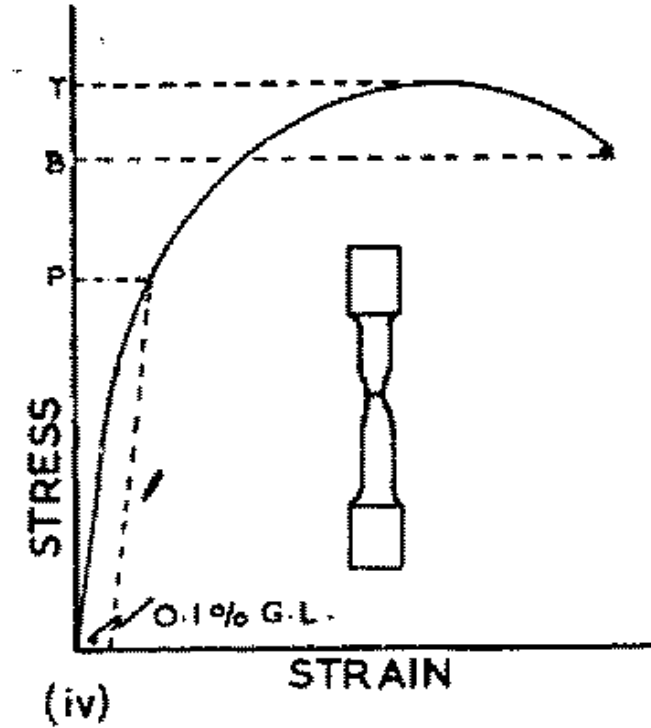
<https://m.youtube.com/watch?v=D8U4G5kcpcM>

*Metallic samples
for tensile test*





Steel



Pure metals

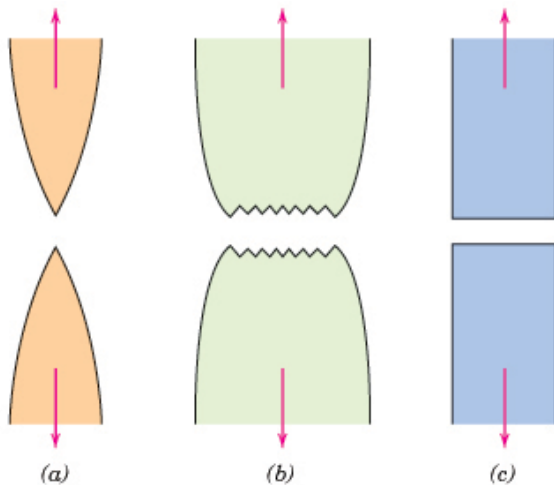


Figura 8.1 (a) Frattura estremamente duttile in cui la provetta riduce per strizione la sua sezione fino ad un punto. (b) Frattura moderatamente duttile dopo una certa strizione. (c) Frattura fragile senza alcuna deformazione plastica.

- a) Au and Pb at RT (100% necking), metals and polymer at high T
- b) Standard ductility
- c) Brittle fracture (no necking)

Yield strength

$(\sigma_s, \sigma_y, \sigma_{0,2})$

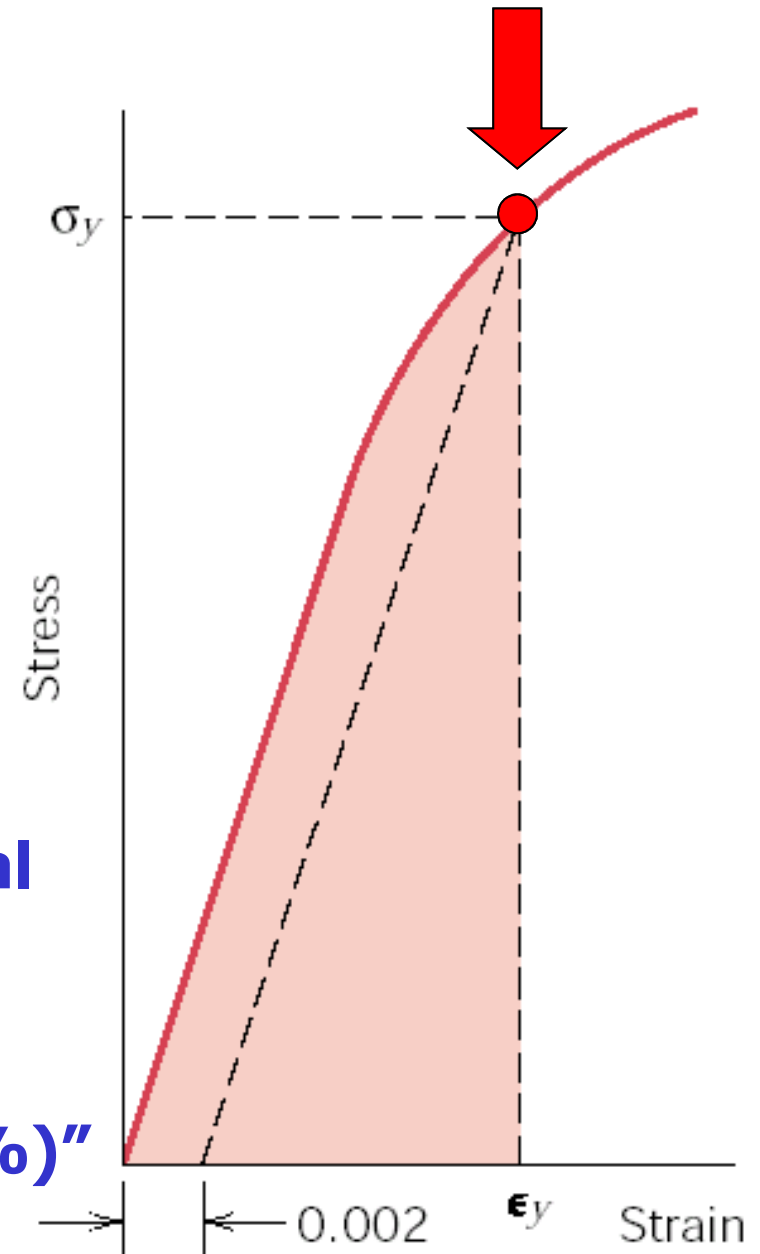
$(\sigma_s, \sigma_{P(0,2)})$



Yield strength or proportional
limit ($R_{p0,2}$)



Yield Strength (YS, S_y),
"Yield Strength (offset = 0.2 %)"



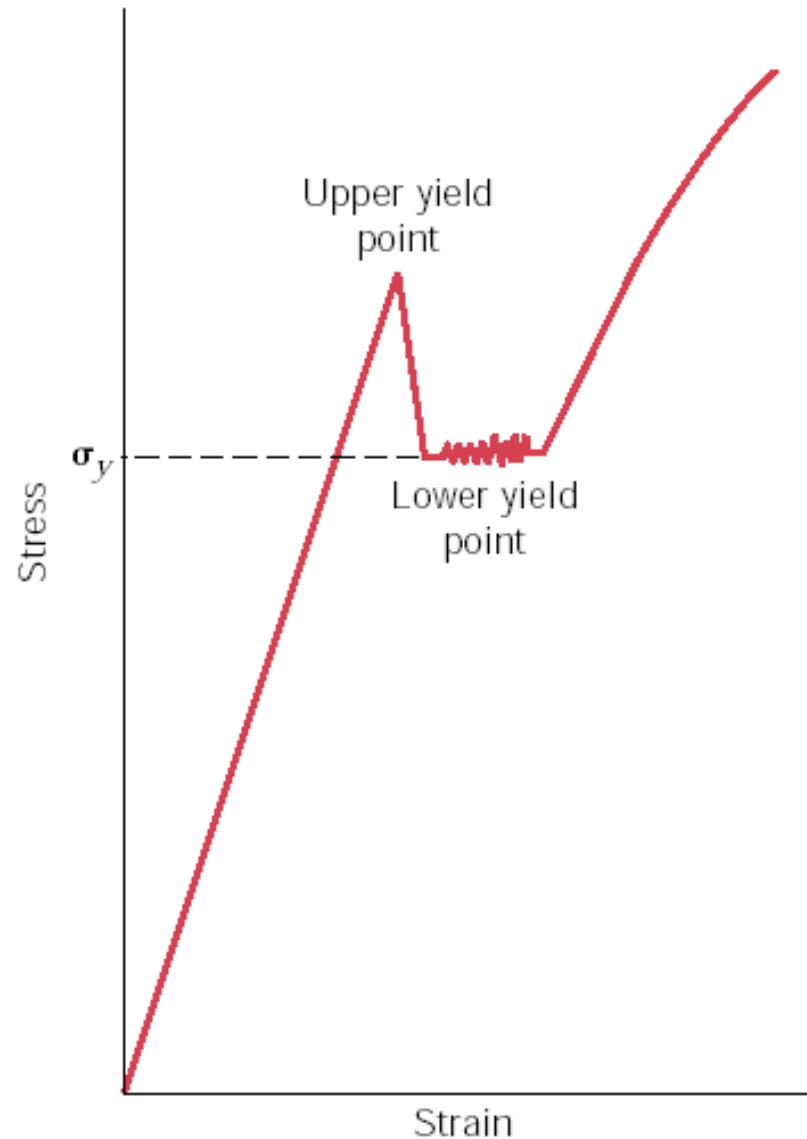
Upper and lower yield strength

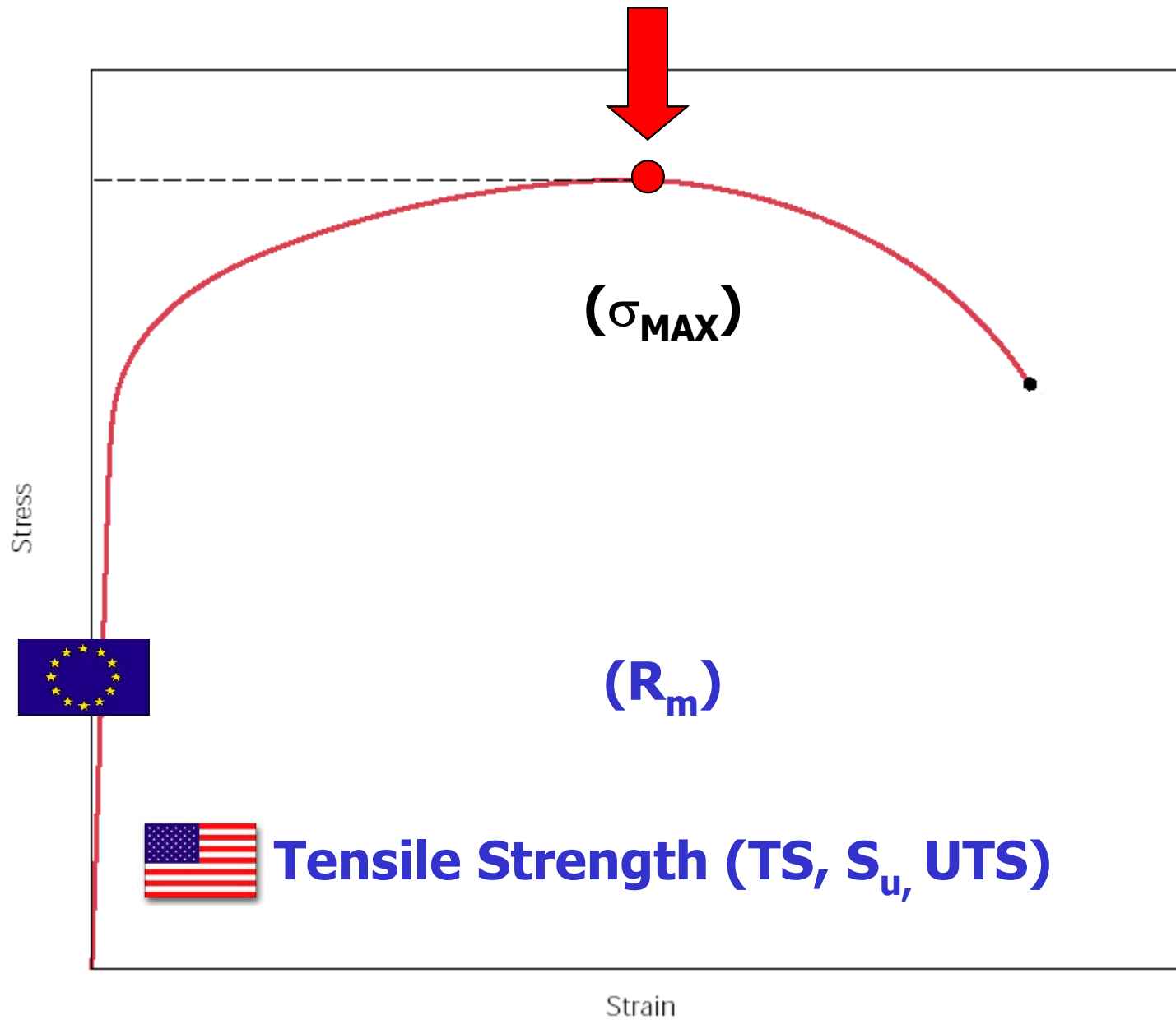
In steel we have the solute (Carbon atoms) in Fe as interstitials. They cause the dislocations to pin. So it becomes difficult for dislocation to move and we need higher load to make them move (upper yield point). Then at some point the pinning is gone and dislocations can move freely and we need smaller load to keep them moving (plastic deformation going) and that would be the lower yield point.

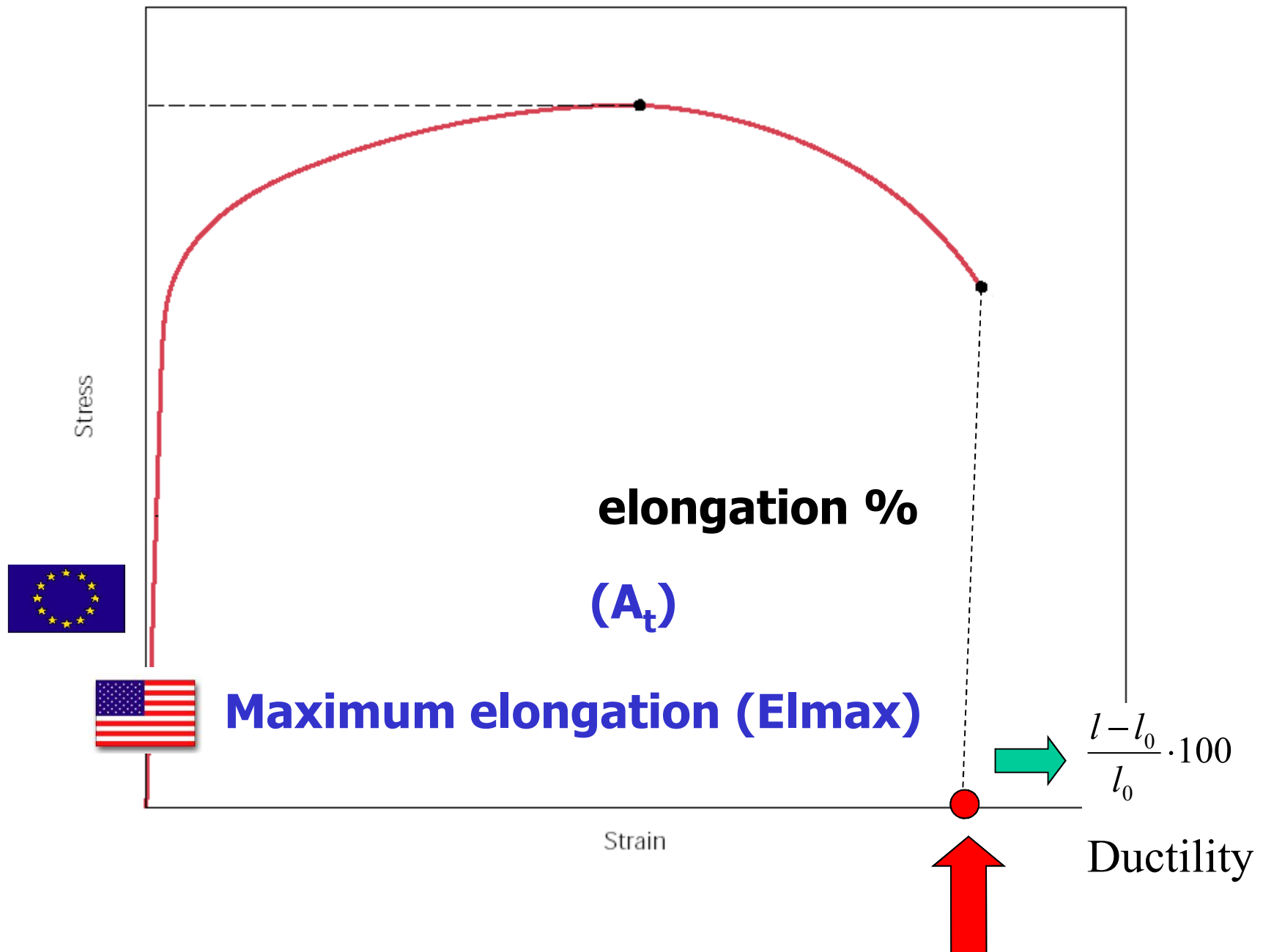
In the beginning the dislocation has to go through the interstitial carbon atom and is struggling, once past that point, it moves fast: video !

<https://youtu.be/zDjbJWlR4jM>

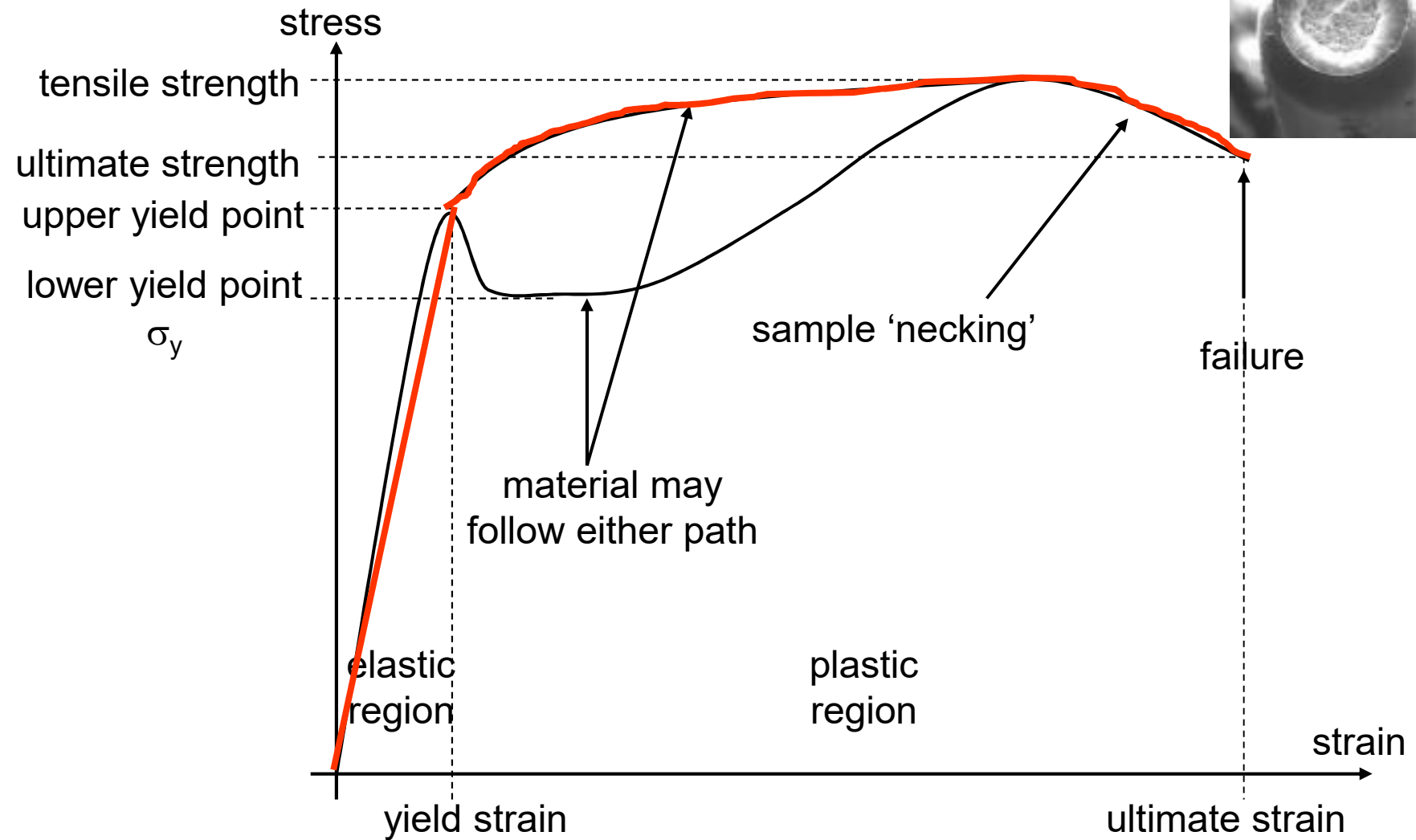
Upper Yield Strength (UYS) and Lower Yield Strength (LYS)

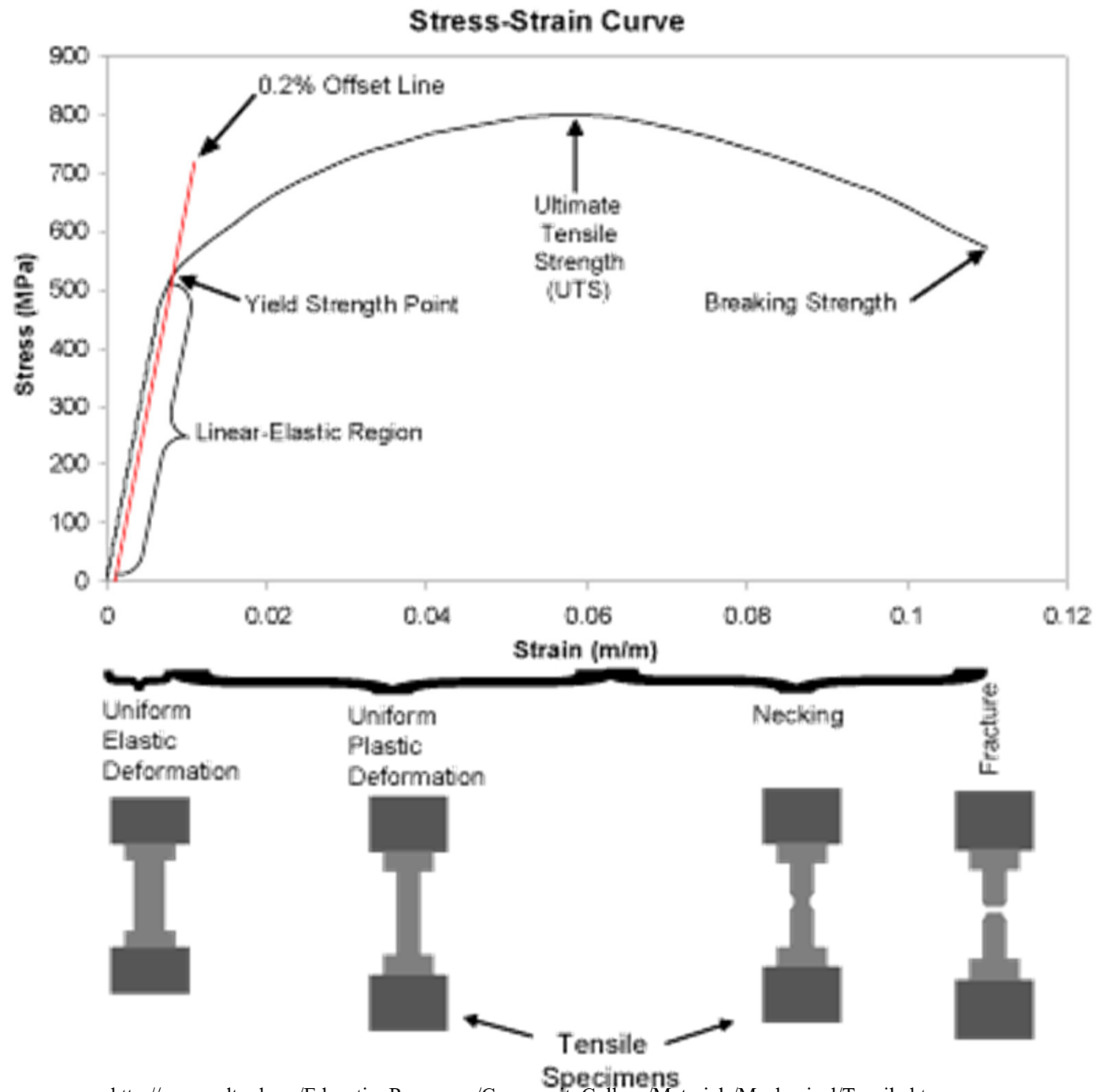






Typical stress-strain curves

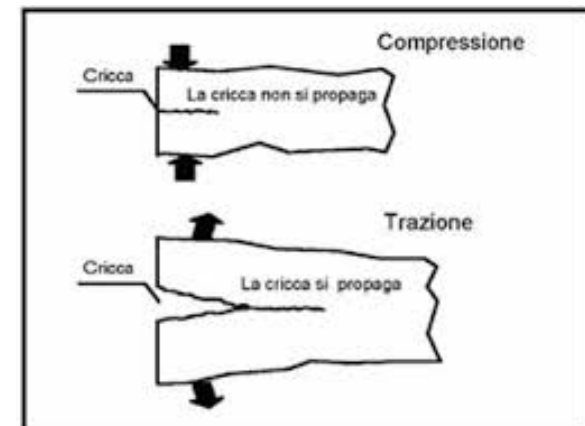
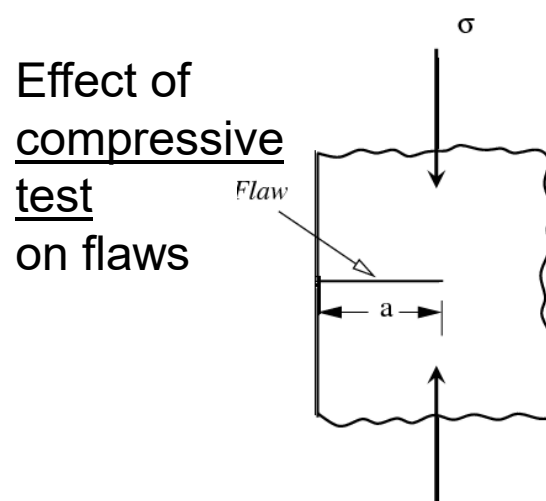
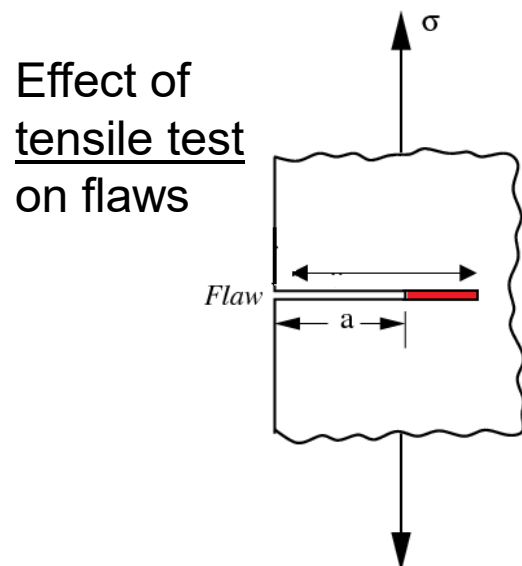


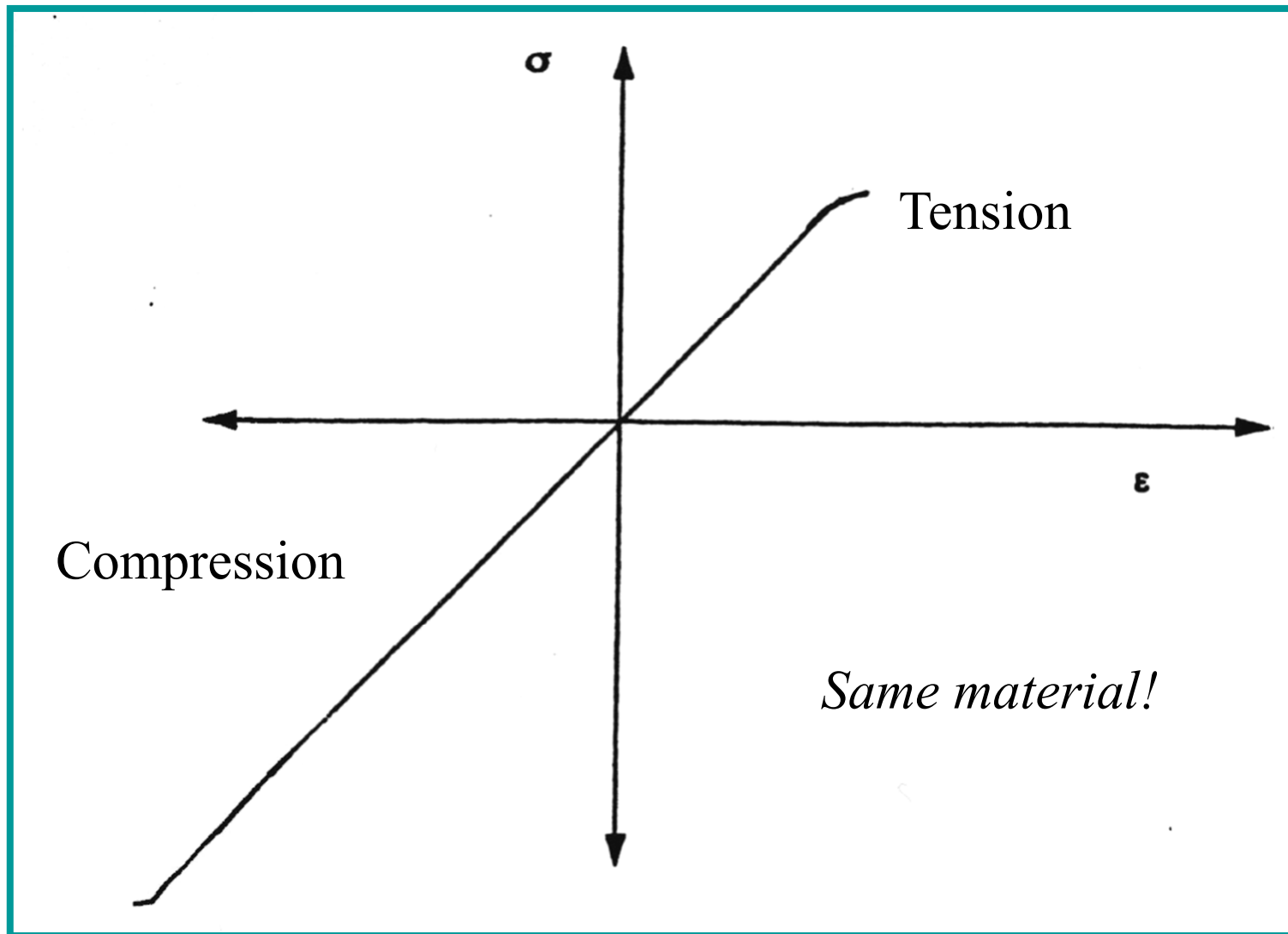


COMPRESSION TEST

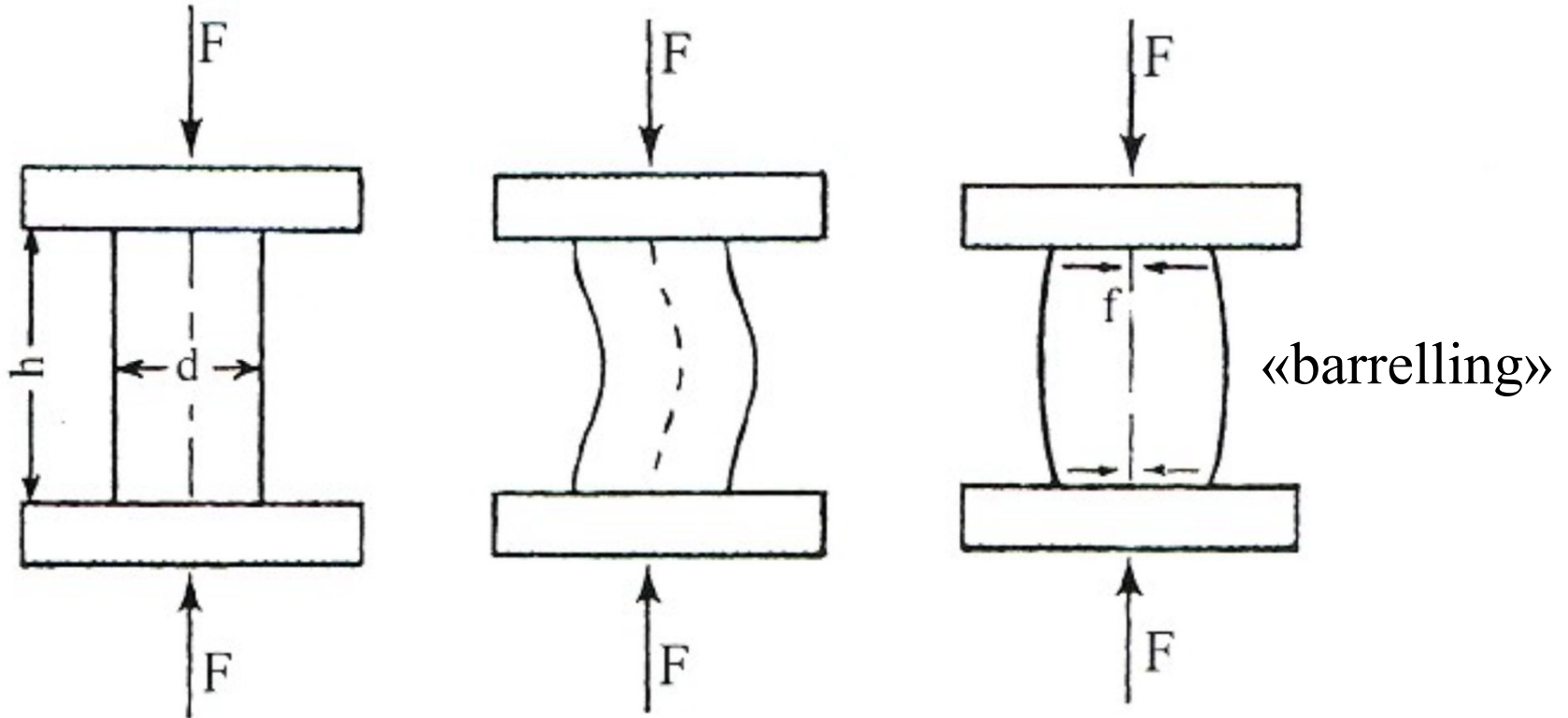
It is adopted for measuring the strength of materials used in compression (the majority of structural materials for Architecture: brick, concrete...) and for calculating the forces required in manufacturing processing

It is useful for **brittle material**: pre-existing **micro- or macro-defects** are not enlarged → higher strength and strain than in tensile test. However, this test overestimates the material resistance: the component can reach premature failure (in comparison to compression data) under real operating conditions if complex stresses (e.g. compression + torsion + bending) are present.



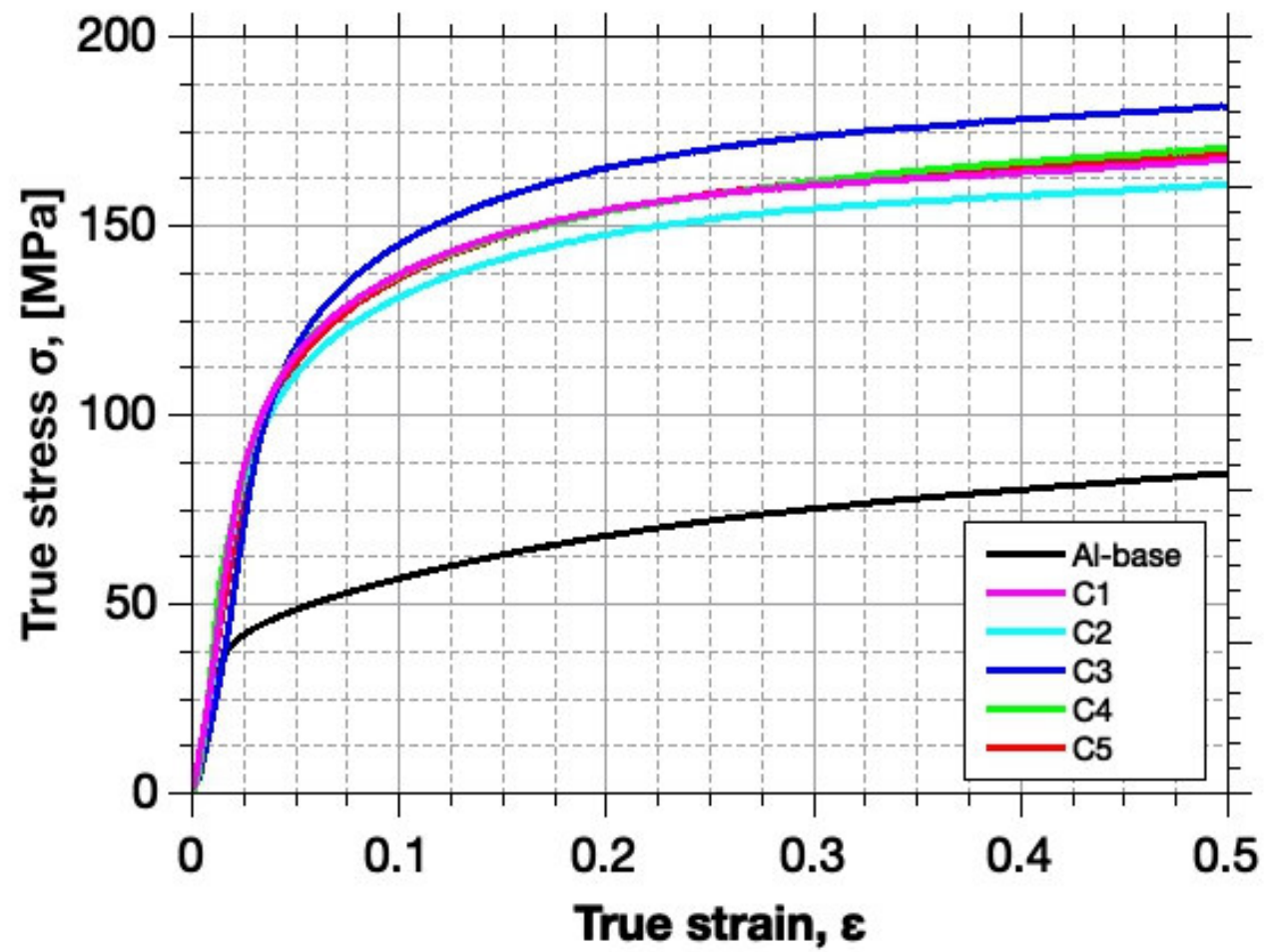


Warning!

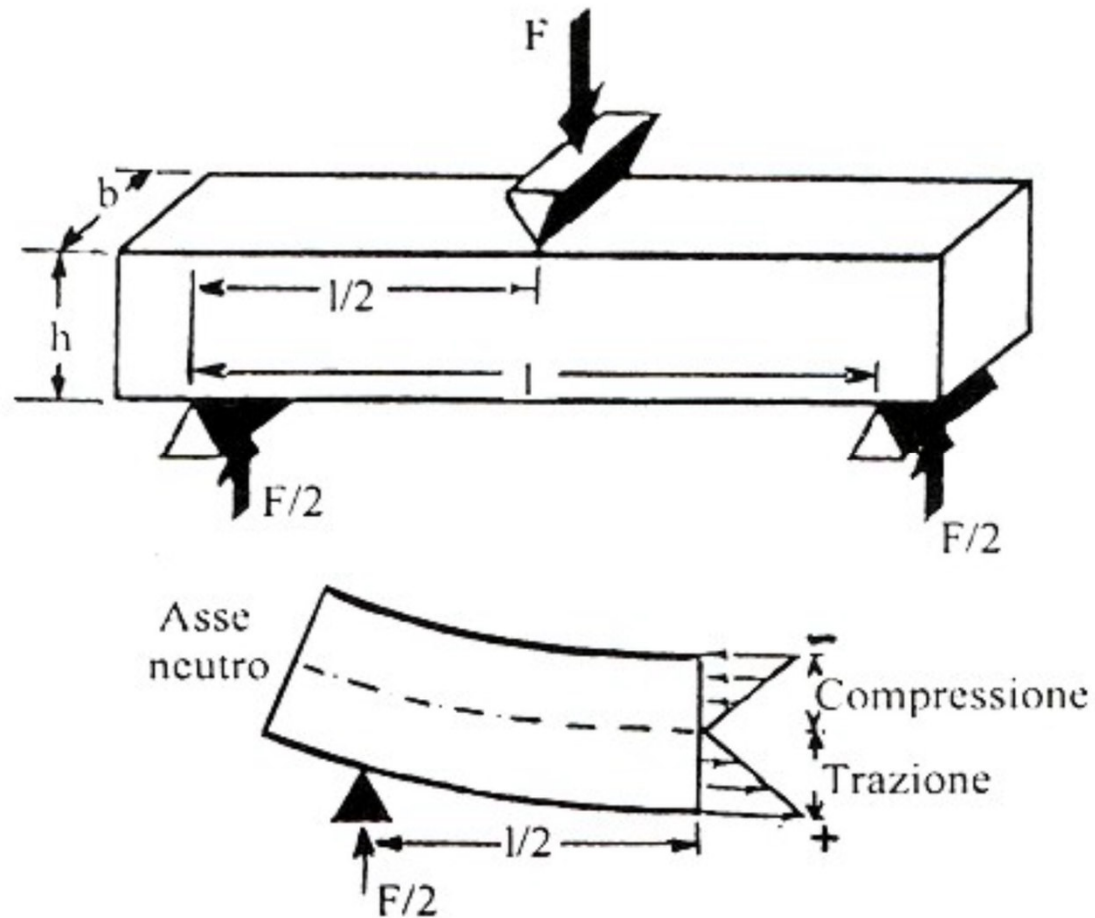


<https://www.instron.de/our-company/library/test-types/compression-test>

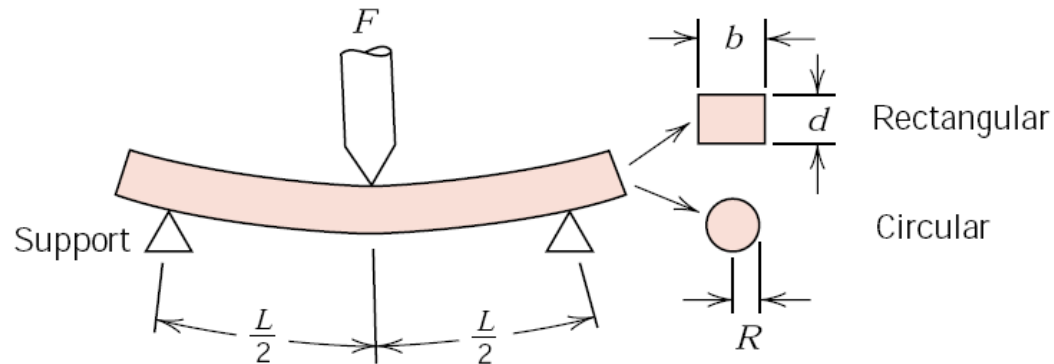
COMPRESSION TEST



Bending or flexural test



Example: 3-point flexural strength



$$\sigma = \text{stress} = \frac{Mc}{I}$$

where M = maximum bending moment

c = distance from center of specimen to outer fibers

I = moment of inertia of cross section

F = applied load

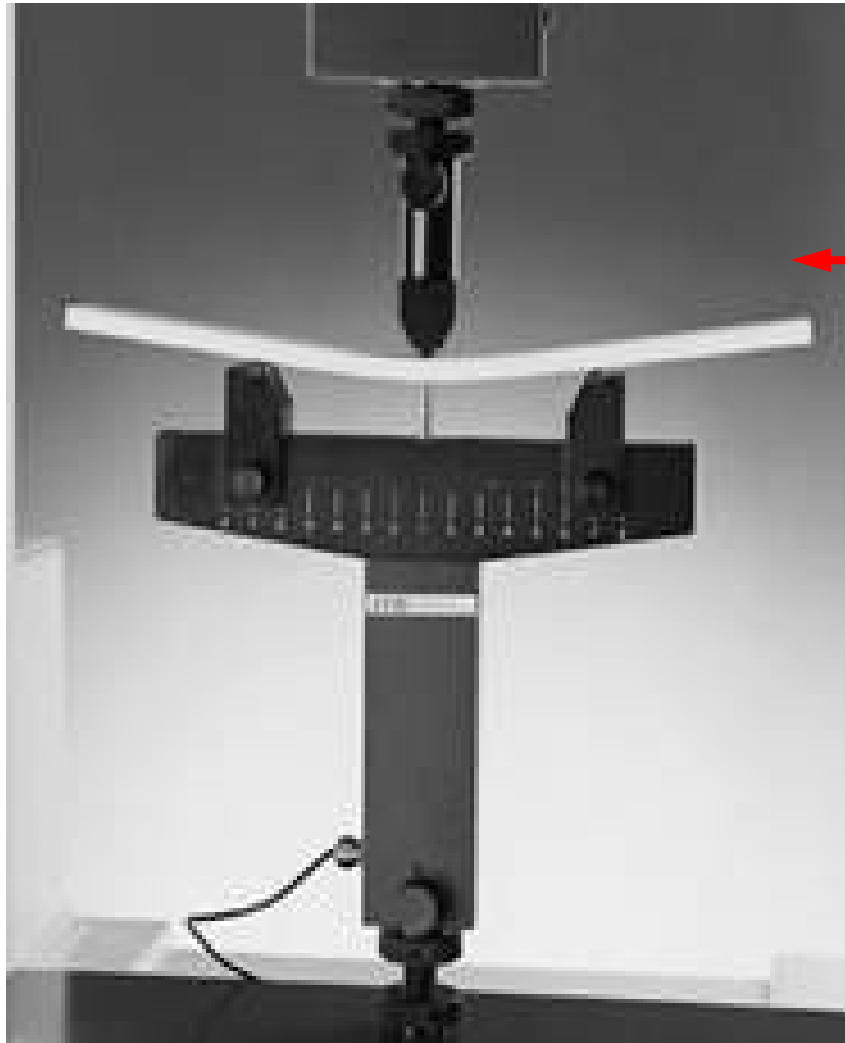
	$\frac{M}{I}$	$\frac{c}{I}$	$\frac{\sigma}{I}$
Rectangular	$\frac{FL}{4}$	$\frac{d}{2}$	$\frac{3FL}{2bd^2}$
Circular	$\frac{FL}{4}$	R	$\frac{FL}{\pi R^3}$

Failure load (F) and sample dimensions are known or can be easily measured!!

It is often used for measuring the strength of **brittle materials**.

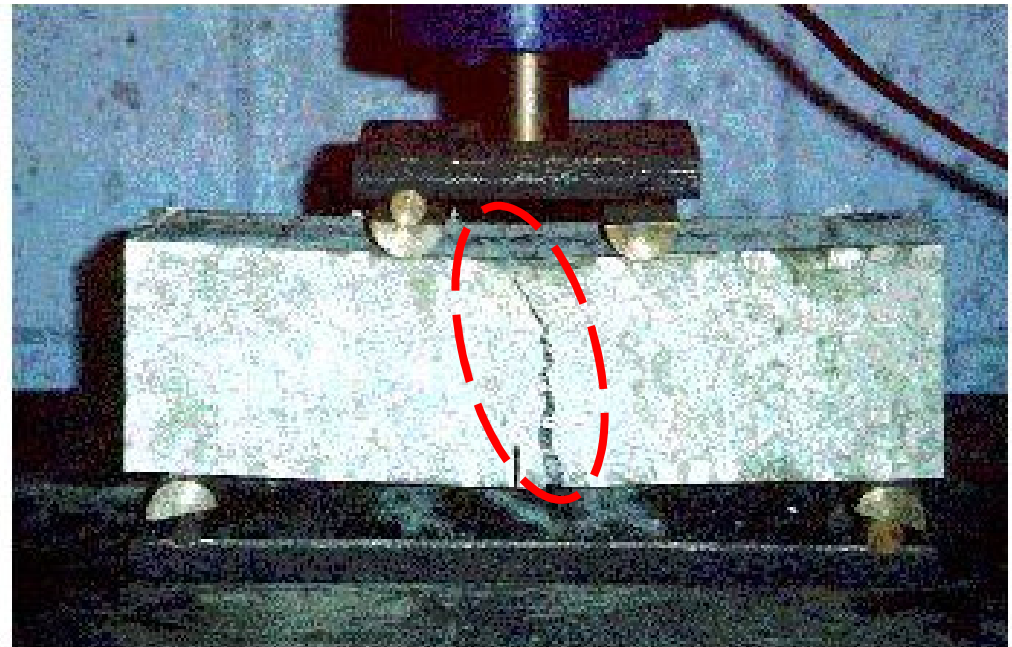
There are compressive stress on the top surface and tensile forces on the bottom surface.

The results of the bending test are similar to the stress-strain curves obtained by tension test; the maximum stress at failure can be analytically calculated.



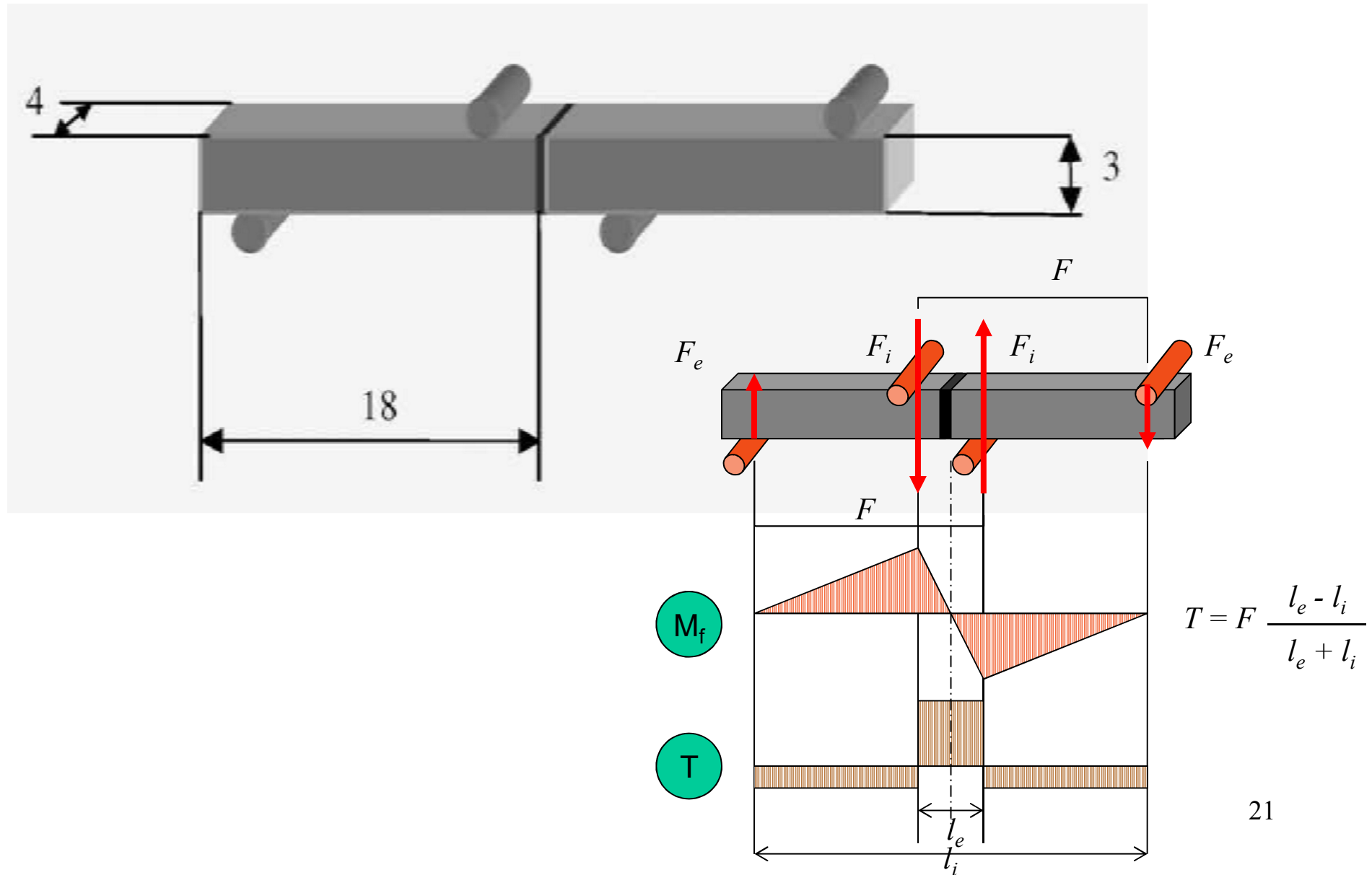
Highly crystalline polymer

← 3-point or
4-point bending test

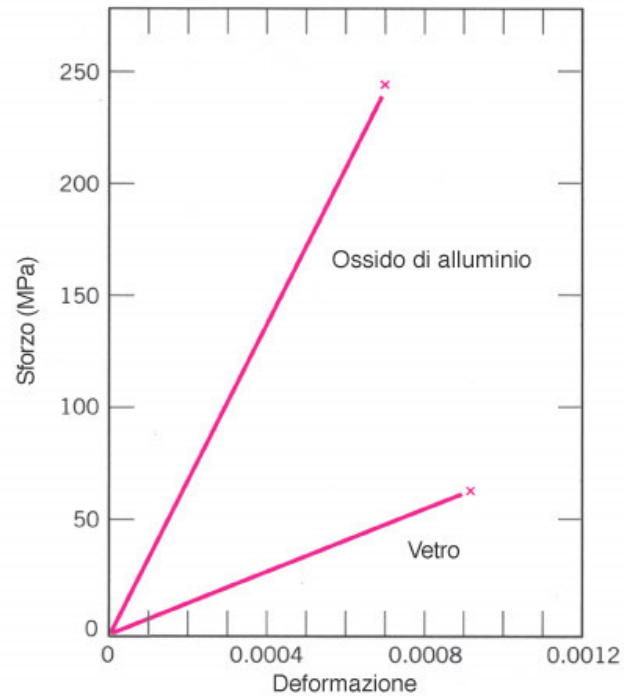


Block of concrete

Asymmetric 4 point bending for ceramic joined materials

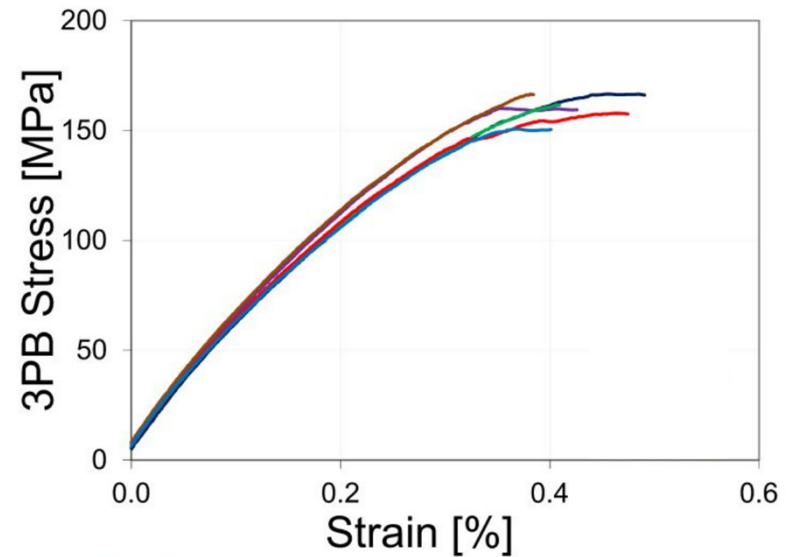


σ (MPa)



Callister
Scienza e Ingegneria dei materiali, una introduzione
EdiSES

ϵ



**BENDING TEST on materials
with plastic deformation**

**BENDING TEST on Al_2O_3
and glass**

Flexural strength of ceramics

Table 13.5 Tabulation of Flexural Strength (Modulus of Rupture) and Modulus of Elasticity for Ten Common Ceramic Materials

<i>Material</i>	<i>Flexural Strength</i>		<i>Modulus of Elasticity</i>	
	<i>MPa</i>	<i>ksi</i>	<i>GPa</i>	<i>10⁶ psi</i>
Silicon nitride (Si ₃ N ₄)	250–1000	35–145	304	44
Zirconia ^a (ZrO ₂)	800–1500	115–215	205	30
Silicon carbide (SiC)	100–820	15–120	345	50
Aluminum oxide (Al ₂ O ₃)	275–700	40–100	393	57
Glass-ceramic (Pyroceram)	247	36	120	17
Mullite (3Al ₂ O ₃ -2SiO ₂)	185	27	145	21
Spinel (MgAl ₂ O ₄)	110–245	16–35.5	260	38
Magnesium oxide (MgO)	105 ^b	15 ^b	225	33
Fused silica (SiO ₂)	110	16	73	11
Soda-lime glass	69	10	69	10

^a Partially stabilized with 3 mol% Y₂O₃.

^b Sintered and containing approximately 5% porosity.

TABLE 8.2 Properties of Various Ceramics at Room Temperature

Material	Symbol	Transverse rupture strength (MPa)	Compressive strength (MPa)	Elastic modulus (GPa)	Hardness (HK)	Poisson's ratio (ν)	Density (kg/m ³)
Aluminum oxide	Al ₂ O ₃	140–240	1000–2900	310–410	2000–3000	0.26	4000–4500
Cubic boron nitride	CBN	725	7000	850	4000–5000	—	3480
Diamond	—	1400	7000	830–1000	7000–8000	—	3500
Silica, fused	SiO ₂	—	1300	70	550	0.25	—
Silicon carbide	SiC	100–750	700–3500	240–480	2100–3000	0.14	3100
Silicon nitride	Si ₃ N ₄	480–600	—	300–310	2000–2500	0.24	3300
Titanium carbide	TiC	1400–1900	3100–3850	310–410	1800–3200	—	5500–5800
Tungsten carbide	WC	1030–2600	4100–5900	520–700	1800–2400	—	10,000–15,000
Partially stabilized zirconia	PSZ	620	—	200	1100	0.30	5800

Note: These properties vary widely depending on the condition of the material.

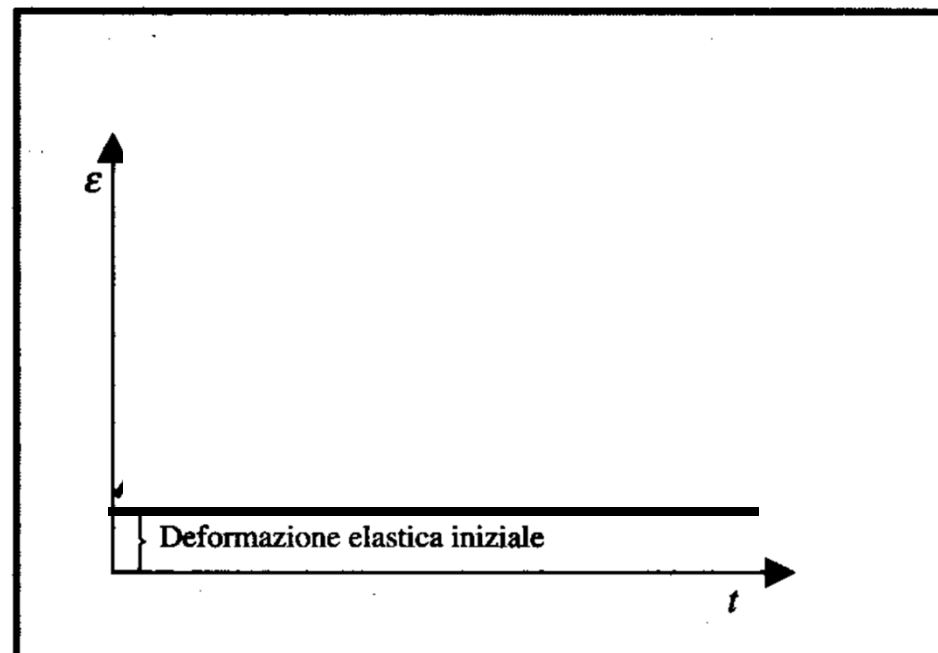
Condition of use!! T ????

TEST:

- Draw ε versus t (strain vs time) when a material is loaded at $\sigma = \text{constant}$, in the elastic field, in two conditions:
- Room temperature
- High temperature (e.g. 50% T_m)

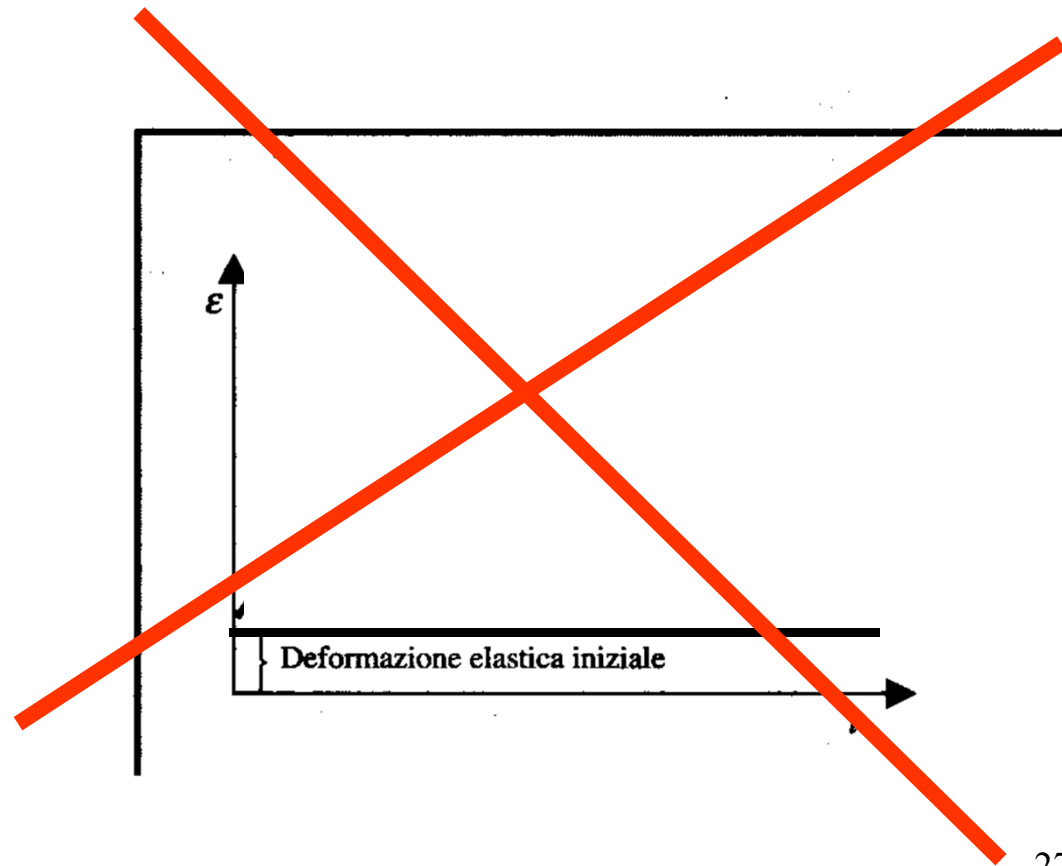
TEST

- Draw ε versus t (strain vs time) when a material is loaded at $\sigma = \text{constant}$, in the elastic field



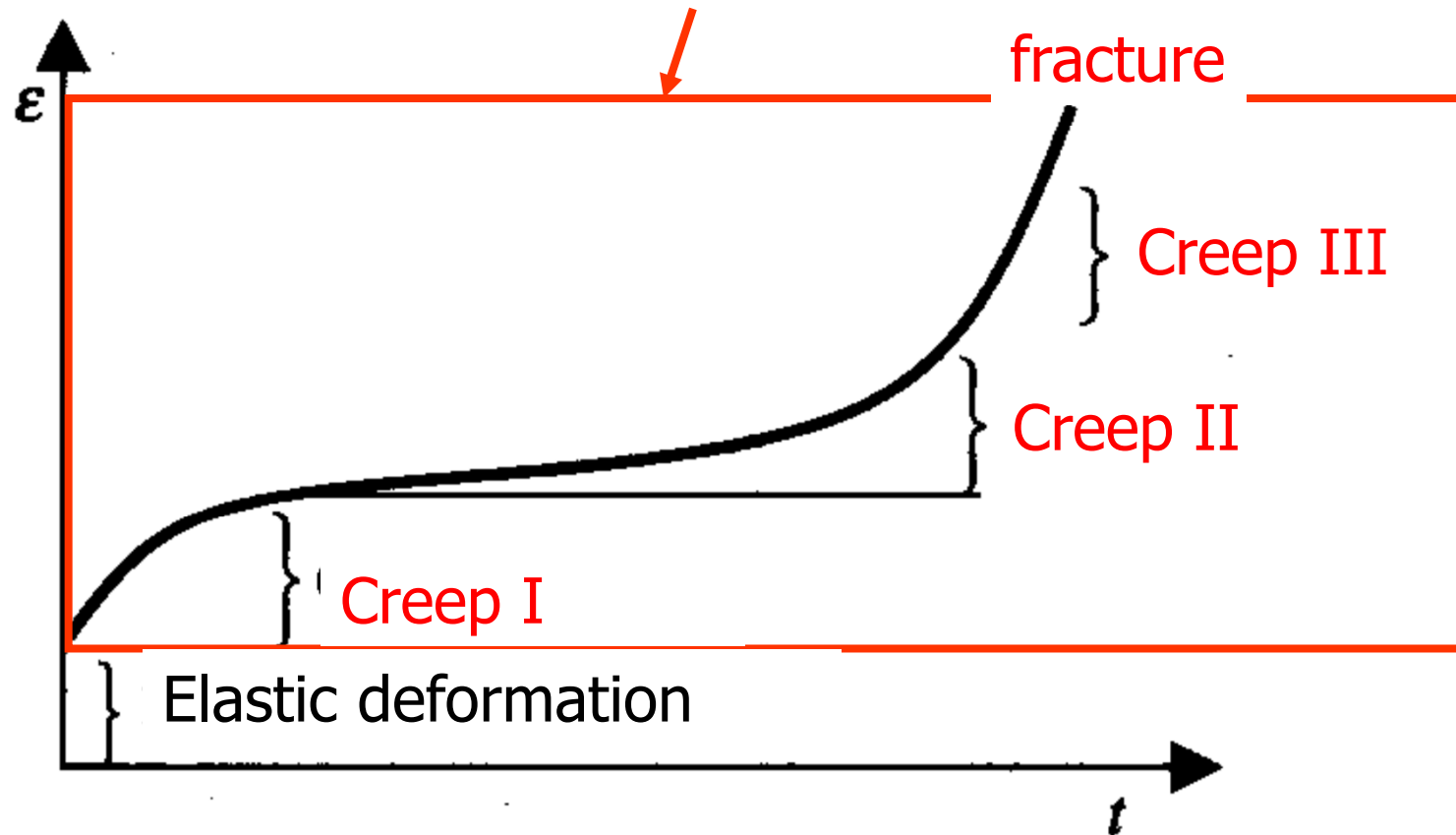
TEST

- Draw ε versus t (strain vs time) when a material is loaded at $\sigma = \text{constant}$, in the elastic field



CREEP

Plastic deformation even if stress is in the elastic field



CREEP

Constant load in the elastic field can give to a plastic deformation, progressive to fracture.

- Thermally activated process

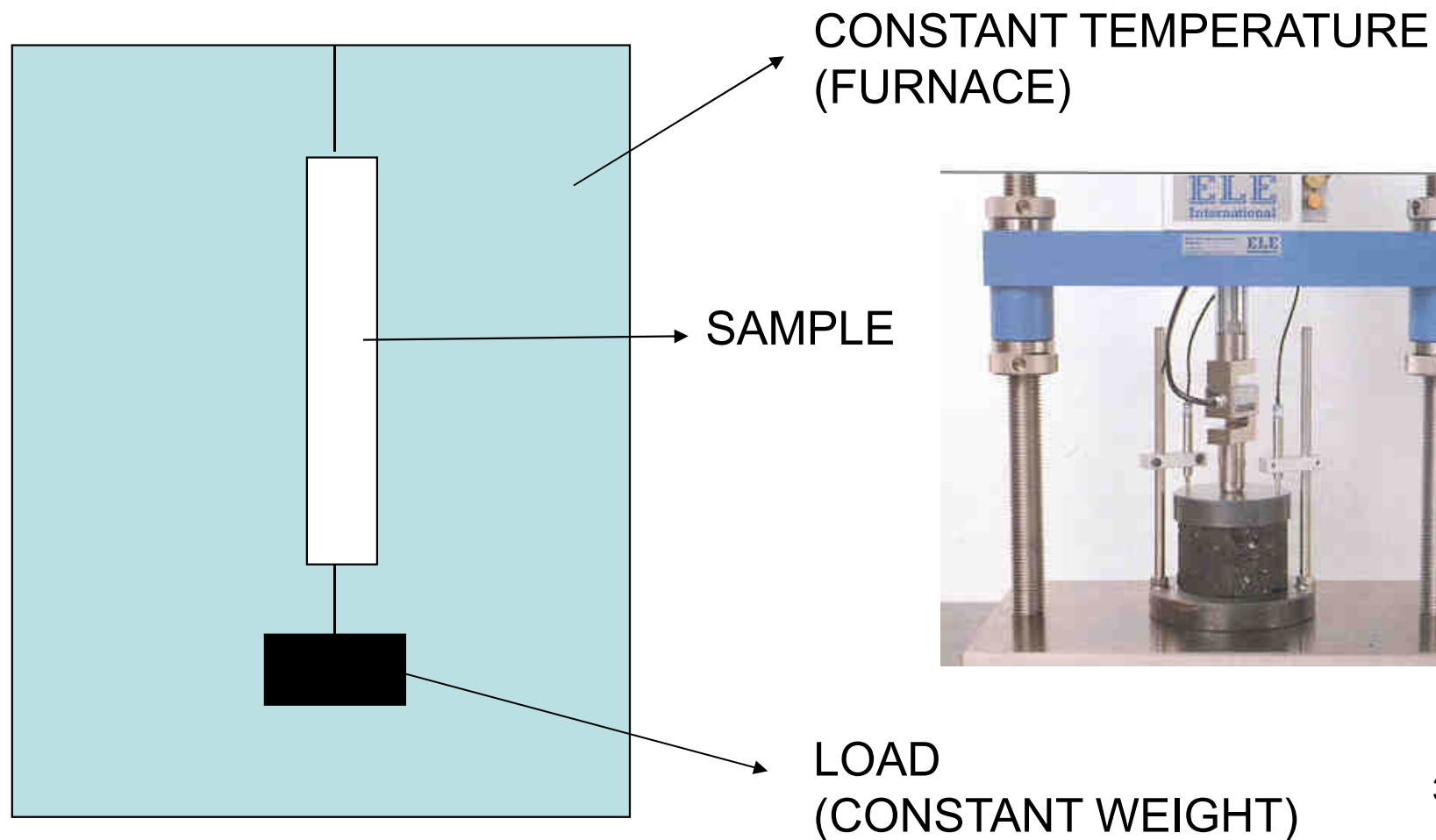
Metals **$T > 0,3-0,4 T_{\text{melting}}$**

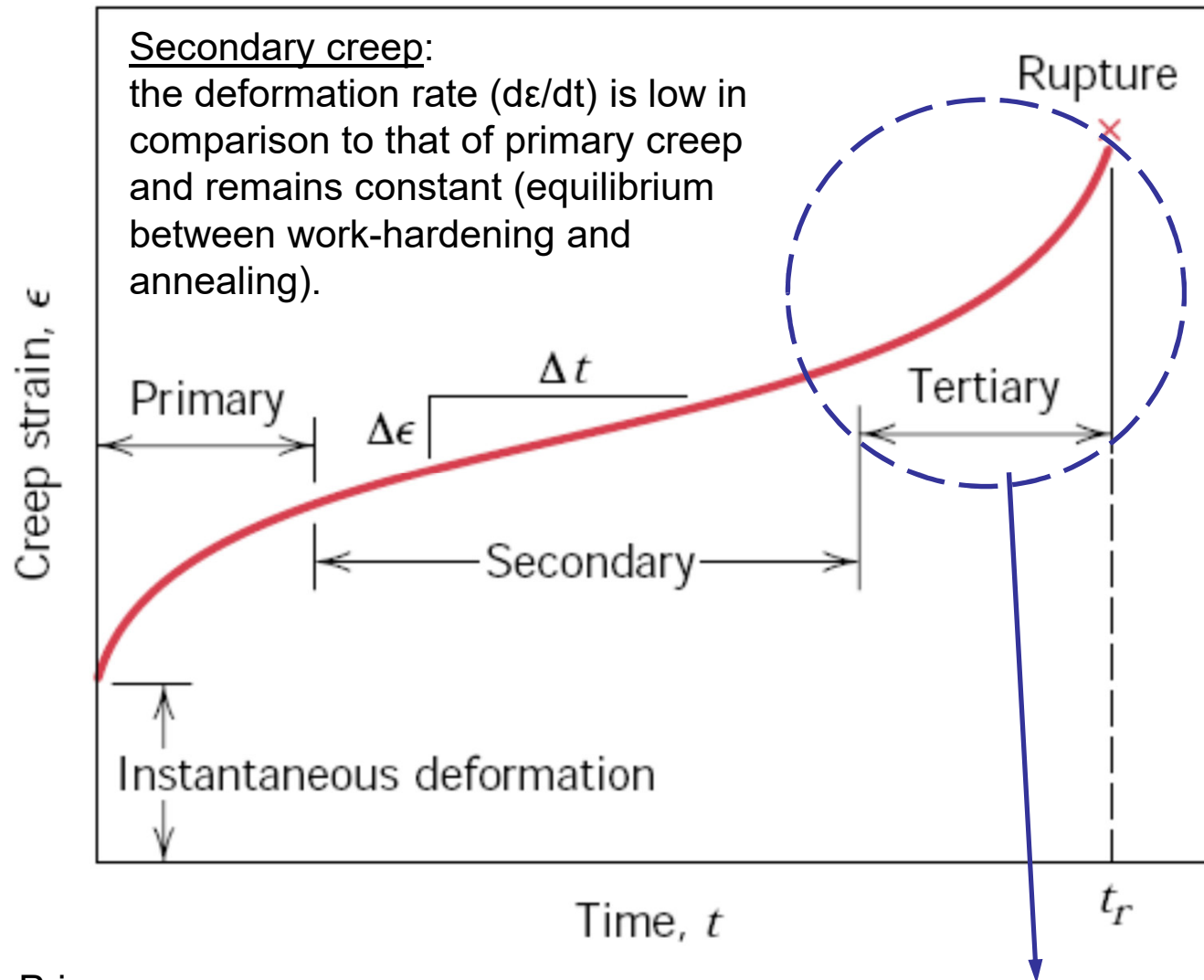
Ceramics **$T > 0,6-0,7 T_{\text{melting}}$**

Amorphous materials **$T > T_{\text{glass}}$**

creep test: a constant load is applied at a given T , strain is recorded versus time.

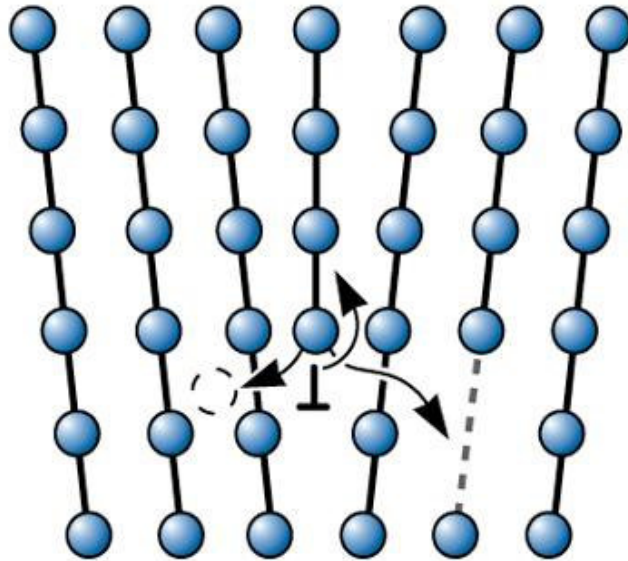
Creep test execution: a specimen is subjected to a **constant load** and **strain is measured during a significant period of time** while maintaining the temperature **high and constant**.



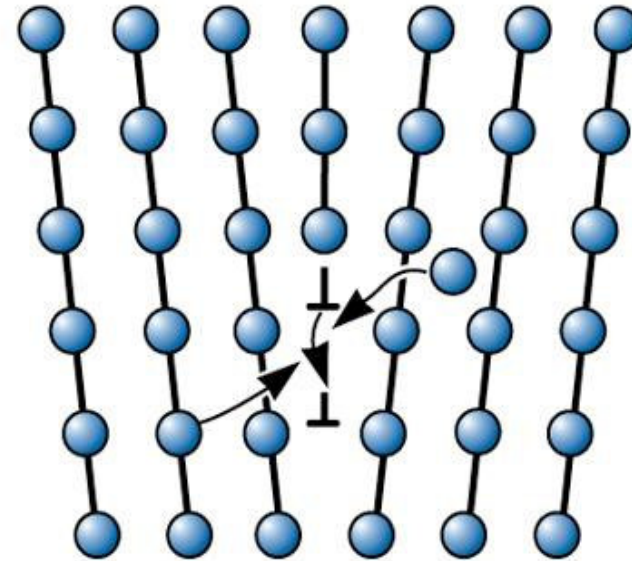


Primary creep:
we have elastic + plastic deformation; the latter is favoured by dislocation movement. The deformation rate ($d\epsilon/dt$) progressively decreases.

Tertiary creep: at grain boundaries some **microcracks and voids** are formed → weakness points → cracks propagation → material rupture



(a)



(b)

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Stage I: primary creep

Dislocations can climb (a) when atoms leave the dislocation line to create interstitials or to fill vacancies or (b) when atoms are attached to the dislocation line by creating vacancies or eliminating interstitials

Stage II : steady state

$$\dot{\mathcal{E}}_{ss} = \frac{d\mathcal{E}}{dt} = C \cdot \mathcal{G}^n \cdot \exp\left(-\frac{E_{CR}}{R \cdot T}\right) = A \cdot \mathcal{G}^n$$

C = constant depending on materials

$\mathcal{G}$ = applied stress

n = coefficient depending on materials $3 < n < 8$

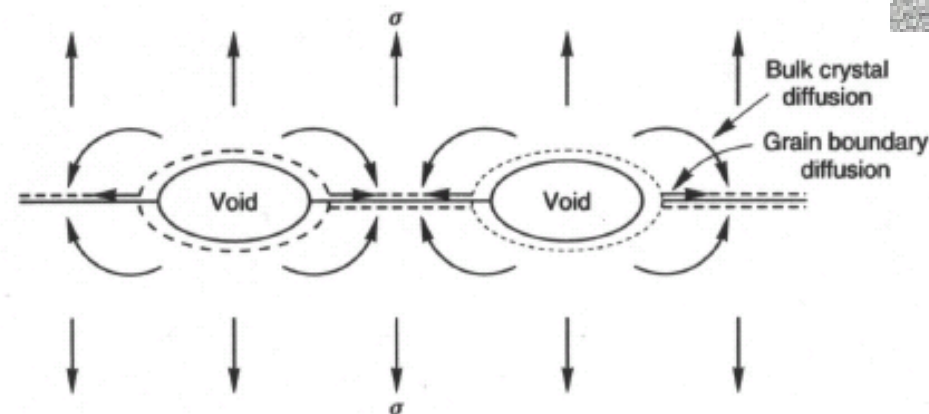
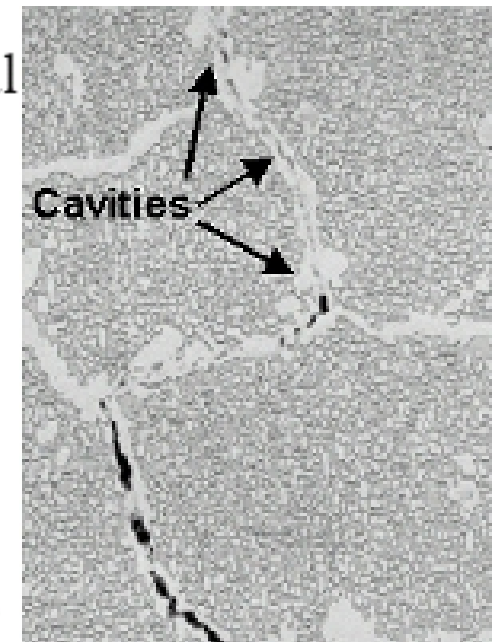
E_{CR} = activation Energy

R = perfect gas constant

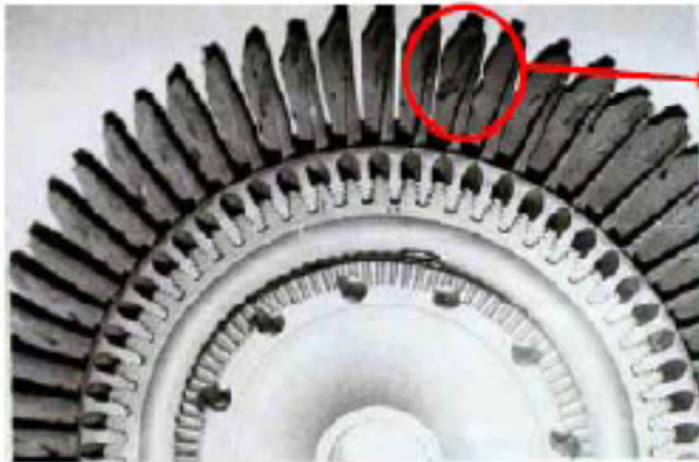
T = test Temperature

Creep fracture Mechanism

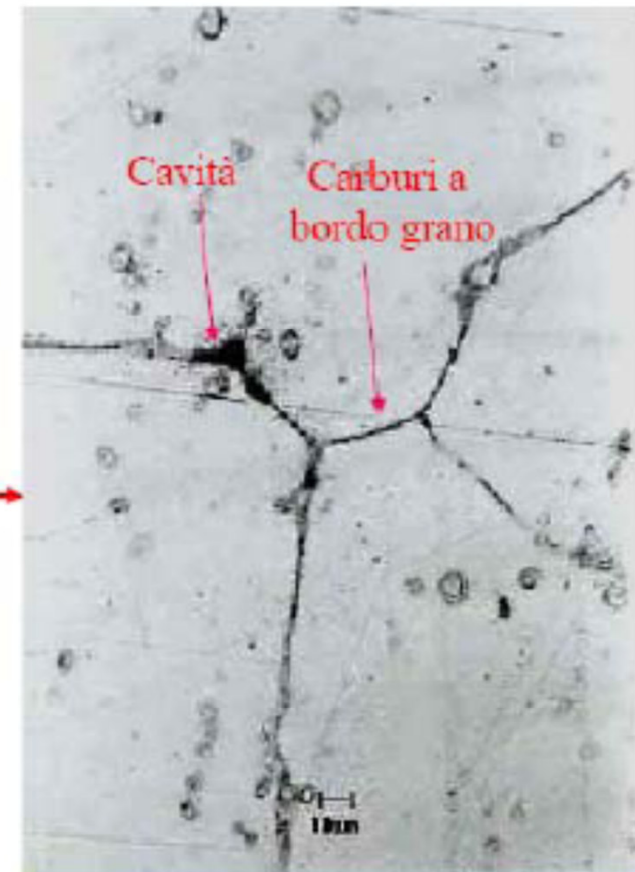
- During creep, damage accumulates in the form of internal voids.
- Voids appear on grain boundaries normal to the tensile stress.
- Atoms diffuse from the void surfaces into grain boundaries, and the voids grow.
- Voids grow faster and faster, until they link, leading to creep fracture.



Stage III: tertiary creep

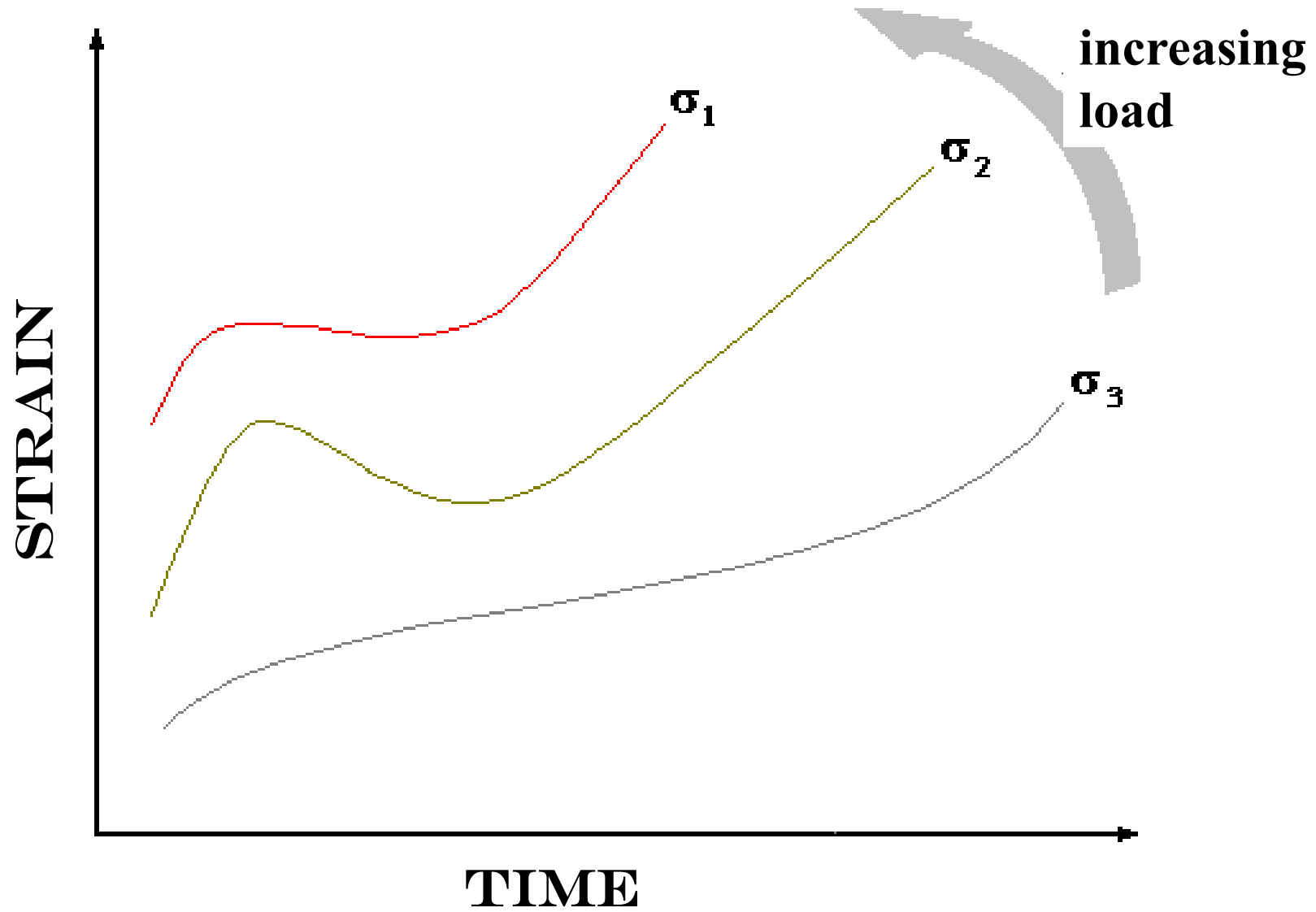


Turbina con palette fratturate

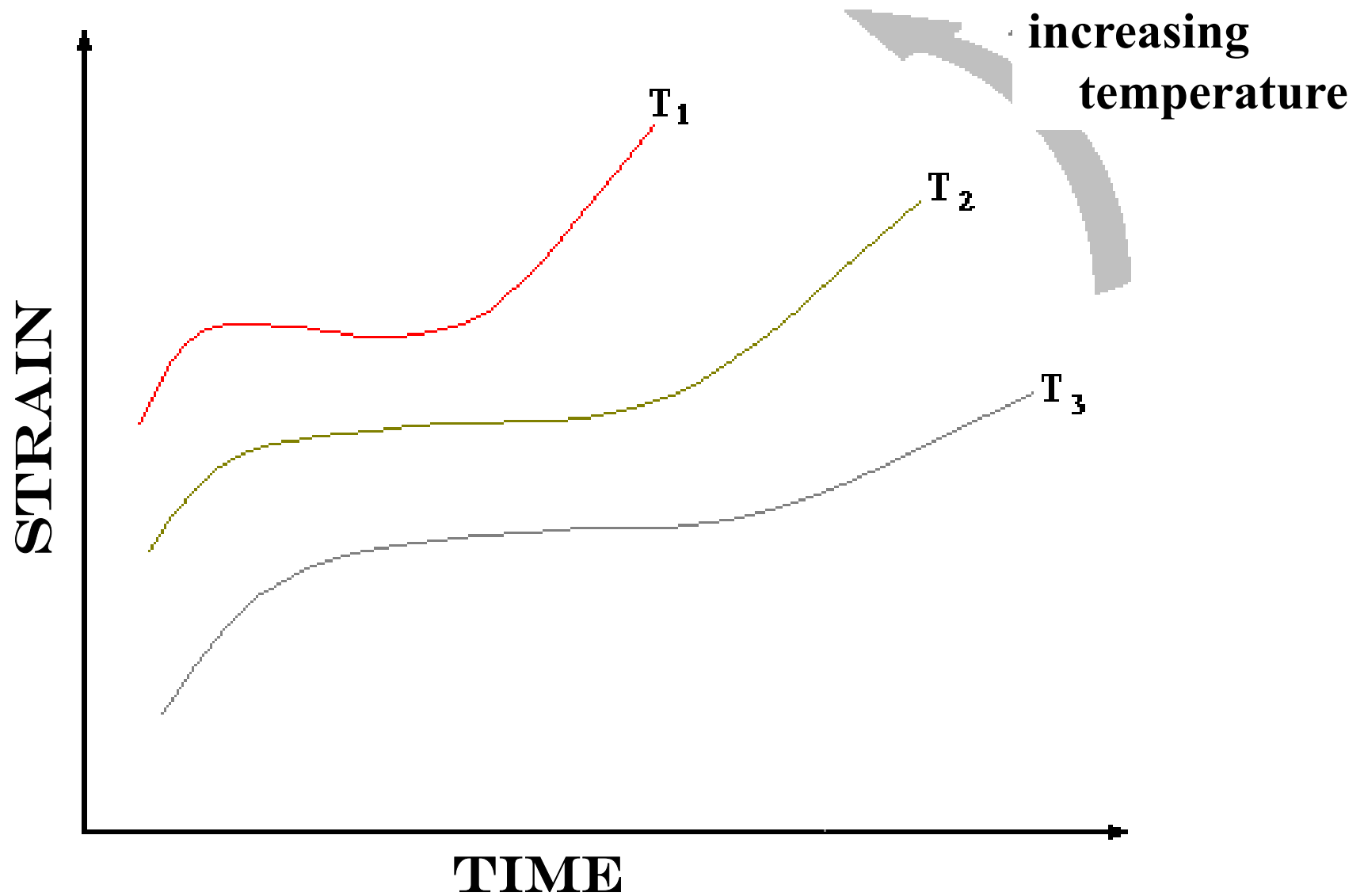


Formazione di cavità dovute a creep e precipitazione di carburi a bordo grano

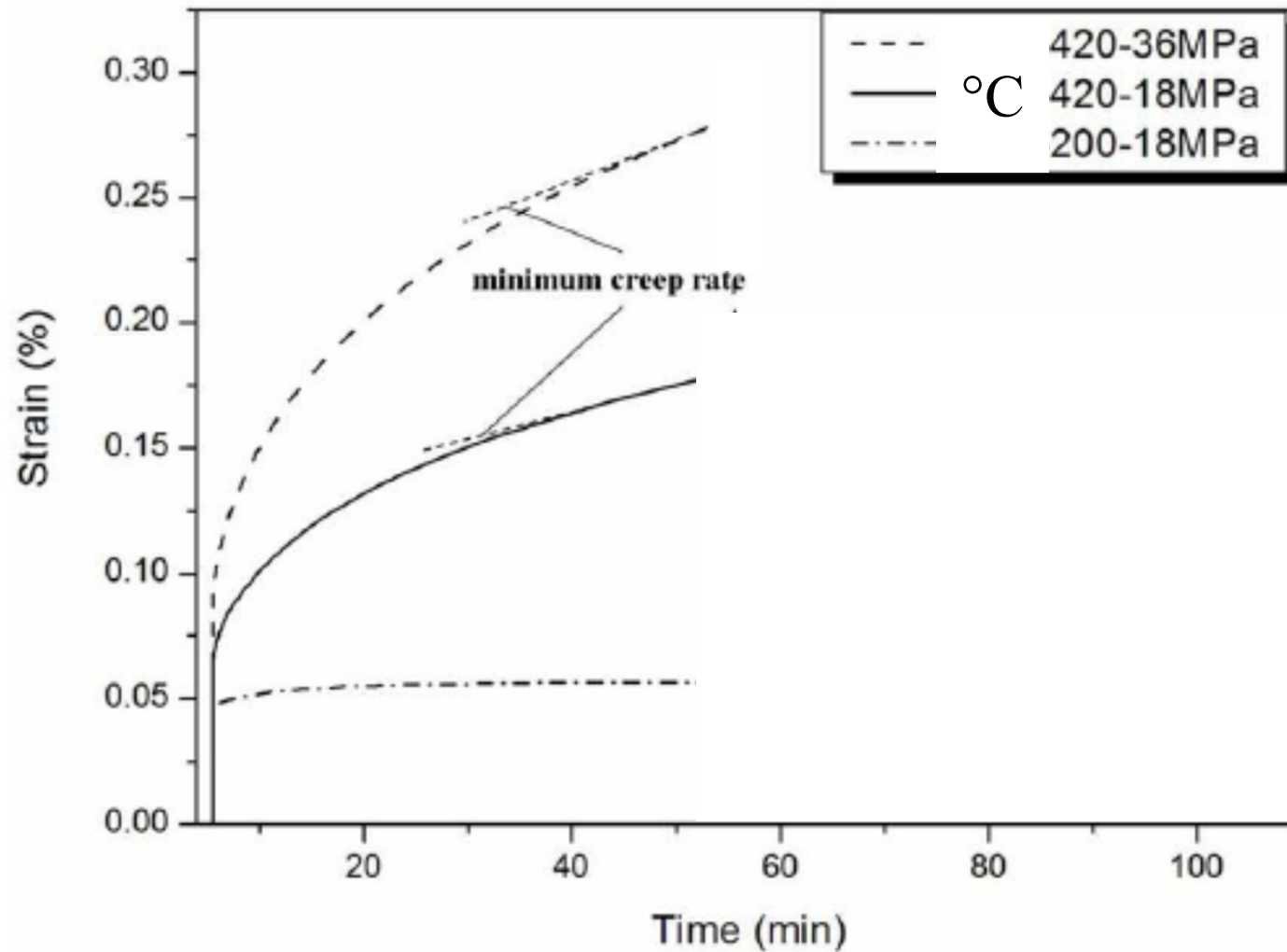
Effect of the applied load



Effect of the working temperature



Experimental creep curves for a borosilicate glass
at 200 and 420 °C, from 18 to 36 MPa)

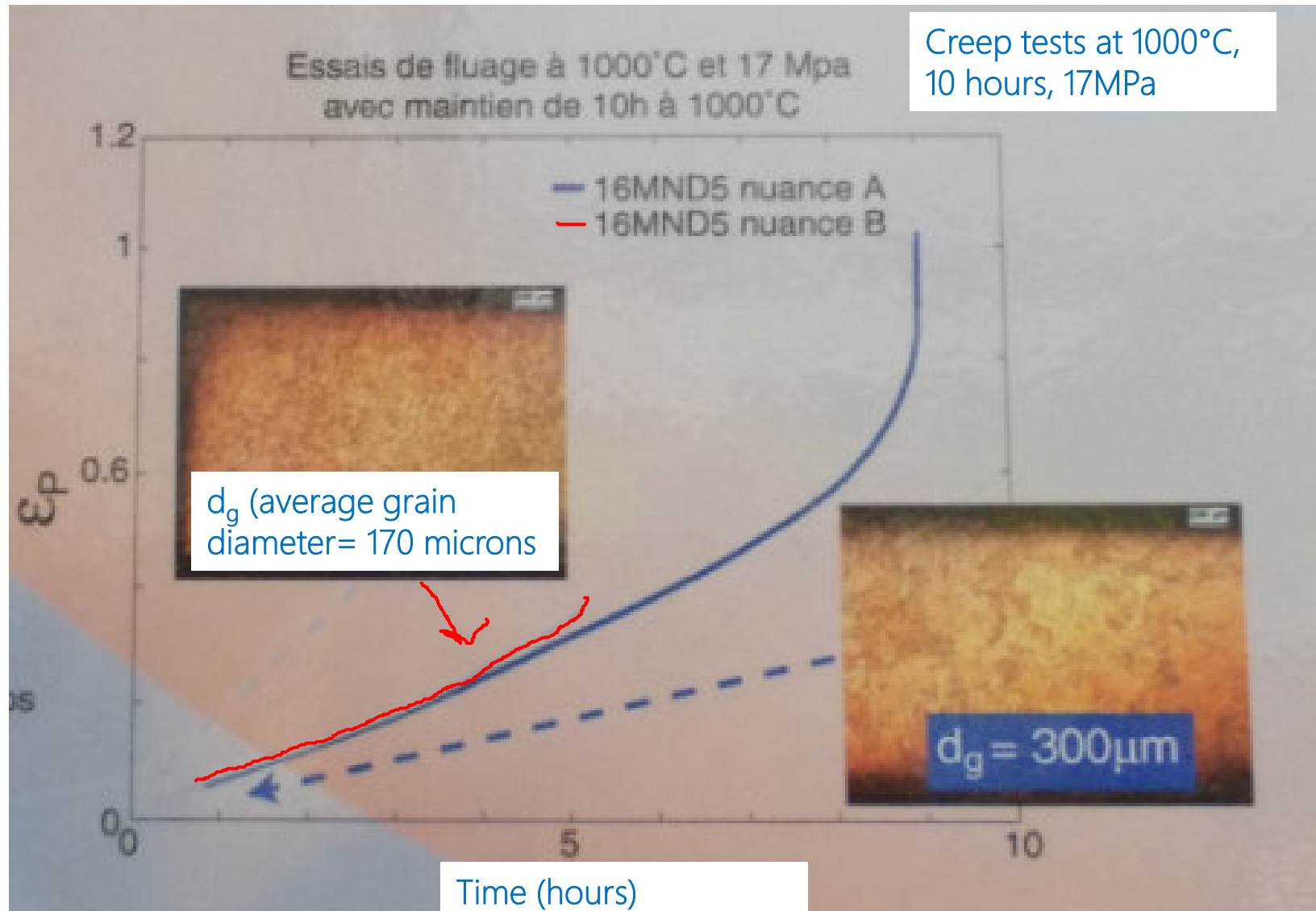


How to increase creep resistance?

- High melting T and E materials
- Large grains or mono-crystals (*small grains increase grain motion at the grain boundaries*)
- Solid solutions (*to restrict dislocation glide*)
- Precipitates (*to restrict dislocation glide*)
- Second phases (composites) (*to restrict dislocation glide*)

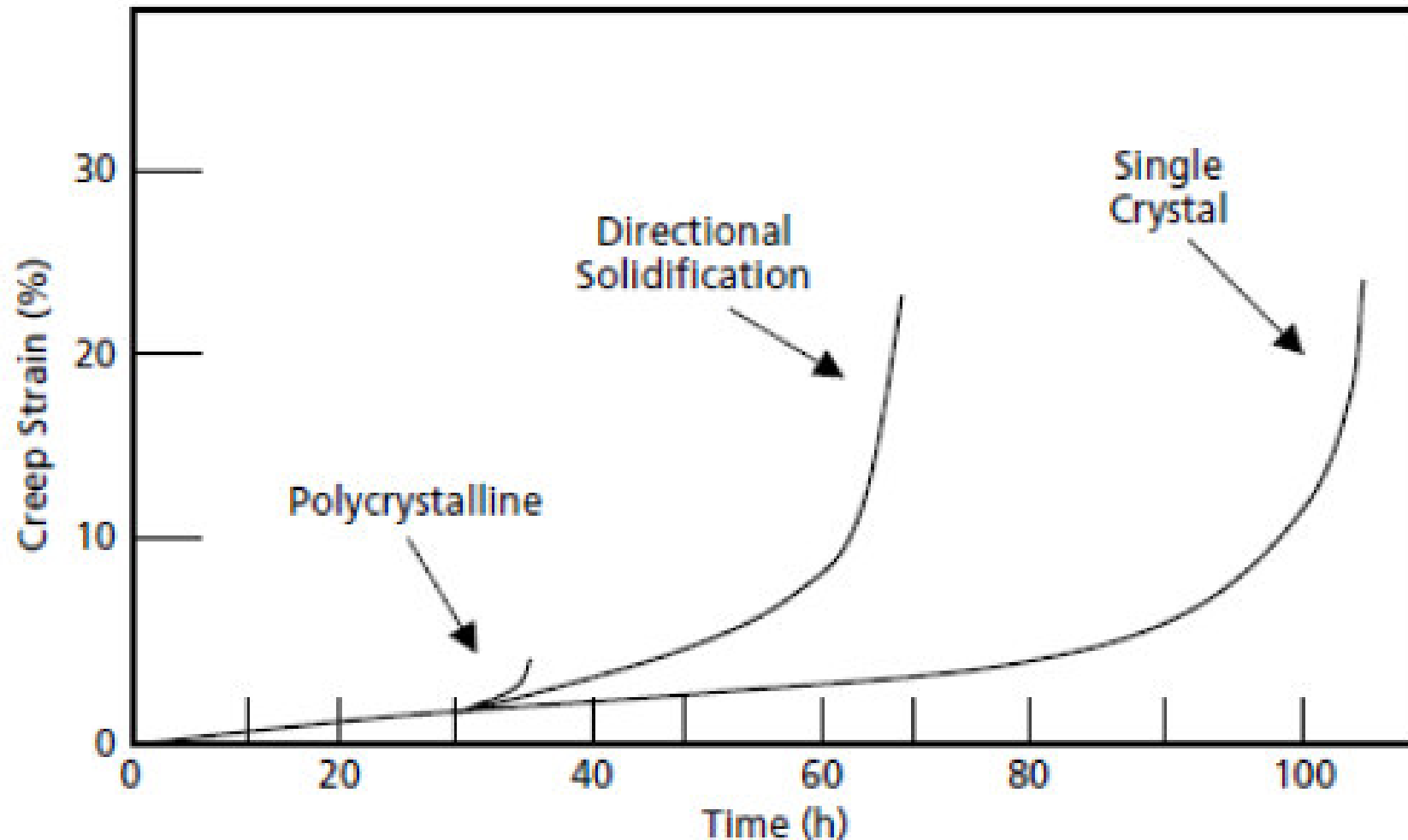
Effect of grain size on yield strength???

Grain size effect on creep



Crystalline structure effect on creep

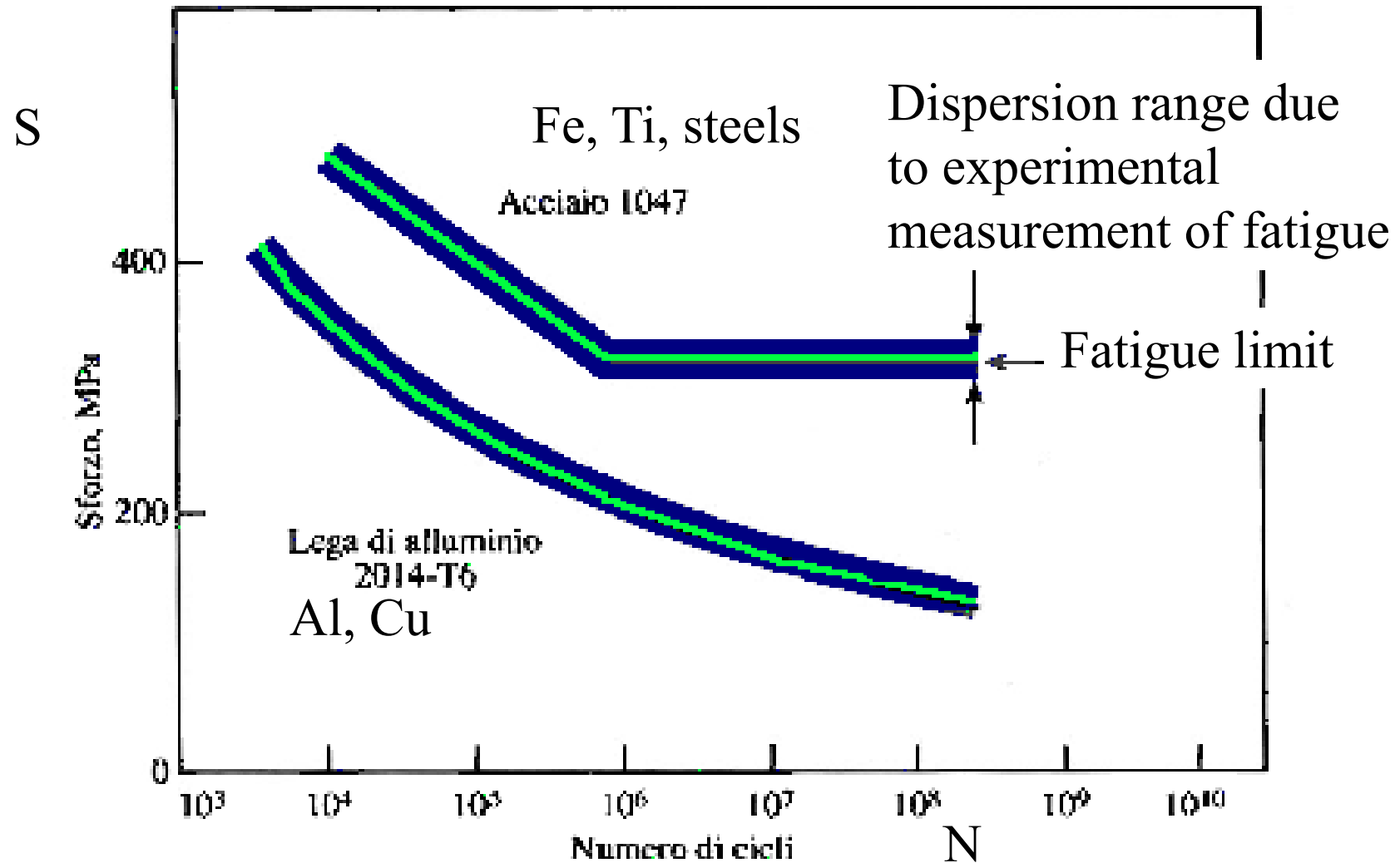
- large grains can increase creep resistance because there is less grain boundary surface able to slide; several small grains slide one on the other one; with a few large grains this is less evident;
- even longer creep life if you have a single crystal with no grains at all



TEST:

- Apply a tensile stress to a material in the elastic field.
- Can this material fail?
- Repeat the test several times.
- Is it possible to have the material failure ?
- If yes, draw a graph with the applied stress (σ) versus the number of cycles (N) necessary to achieve the material failure

S-N curves: stress (S) vs number of cycles (N) to obtain failure



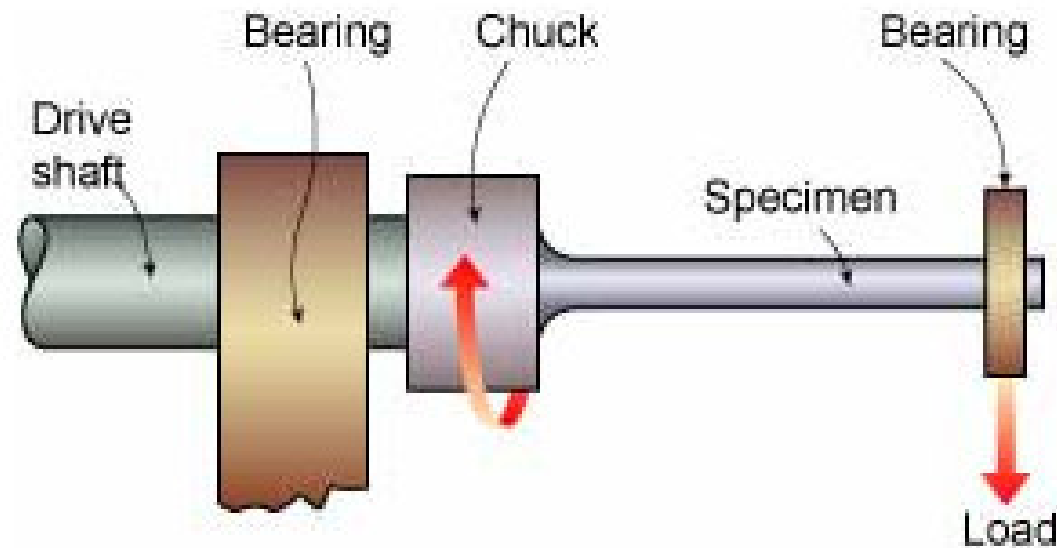
Wohler curves

Fatigue

- Failure of materials due to cyclic loading.
- Main reason of mechanical failure of materials
- Failure happens at stress lower than σ_R or σ_Y
- Catastrophic failure of materials (ALSO FOR DUCTILE MATERIALS !)
- Fatigue tests: materials are cyclically loaded at different stresses up to failure.
- Fatigue limit: when cyclically loaded below this limit, materials do not fail (time considered by the graph)

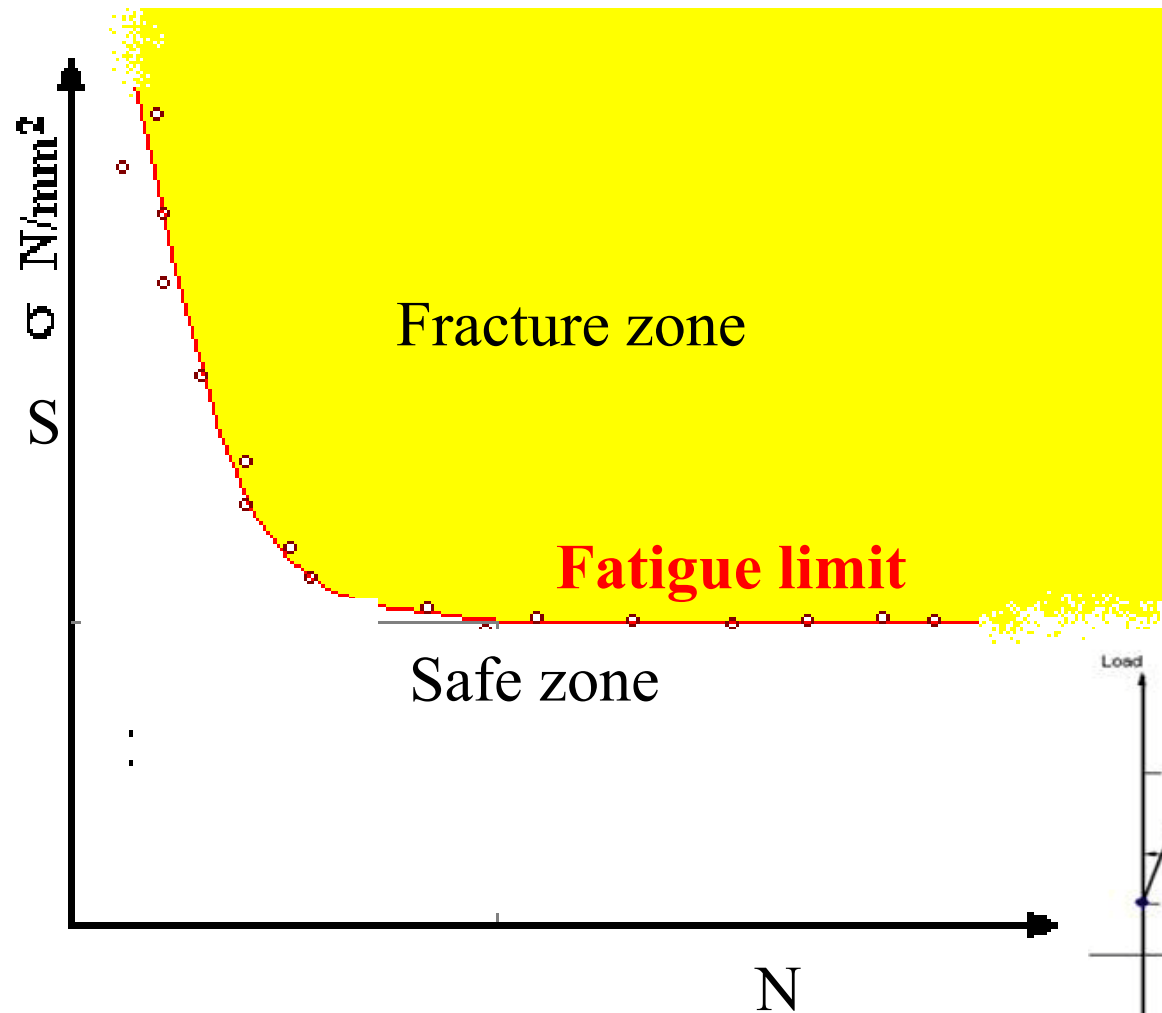
Experimental measurement: fatigue testing

Rotating-bending test: the specimen is rotated and a load perpendicular to the rotational axis is imposed.



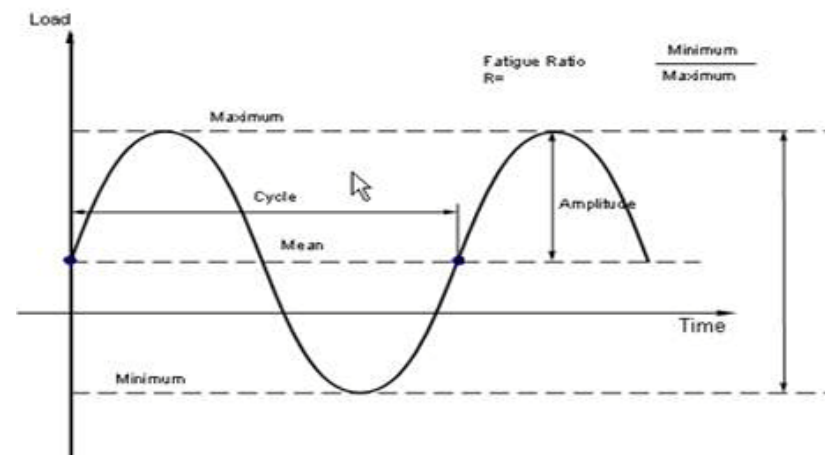
This test is very time-consuming → the specimens are tested under several millions of stress cycles!

S-N curve



$$S = \sigma_a = \frac{\sigma_{\max} - \sigma_{\min}}{2}$$

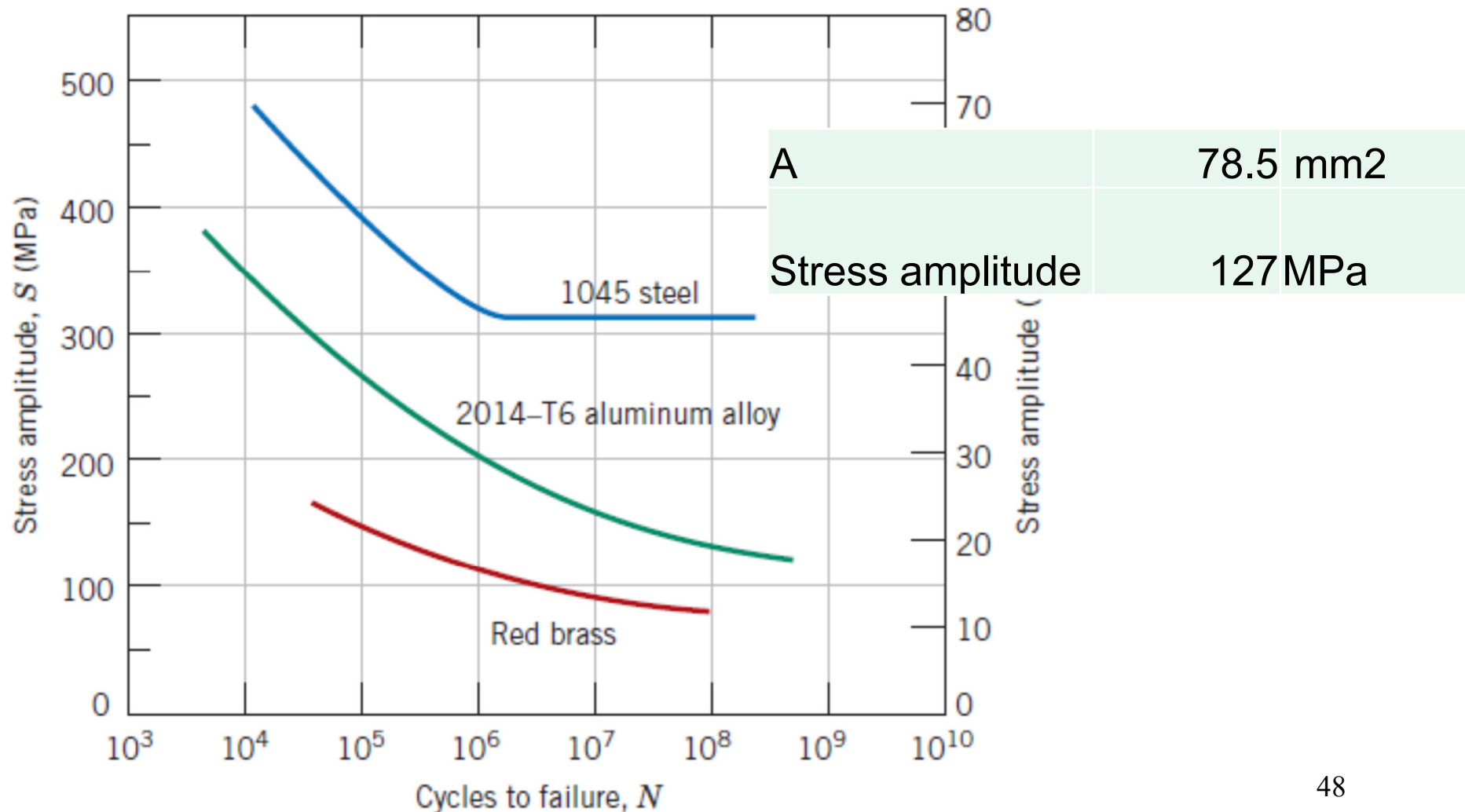
$\sigma_{\max}$, $\sigma_{\min}$: applied stresses during tests



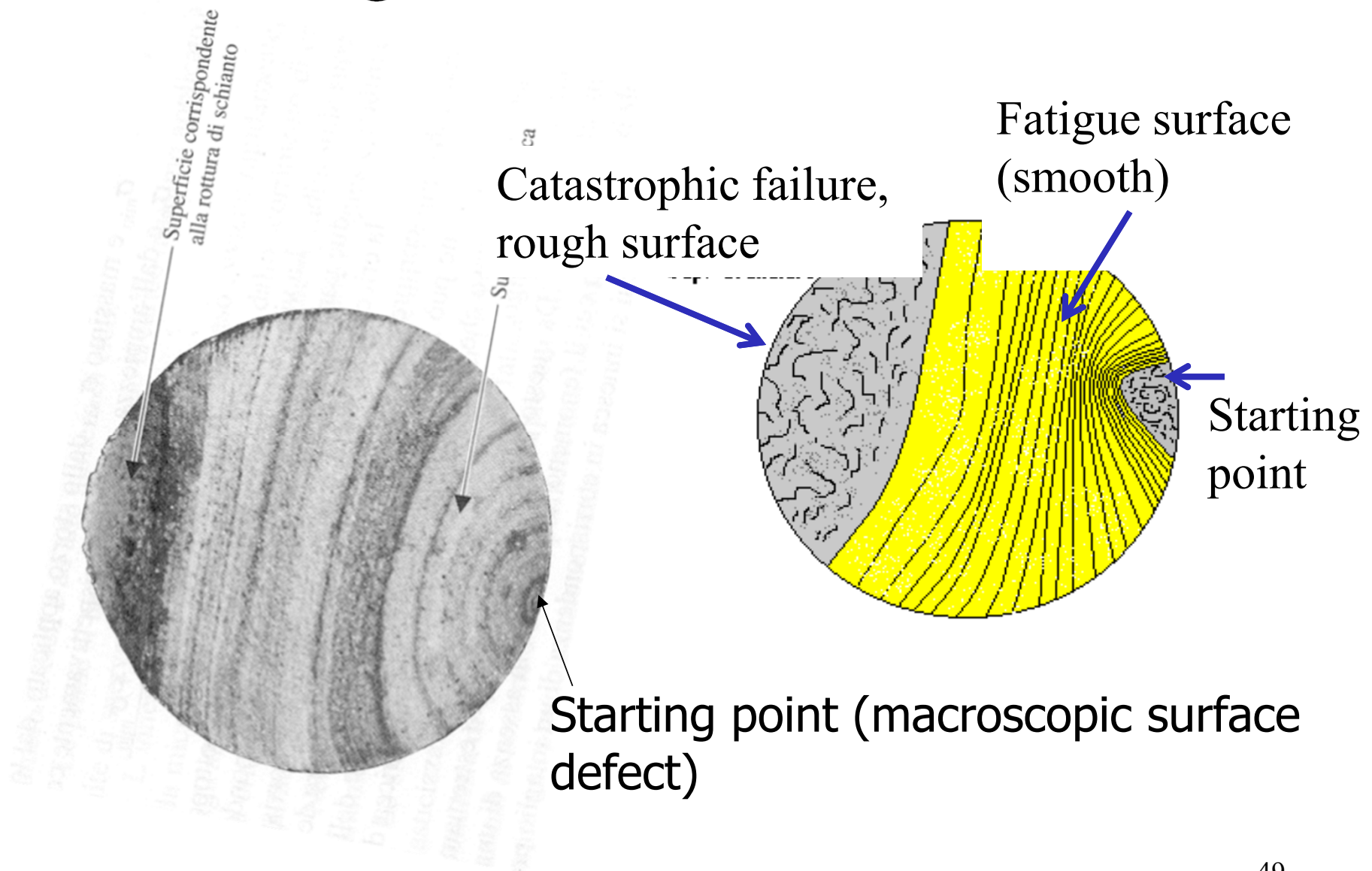
Materials and Fatigue

- Between 35-65% of their tensile strength most of metals fail because of fatigue (e.g. Fe, Ti alloys, intrinsic fatigue limit)
- Other metals fail in any case after a given limit (e.g. Al, no intrinsic fatigue limit)
- Fatigue resistance: stress necessary to fracture the material after a given number of cycles at this stress
- Fatigue life: number of cycles necessary to fracture the materials at a given load.

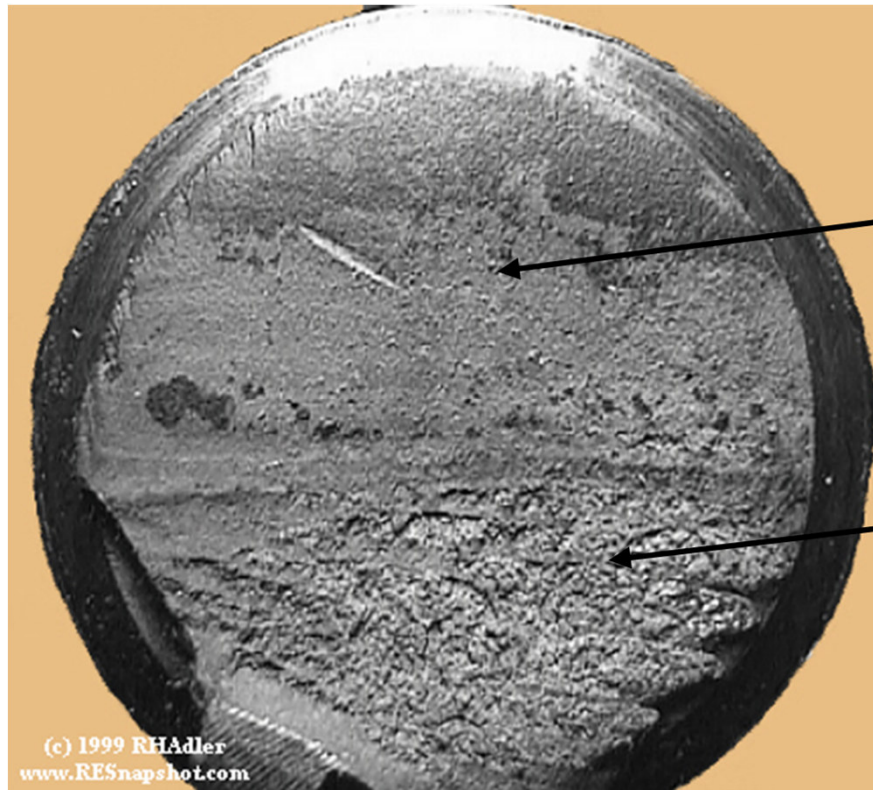
A cylindrical component (10 mm diameter) is cyclically loaded at 10 kN (compression, tensile stresses). Calculate its fatigue life if the component is made of brass, aluminum alloy or steel



Fatigue: fracture surface



Fatigue: fracture surface



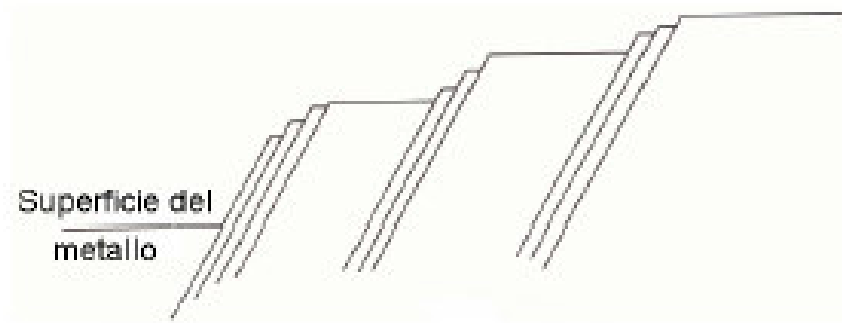
Fatigue surface, smooth

Catastrophic brittle failure,
also on ductile materials,
rough surface

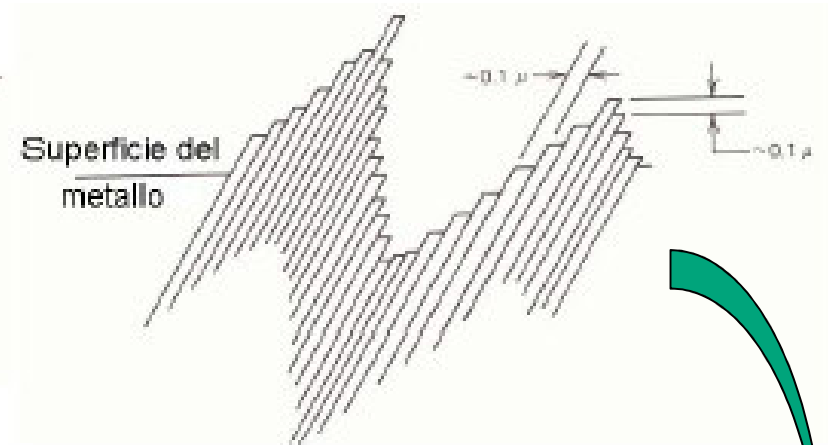
<http://www.resnapshot.com/MP1198-2.jpg>

Ductile and brittle fracture

- Ductile fracture: high plastic deformation at the crack tip, slow crack propagation (i.e. stable propagation)
- Brittle fracture: no (or low) plastic deformation at the crack tip, quick crack propagation (i.e. unstable propagation), catastrophic failure

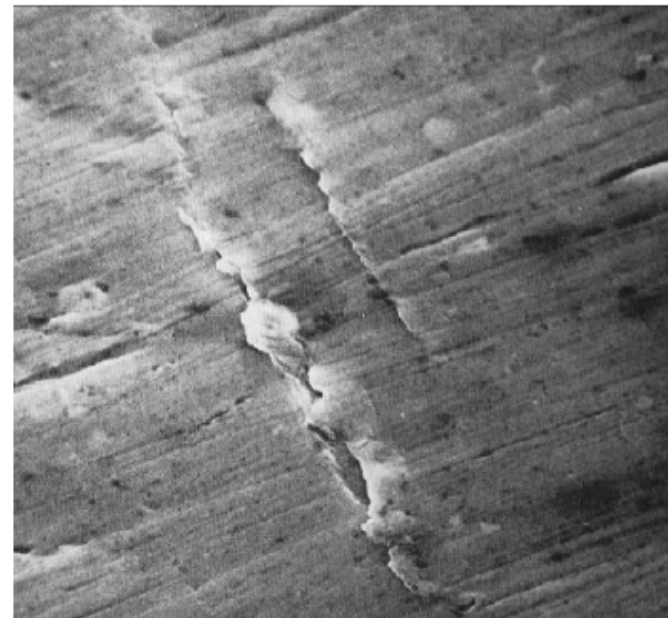
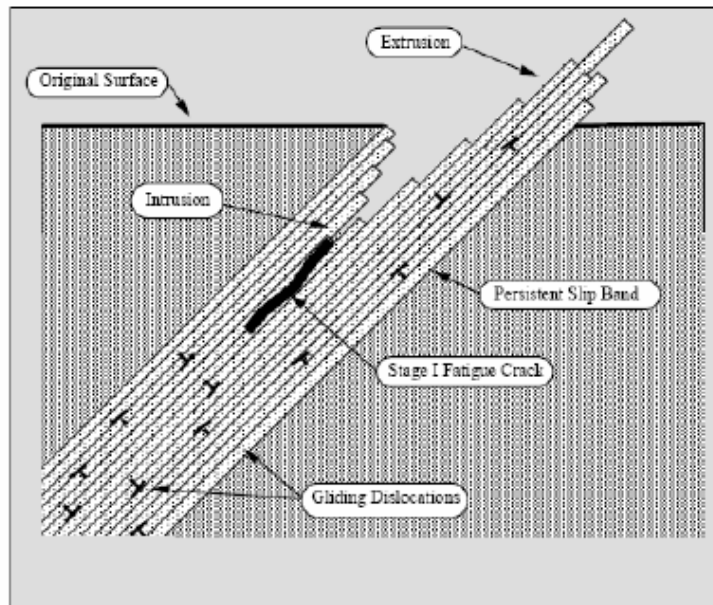


Static loads



Cyclic loads

Intrusion bands



Stress intensity factor

- Macro-defects (pores, cracks) in **all materials** act as stress concentration factors
- True stress on the material at the tip of the crack (σ_m) is higher than the *nominal stress* (σ_0)

ρ_t = radius of the crack

a = length of a crack on the surface

- Critical Defects (Griffith Theory, Fracture mechanics)**

- Without defects, mechanical strength would be close to the theoretical values (as it is for monocrystalline materials or small brittle materials)

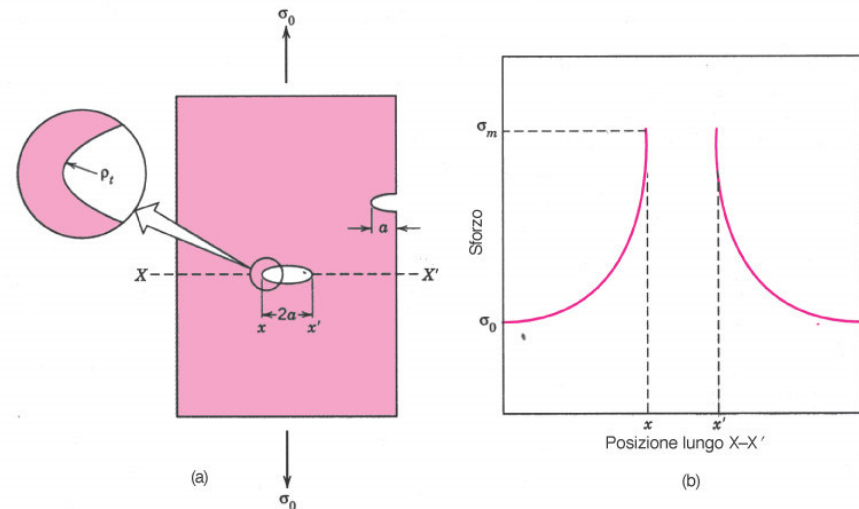


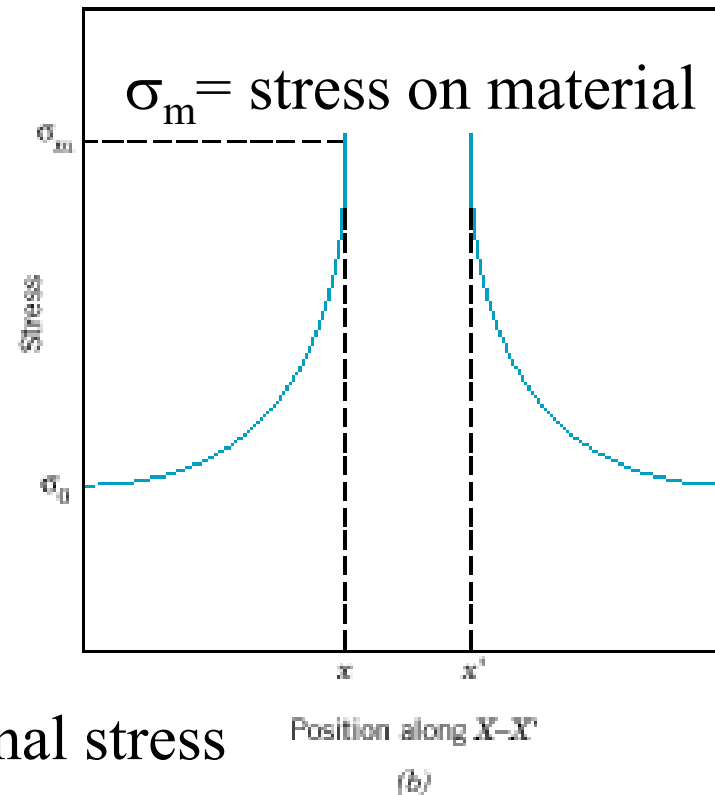
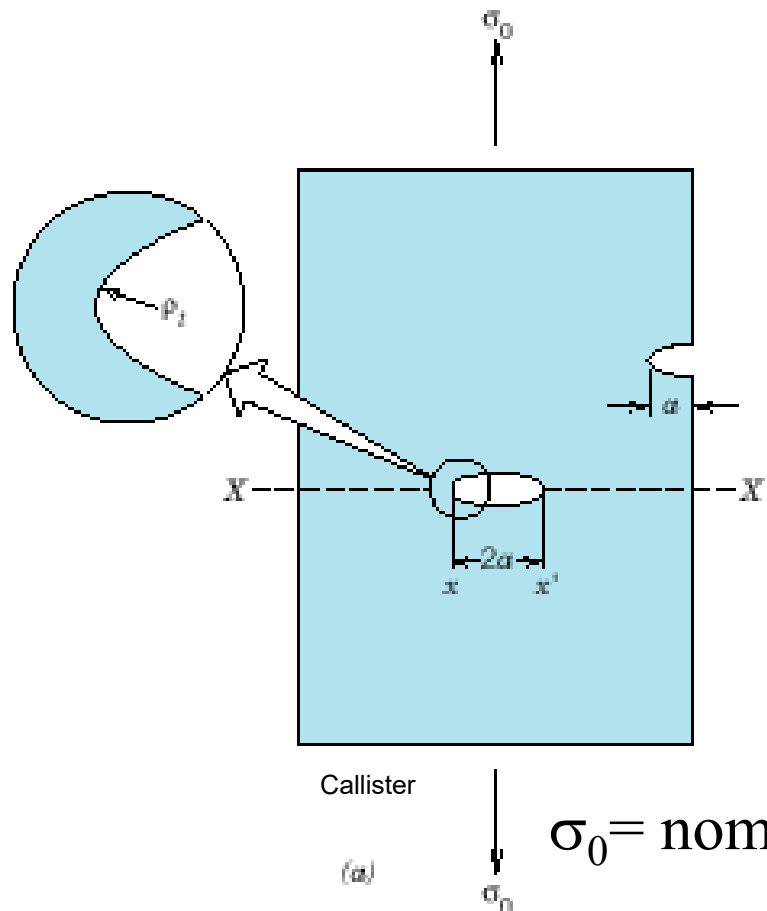
FIGURA 8.7 (a) Geometria di cricche interne e superficiali. (b) Profilo di sforzo lungo la linea $X-X'$ in (a), da cui si evince l'amplificazione dello sforzo all'apice della cricca.

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Stress intensity factor for elliptical cracks with length = $2a$ and small curvature radius:

$K = \sigma_m / \sigma_0 = \text{stress intensity factor}, K = 2(a/\rho)^{1/2}$

$$\sigma_m = 2\sigma_0 \left(\frac{a}{\rho_t} \right)^{1/2}$$



Crack propagation

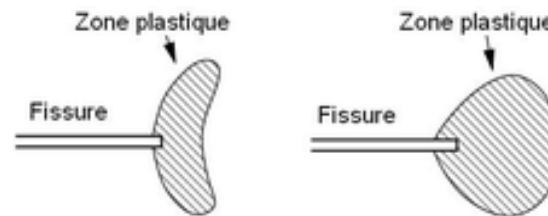
$$\sigma_m = 2\sigma_0 \left(\frac{a}{\rho_t} \right)^{1/2}$$

Role of :

ρ_t = radius of the crack

a = length of a crack on the material surface

- If plastic deformation is possible, ρ_t can increase $\rightarrow$ decrease of σ_m



- If plastic deformation is not possible $\rightarrow$ catastrophic failure.
- **Griffith Theory (with subsequent Irwin modifications)** quantifies these issues

<https://ceramicsandglass.wordpress.com/2021/05/14/a-castle-of-glass-explanation-of-griffiths-fracture-theory/>

Crack propagation and critical parameters

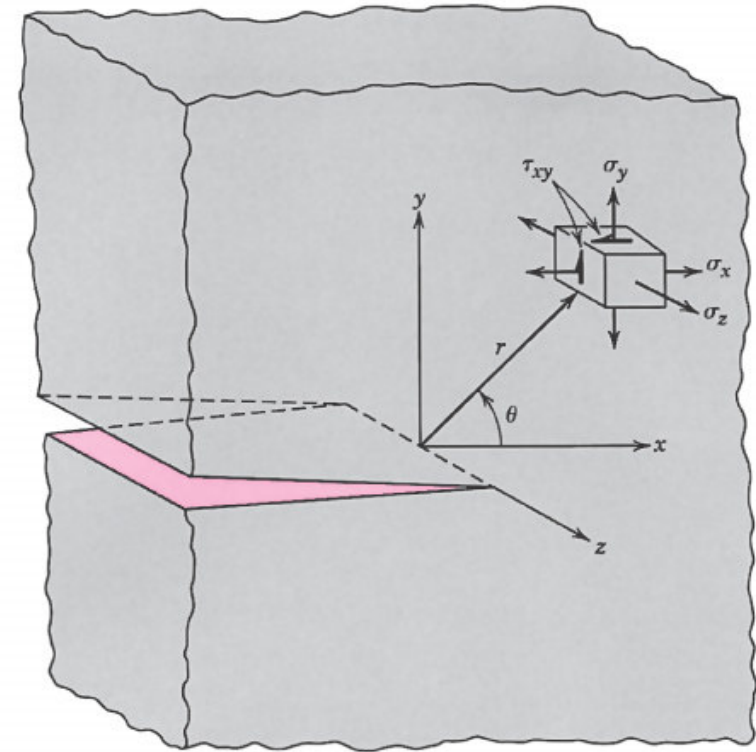
σ_c = critical stress, crack propagation
for $\sigma > \sigma_c$

$$K_{Ic} = Y \sigma_c (\pi a)^{1/2} \text{ (MPa m}^{1/2}\text{)}$$

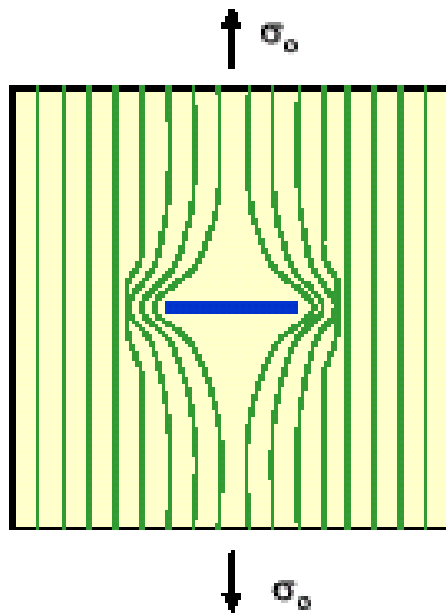
**Fracture toughness
(mode I)**

a = half length of the crack

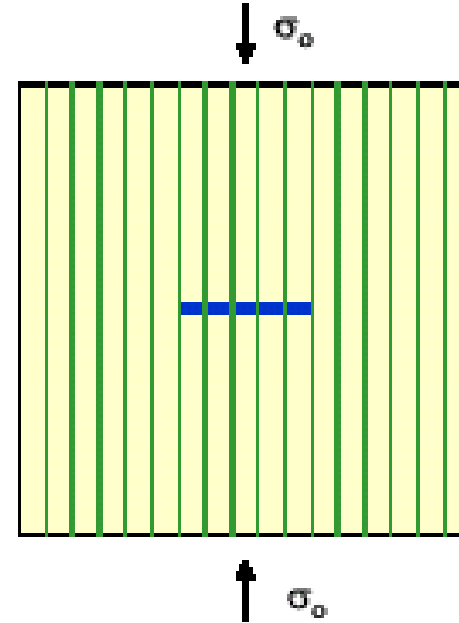
Y = geometrical parameter



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Stress concentration at crack tip



No stress concentration at crack tips

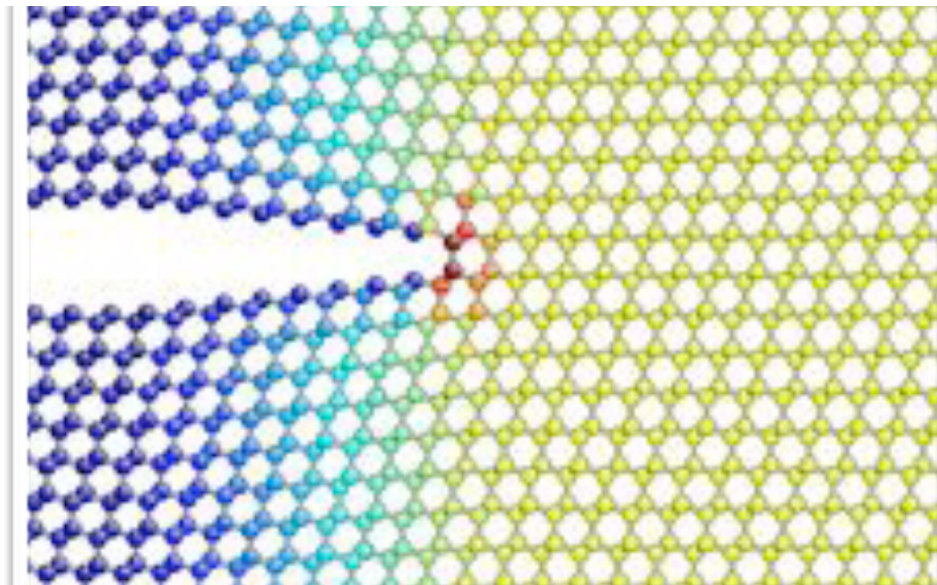
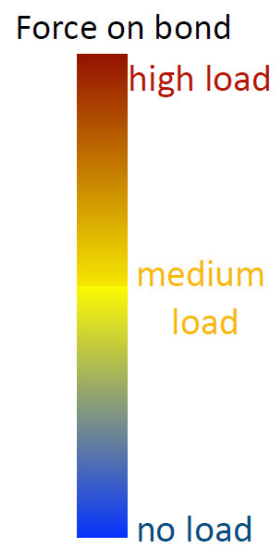


Table 9.1 Room-Temperature Yield Strength and Plane Strain Fracture Toughness Data for Selected Engineering Materials

Material	Yield Strength		K_{Ic}	
	MPa	ksi	MPa $\sqrt{m}$	ksi $\sqrt{in.}$
Metals				
Aluminum Alloy ^a (7075-T651)	495	72	24	22
Aluminum Alloy ^a (2024-T3)	345	50	44	40
Titanium Alloy ^a (Ti-6Al-4V)	910	132	55	50
Alloy Steel ^a (4340 tempered @ 260°C)	1640	238	50.0	45.8
Alloy Steel ^a (4340 tempered @ 425°C)	1420	206	87.4	80.0
Ceramics				
Concrete	—	—	0.2–1.4	0.18–1.27
Soda-Lime Glass	—	—	0.7–0.8	0.64–0.73
Aluminum Oxide	—	—	2.7–5.0	2.5–4.6
Polymers				
Polystyrene (PS)	—	—	0.7–1.1	0.64–1.0
Polymethyl Methacrylate (PMMA)	53.8–73.1	7.8–10.6	0.7–1.6	0.64–1.5
Polycarbonate (PC)	62.1	9.0	2.2	2.0

^a **Source:** Reprinted with permission, *Advanced Materials and Processes*, ASM International, © 1990.

K_{Ic} critical parameters
(defect length and stress)
above which there is failure (*all materials*)

$$K_{Ic} = Y \sigma_c (\pi a)^{1/2}$$

(ASTM E 399)

PS-ZrO₂

8-12

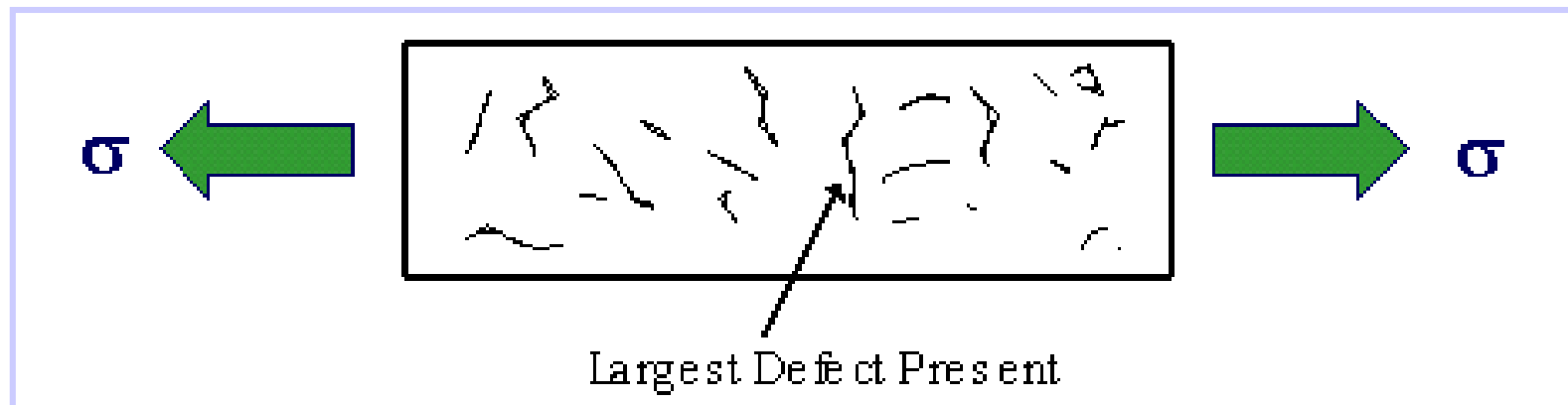
Toughness, Crack Size and Strength

	Tough Metal	Brittle Metal	Typical Ceramic	Brittle Materials
Example	Good steel	Cast Iron	Alumina, SiC	Glass, Silicon
K_{Ic} (MPa $m^{1/2}$)	100	20	4	1
a for $\sigma_F = 100$ MPa	320 mm	13 mm	510 μm	32 μm
a for $\sigma_F = 500$ MPa	13 mm	510 μm	20 μm	1.2 μm

The strength of materials (for allowing crack propagation) is given by the combination of:

- fracture toughness
- crack size
- sample geometry

$$\sigma_f = \frac{K_{Ic}}{Y\sqrt{\pi a}}$$



WEIBULL STATISTICS

The PROBABILITY of fracture is linked to the dimensional distribution of the defects in the material volume and to the probability of finding cracks of certain dimensions in a sample

Larger samples statistically fail under lower loads → greater probability of containing defects of critical size

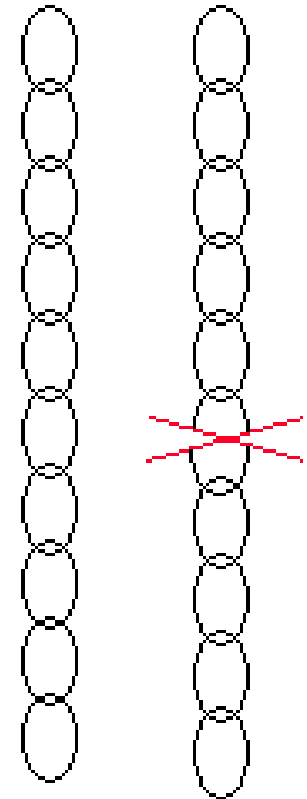
SCALE EFFECT!

Mechanical strength - «chain analogy»

- ❖ The resistance of a chain depends on the weakest element
- ❖ If the resistance of each element is randomly distributed and the probability of NOT BREAKING of a chain of length L is $P_s(L)$ then the probability of not breaking of a chain of length $2L$ will be:

$$P_s(2L) = P_s(L) \cdot P_s(L)$$

Note on the *scale effect* (see previous slide):
as $P_s(L)$ ranges between 0 and 1, then $P_s(2L) < P_s(L)$



Example: glass fibres

- Tensile strength of a glass slab around 100 MPa
- For glass fibres → 900-1000 MPa
- Scale or volume effect → Weibull statistics
- Tensile strength can reach values comparable to the compressive strength

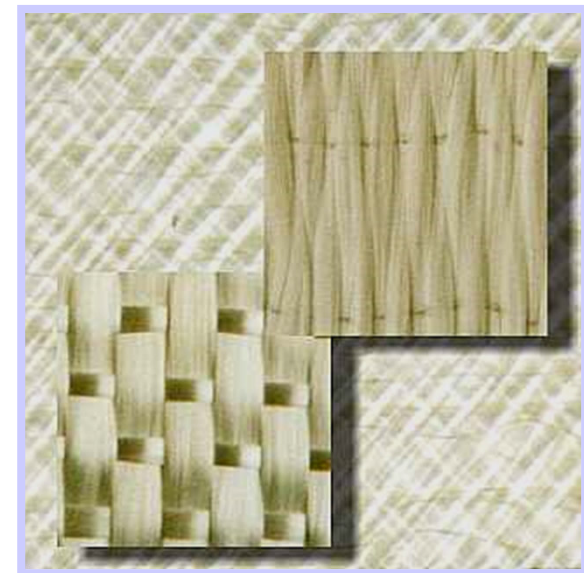
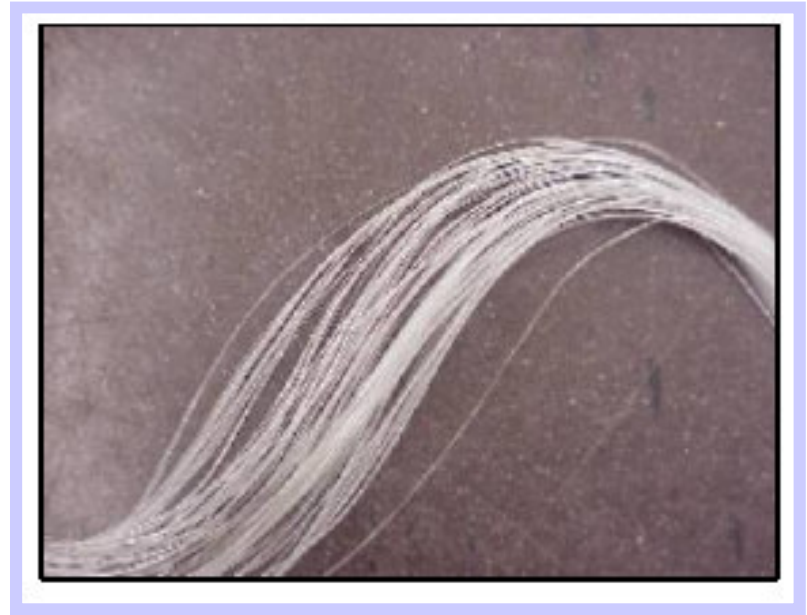


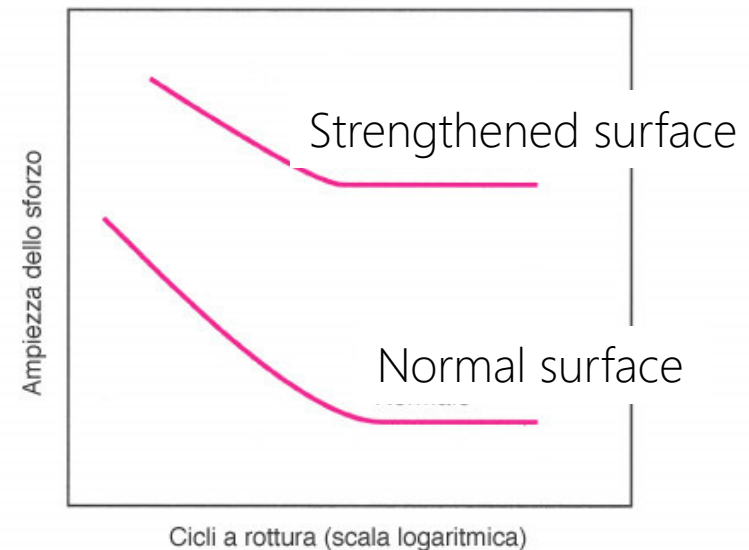
Table 15.4 Characteristics of Several Fiber-Reinforcement Materials

<i>Material</i>	<i>Specific Gravity</i>	<i>Tensile Strength</i> [GPa (10^6 psi)]	<i>Specific Strength</i> (GPa)	<i>Modulus of Elasticity</i> [GPa (10^6 psi)]	<i>Specific Modulus</i> (GPa)
<i>Whiskers</i>					
Graphite	2.2	20 (3)	9.1	700 (100)	318
Silicon nitride	3.2	5–7 (0.75–1.0)	1.56–2.2	350–380 (50–55)	109–118
Aluminum oxide	4.0	10–20 (1–3)	2.5–5.0	700–1500 (100–220)	175–375
Silicon carbide	3.2	20 (3)	6.25	480 (70)	150
<i>Fibers</i>					
Aluminum oxide	3.95	1.38 (0.2)	0.35	379 (55)	96
Aramid (Kevlar 49)	1.44	3.6–4.1 (0.525–0.600)	2.5–2.85	131 (19)	91
Carbon ^a	1.78–2.15	1.5–4.8 (0.22–0.70)	0.70–2.70	228–724 (32–100)	106–407
E-Glass	2.58	3.45 (0.5)	1.34	72.5 (10.5)	28.1
Boron	2.57	3.6 (0.52)	1.40	400 (60)	156
Silicon carbide	3.0	3.9 (0.57)	1.30	400 (60)	133
UHMWPE (Spectra 900)	0.97	2.6 (0.38)	2.68	117 (17)	121
<i>Metallic Wires</i>					
High-strength steel	7.9	2.39 (0.35)	0.30	210 (30)	26.6
Molybdenum	10.2	2.2 (0.32)	0.22	324 (47)	31.8
Tungsten	19.3	2.89 (0.42)	0.15	407 (59)	21.1

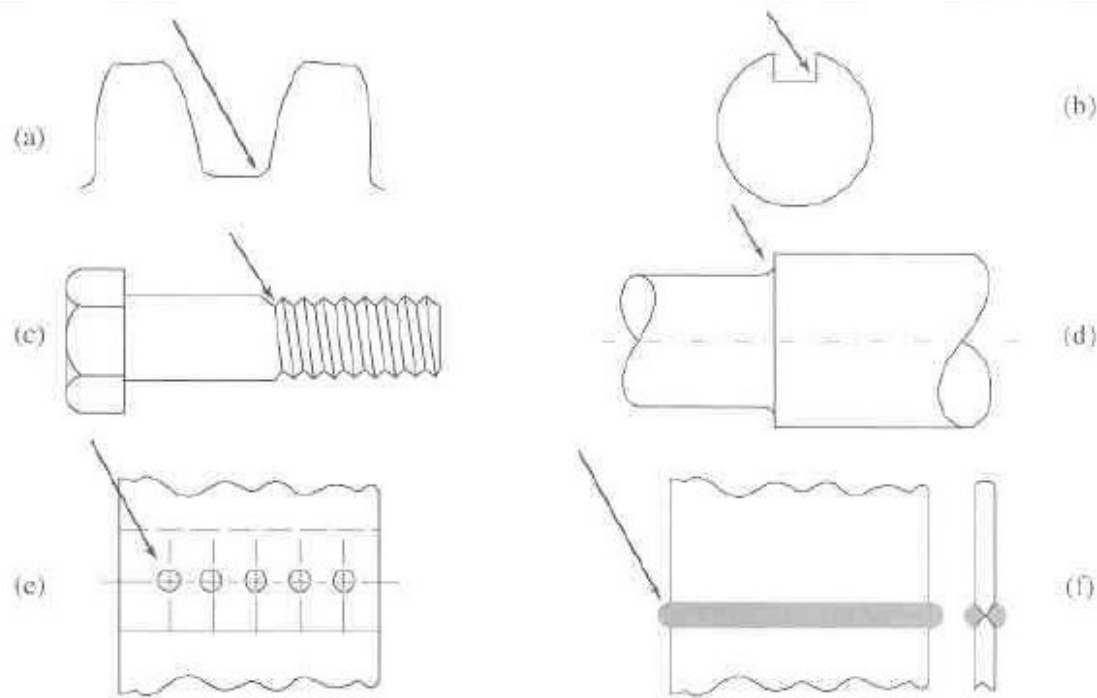
^aThe term “carbon” instead of “graphite” is used to denote these fibers, since they are composed of crystalline graphite regions, and also of noncrystalline material and areas of crystal misalignment.

How to increase materials fatigue resistance?

- Surface *strengthening methods*
- Environment (corrosion?) → coatings
- Suitable mechanical design
- **Fatigue and fracture mechanics to model and predict components *life* !**



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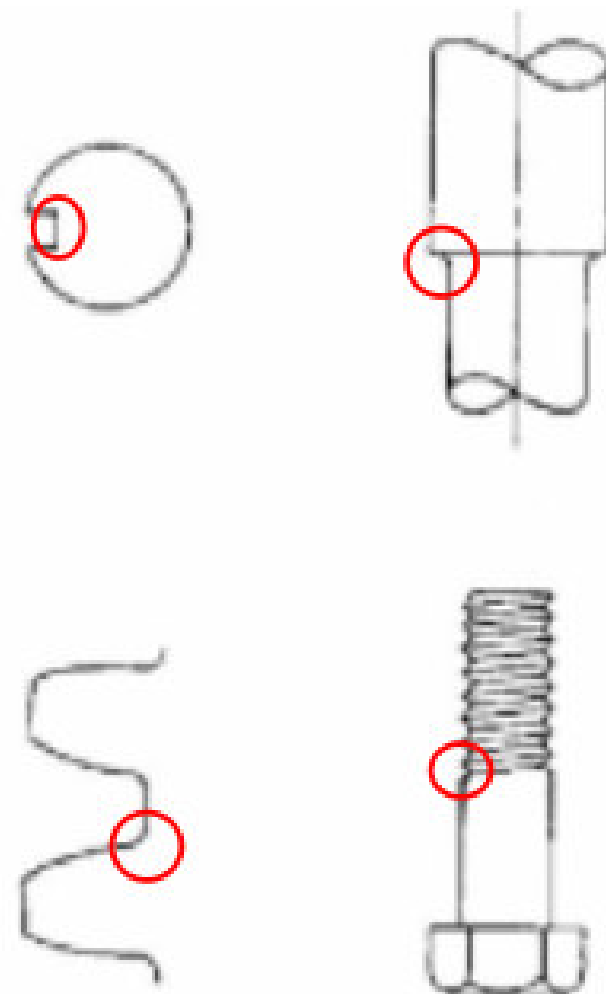


Esempi di intagli negli elementi di macchine: a) ruota dentata; b) sede di una linguetta; c) filettatura di una vite; d) spallamento di un albero; e) fori per il passaggio delle viti; f) giunzione saldata.

Fig. 3 – Esempi di intaglio negli elementi di macchine

Notches in engineering components

How to solve?



Finite Element Analysis (FEA)

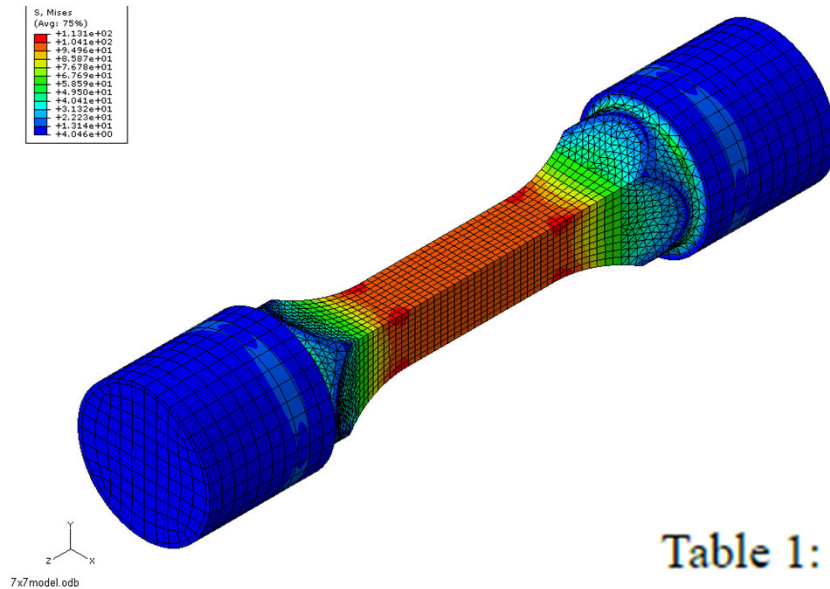
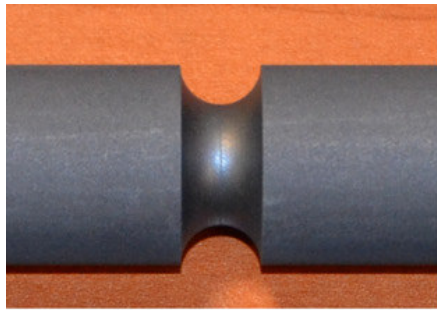


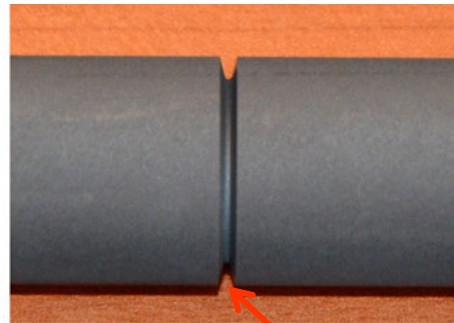
Table 1: Finite element modeling parameters.

	Ag alloy	Cu alloy	Ti-Si compound	TiC
Young modulus, E (GPa)	71	129.8	234 ^[23]	439 ^[24]
Poisson ratio, ν	0.37	0.343	0.22 ^[23]	0.188 ^[24]
Yield strength, σ (MPa)	308	469	-	-

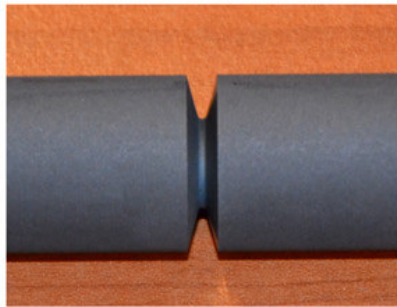
Examples of different ρ and stress concentration by Finite Element Analysis (FEA)



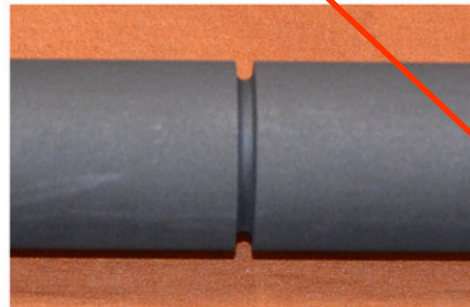
Semi-circular notch



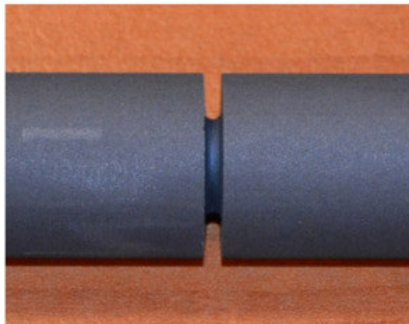
V-notch, $2\alpha=30^\circ$



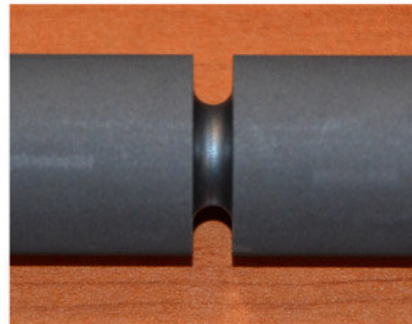
V-notch, $2\alpha=30^\circ$



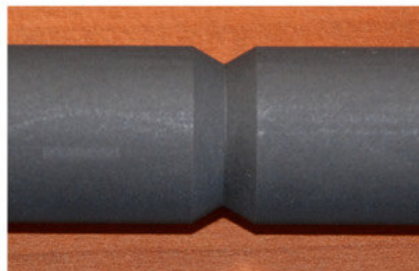
U-notch



U-notch



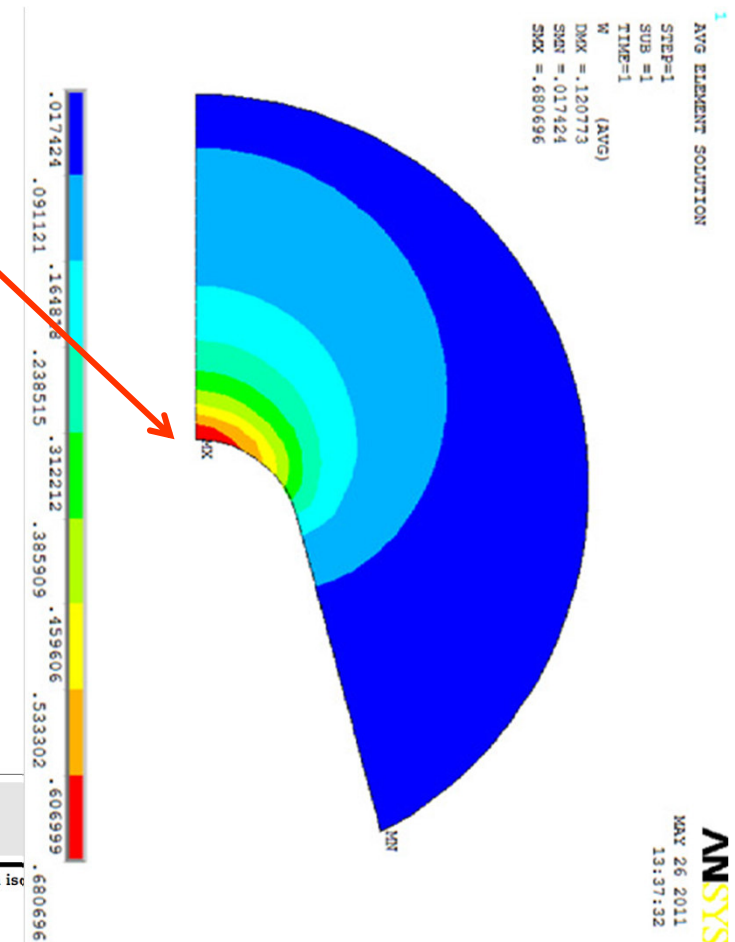
U-notch

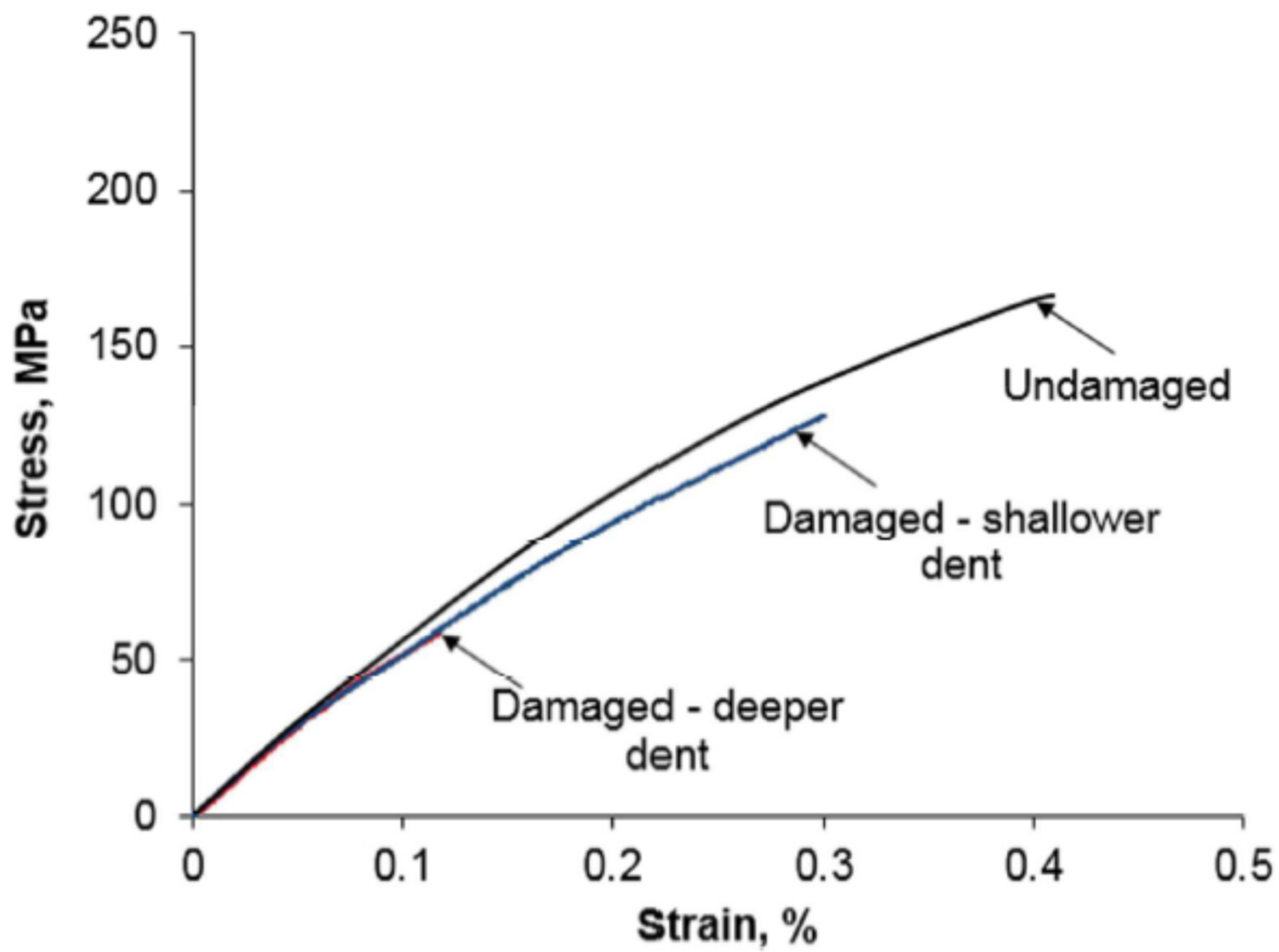


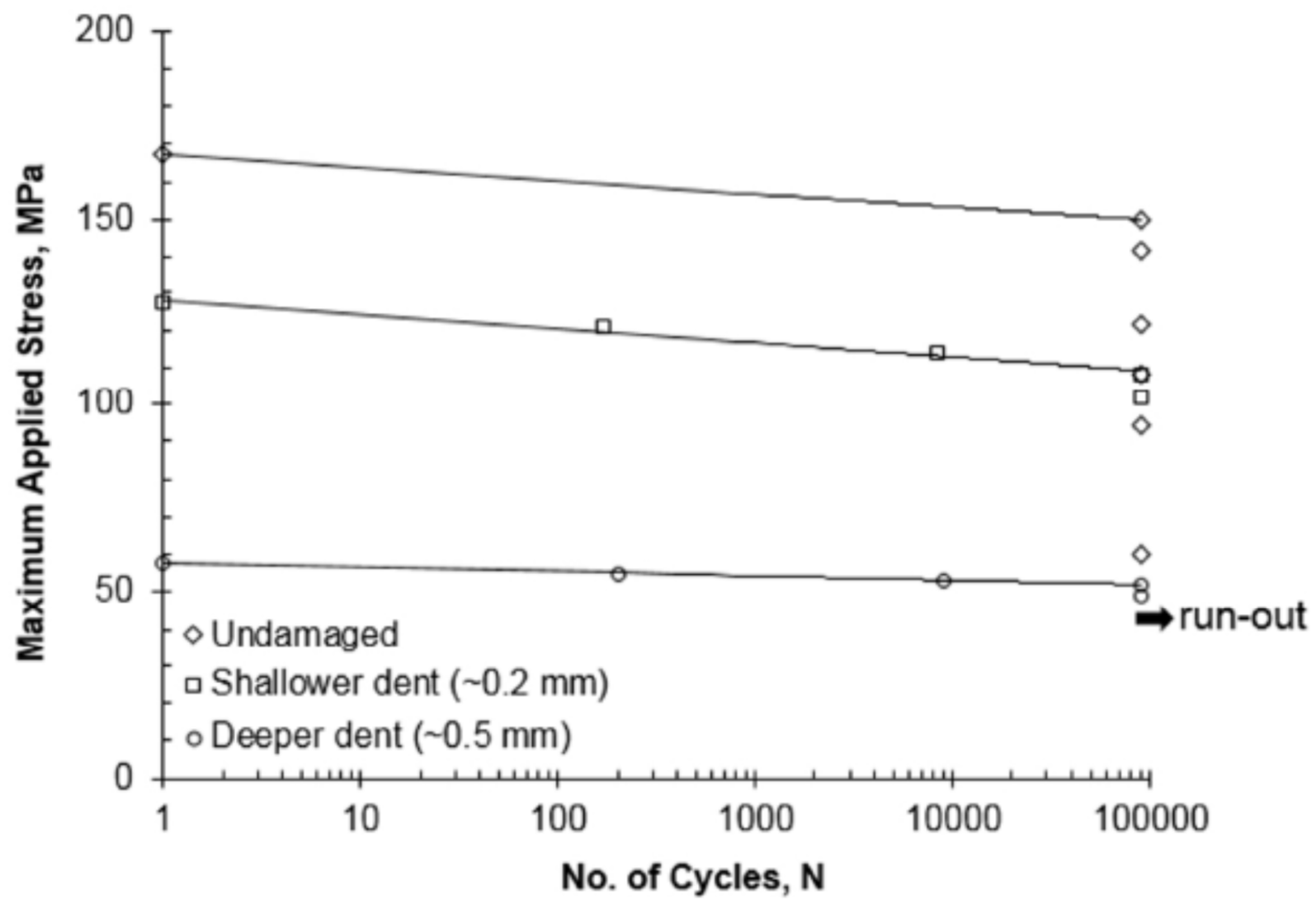
V-notch, $2\alpha=120^\circ$



V-notch, $2\alpha=120^\circ$







HARDNESS

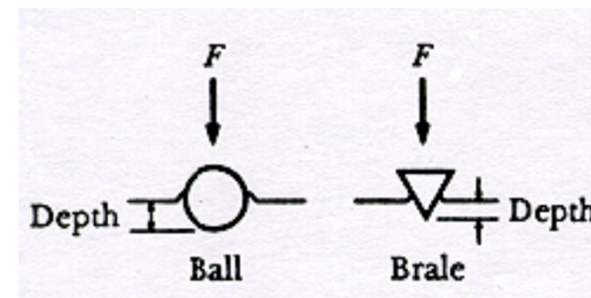
It is a measure of the material resistance to **localized compressive stress** due to **INDENTATION**.

The indenter is subjected to a load and causes a localized elastic deformation, followed by a plastic deformation which leads to an indentation (**imprint**).

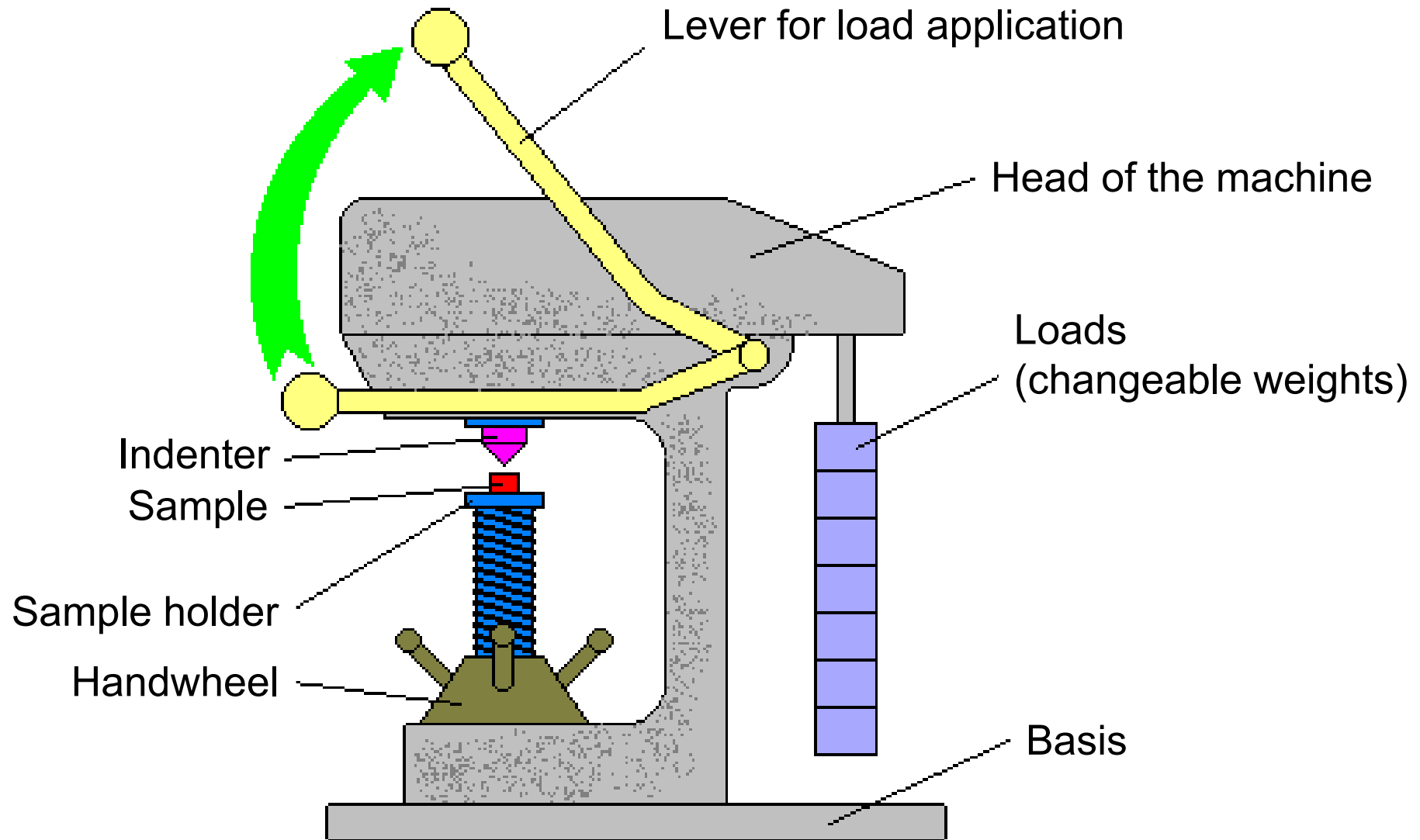
Resulting indentations are observed and measured.

Different scales of hardness exist; the hardness **H** can be expressed as **F/A** , where **A** is the area **of the indentation** (note that **H** is formally expressed in **GPa**), or as the **depth of the indentation** (**H** is expressed in **μm or mm**).

Shape of the indenter:
spherical or pyramidal



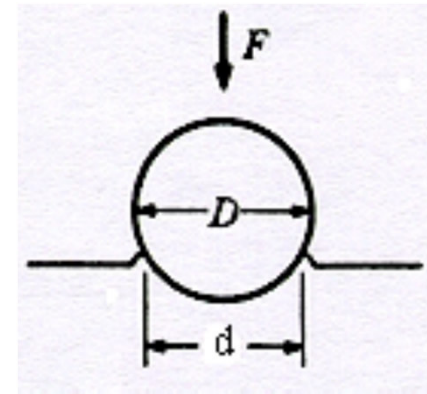
Test execution



Examples of hardness tests

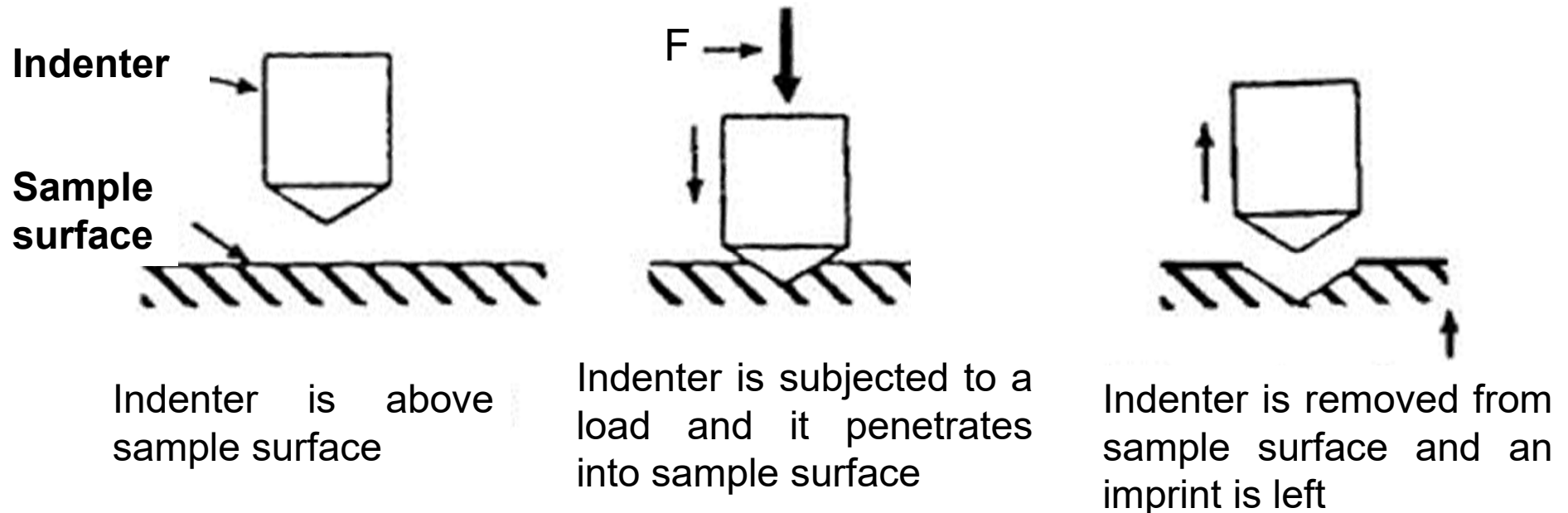
Brinell test : spherical indenter

$$HB = F/(\pi Dd) = \text{load/indentation area}$$

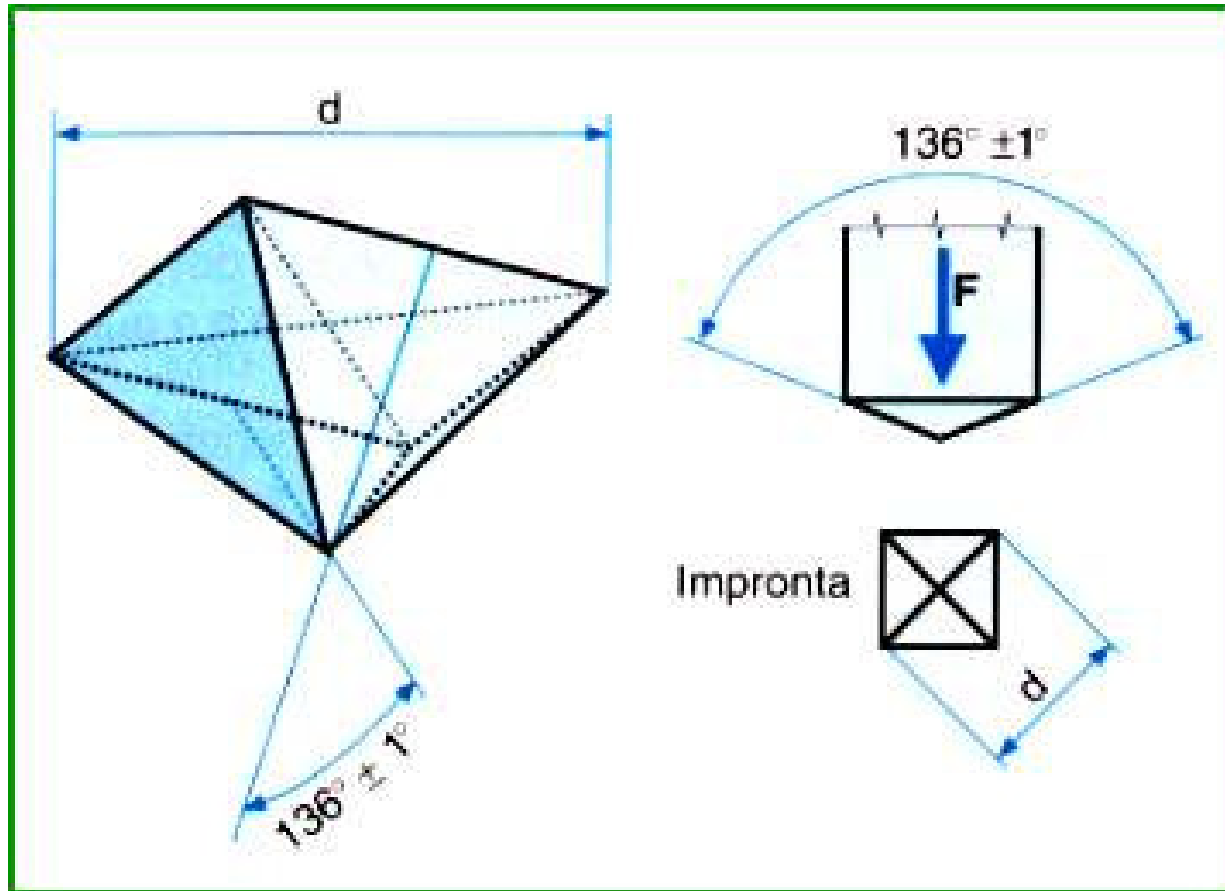


Vickers test: pyramidal indenter

$$HV = 1.854(F/L^2), \text{ where } L \text{ is the imprint diagonal length}$$



Vickers



Brinell

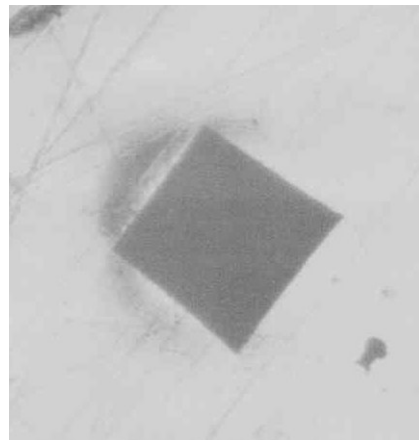
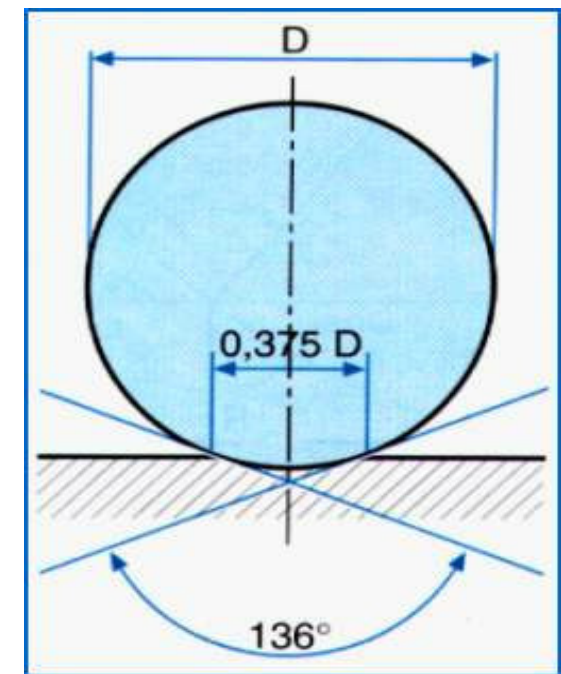
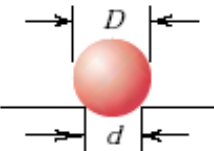
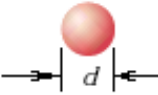
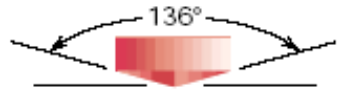
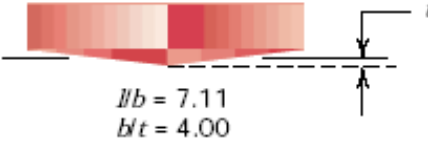
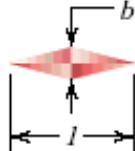
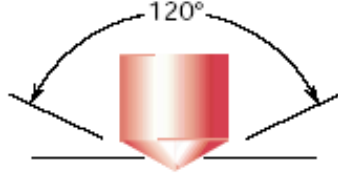
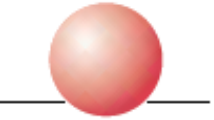


Table 7.4 Hardness Testing Techniques

Test	Indenter	Shape of Indentation		Load	Formula for Hardness Number ^a
		Side View	Top View		
Brinell	10-mm sphere of steel or tungsten carbide			P	$HB = \frac{2P}{\pi D[D - \sqrt{D^2 - d^2}]}$
Vickers microhardness	Diamond pyramid			P	$HV = 1.854P/d_1^2$
Knoop microhardness	Diamond pyramid			P	$HK = 14.2P/l^2$
Rockwell and Superficial Rockwell	{ Diamond cone $\frac{1}{16}, \frac{1}{8}, \frac{1}{4}, \frac{1}{2}$ in. diameter steel spheres	 	 	60 kg 100 kg 150 kg } Rockwell 15 kg 30 kg 45 kg } Superficial Rockwell	

Rockwell Hardness
measures penetration depth

^a For the hardness formulas given, P (the applied load) is in kg, while D , d , d_1 , and l are all in mm.

Source: Adapted from H. W. Hayden, W. G. Moffatt, and J. Wulff, *The Structure and Properties of Materials*, Vol. III, *Mechanical Behavior*. Copyright © 1965 by John Wiley & Sons, New York. Reprinted by permission of John Wiley & Sons, Inc.

Table 7.5b Superficial Rockwell Hardness Scales

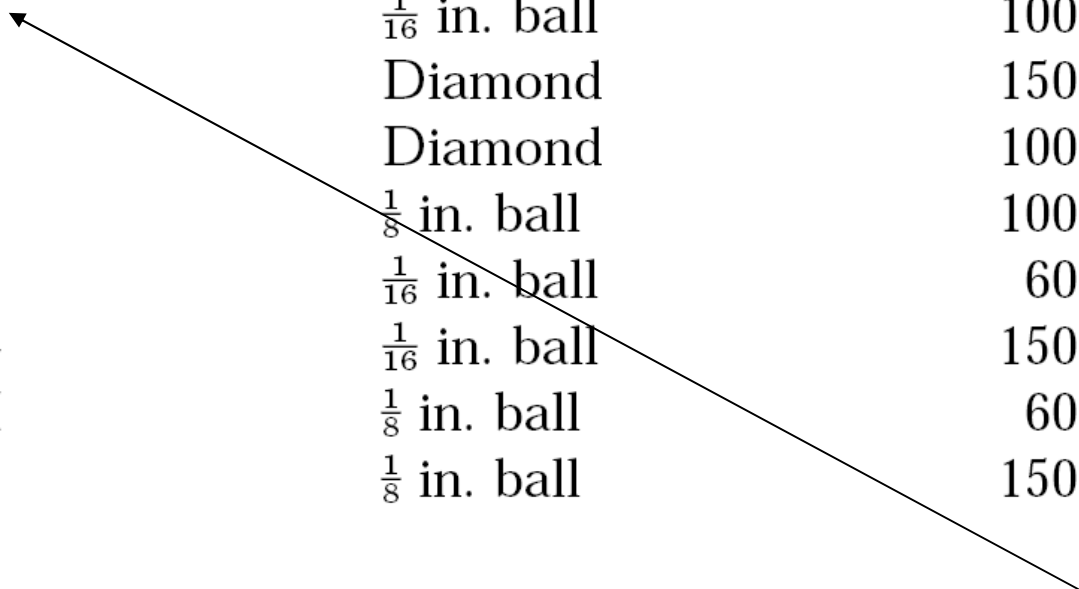
<i>Scale Symbol</i>	<i>Indenter</i>	<i>Major Load (kg)</i>
15N	Diamond	15
30N	Diamond	30
45N	Diamond	45
15T	$\frac{1}{16}$ in. ball	15
30T	$\frac{1}{16}$ in. ball	30
45T	$\frac{1}{16}$ in. ball	45
15W	$\frac{1}{8}$ in. ball	15
30W	$\frac{1}{8}$ in. ball	30
45W	$\frac{1}{8}$ in. ball	45

example: **60 HR30W**= superficial Rockwell hardness =

You measure 60 on a scale 30W, with a Rockwell hardness test

Table 7.5a Rockwell Hardness Scales

<i>Scale Symbol</i>	<i>Indenter</i>	<i>Major Load (kg)</i>
A	Diamond	60
B	$\frac{1}{16}$ in. ball	100
C	Diamond	150
D	Diamond	100
E	$\frac{1}{8}$ in. ball	100
F	$\frac{1}{16}$ in. ball	60
G	$\frac{1}{16}$ in. ball	150
H	$\frac{1}{8}$ in. ball	60
K	$\frac{1}{8}$ in. ball	150



Example: 80 HRB= Rockwell hardness = 80 scale B

Values lower than 20 or higher than 100 are not acceptable

Correlation hardness/tensile properties

Steel TS: about= 3.45 HB

This correlation is useful because hardness test is easier to be performed than tensile test and it does not fracture the sample.

This is the reason why the hardness test is widely used as **quality control** test, as well as for a preliminary evaluation of the mechanical properties of materials.

By knowing ***H***, we can graphically estimate the tensile strength.

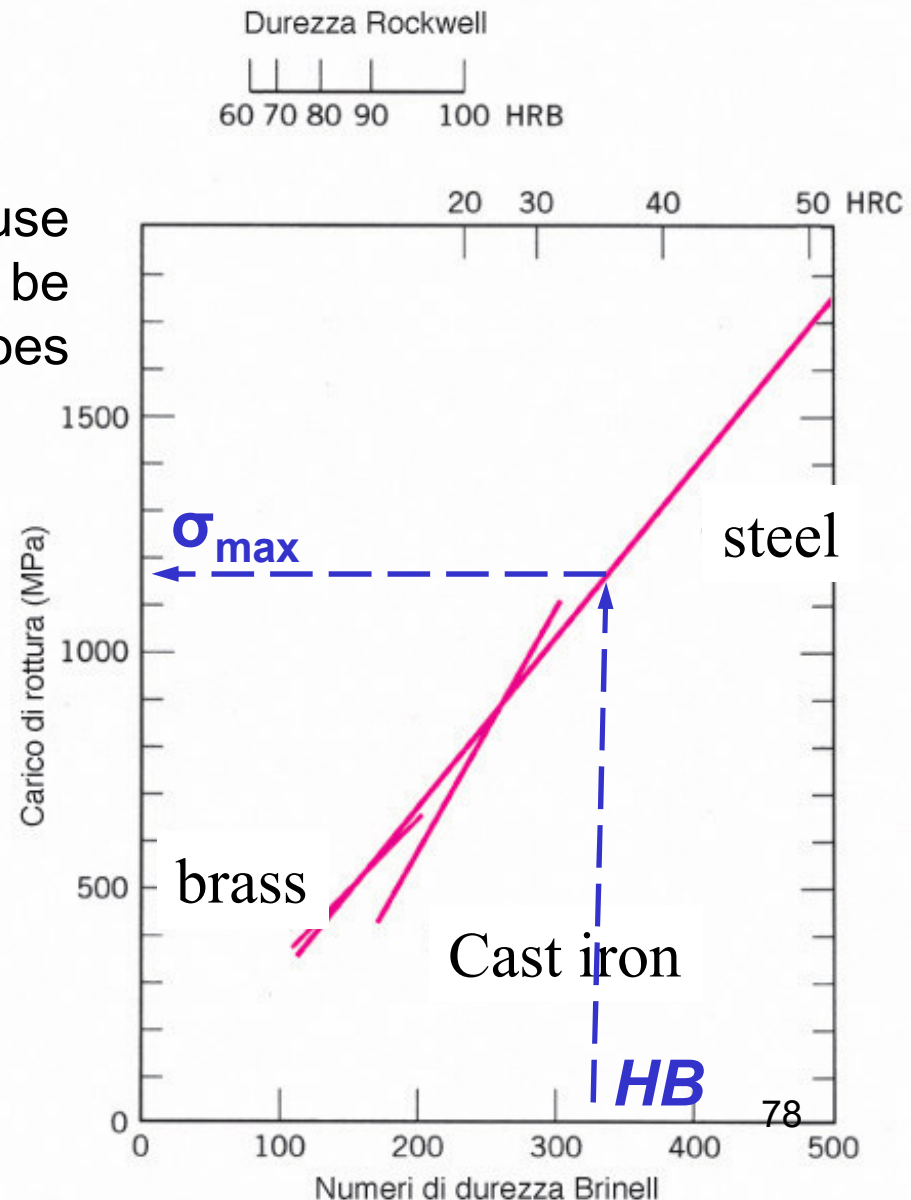
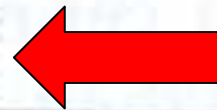
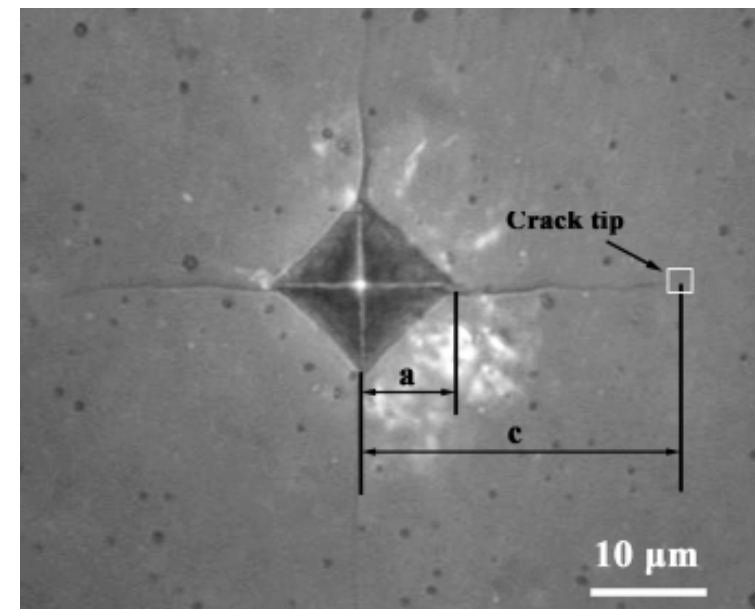
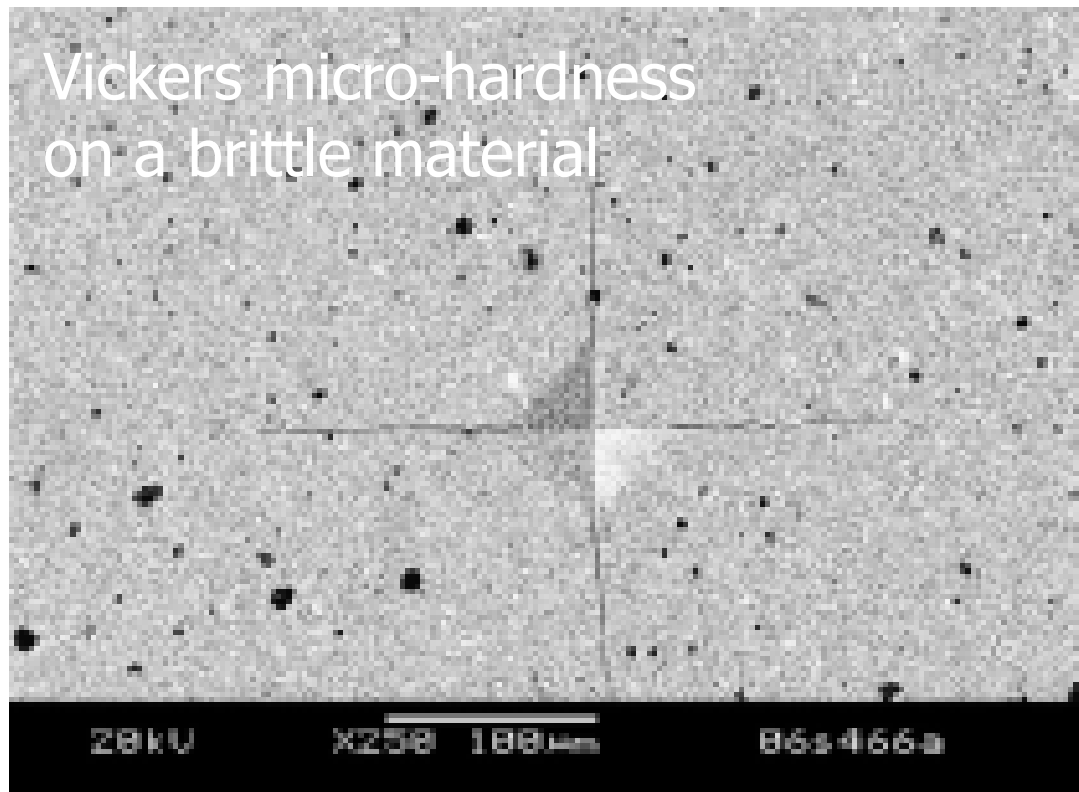


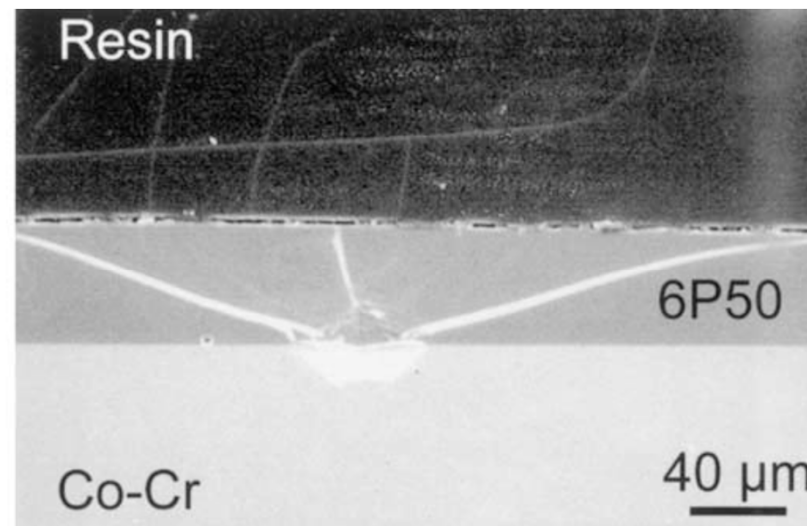
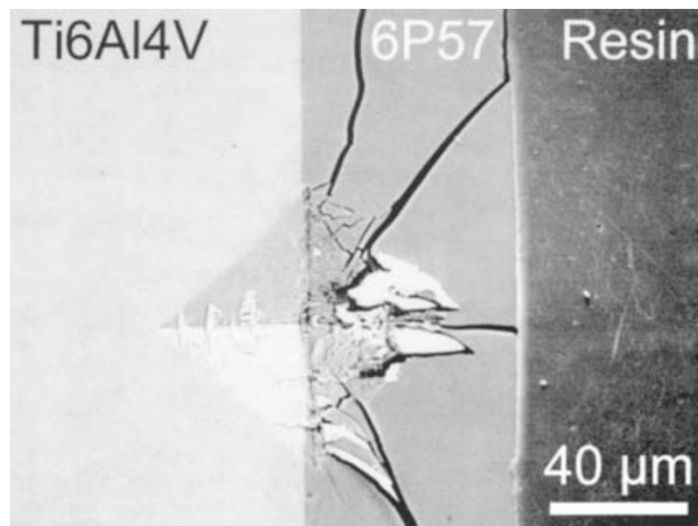
Table 13.1 *Hardness of Some Ceramics*

<i>Material</i>	<i>Knoop hardness number</i>
Diamond	7000
Cubic boron nitride	6900
Titanium diboride	3300
Boron carbide	2900
Silicon carbide	2600
Silicon nitride	2600
Alumina	2000
Beryllium oxide	1220
Hardened steel	700

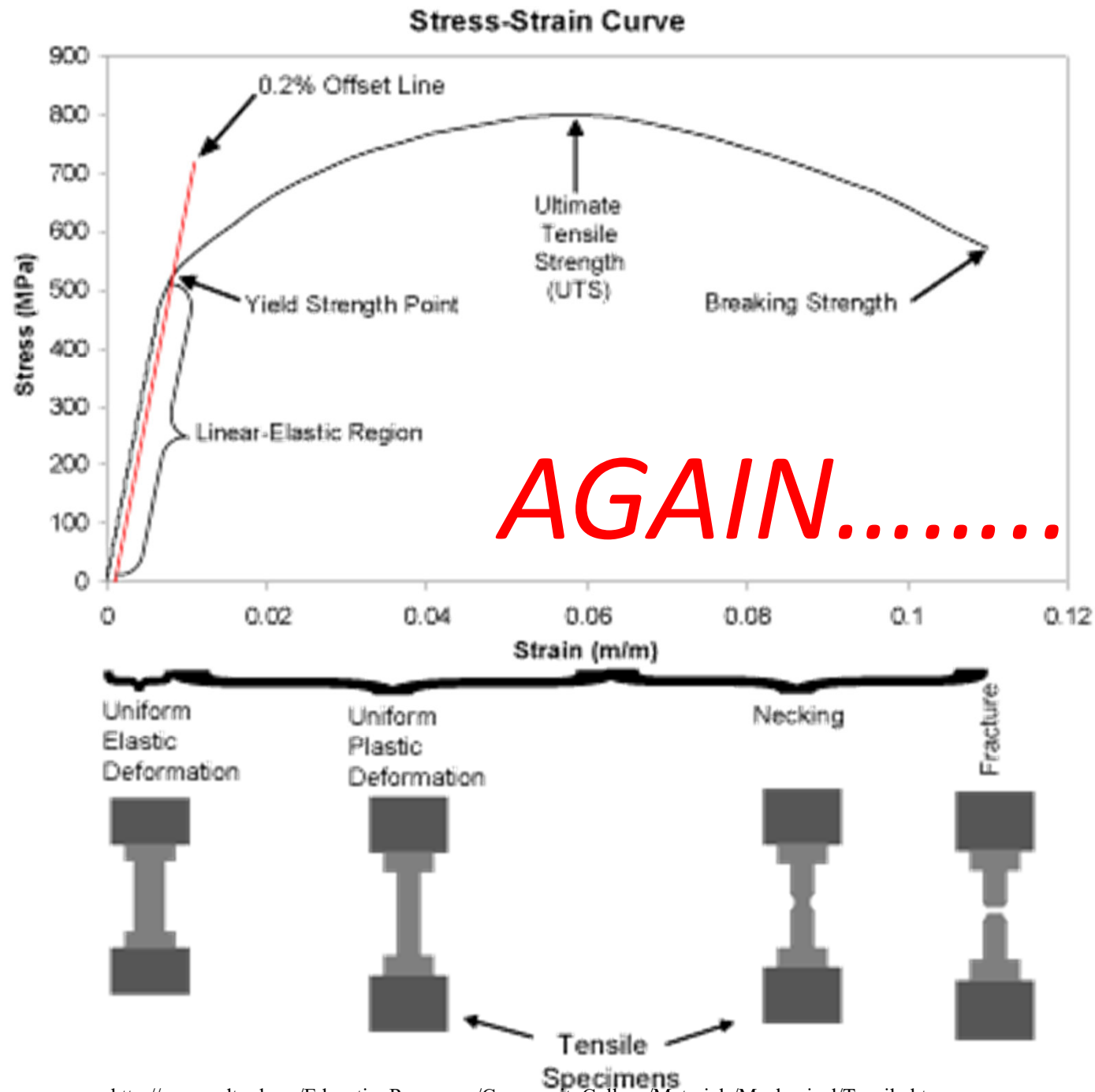




$$K_{IC} = 0.016F_H \frac{(E/HV)^{1/2}}{c^{3/2}}$$

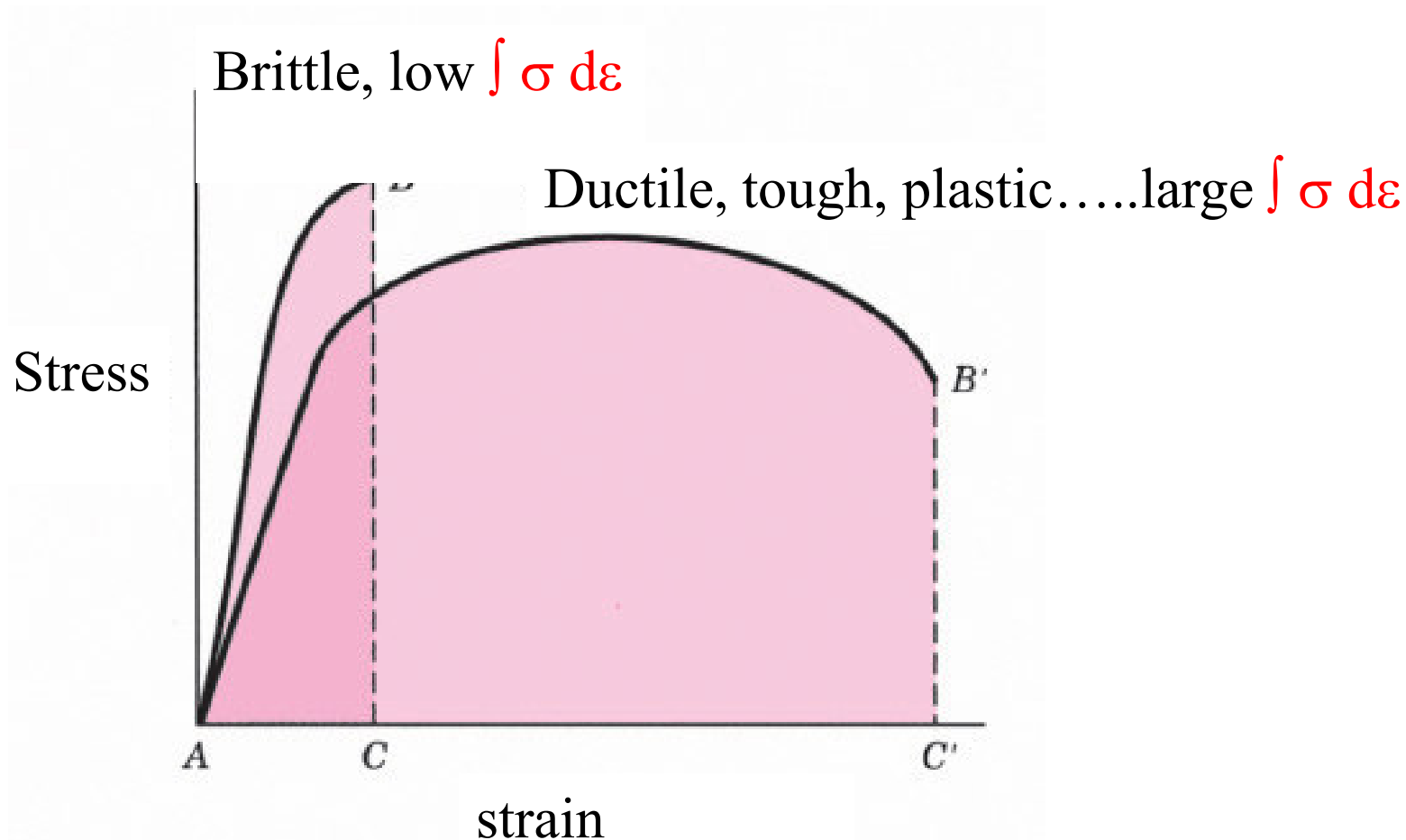


Quality of
joining

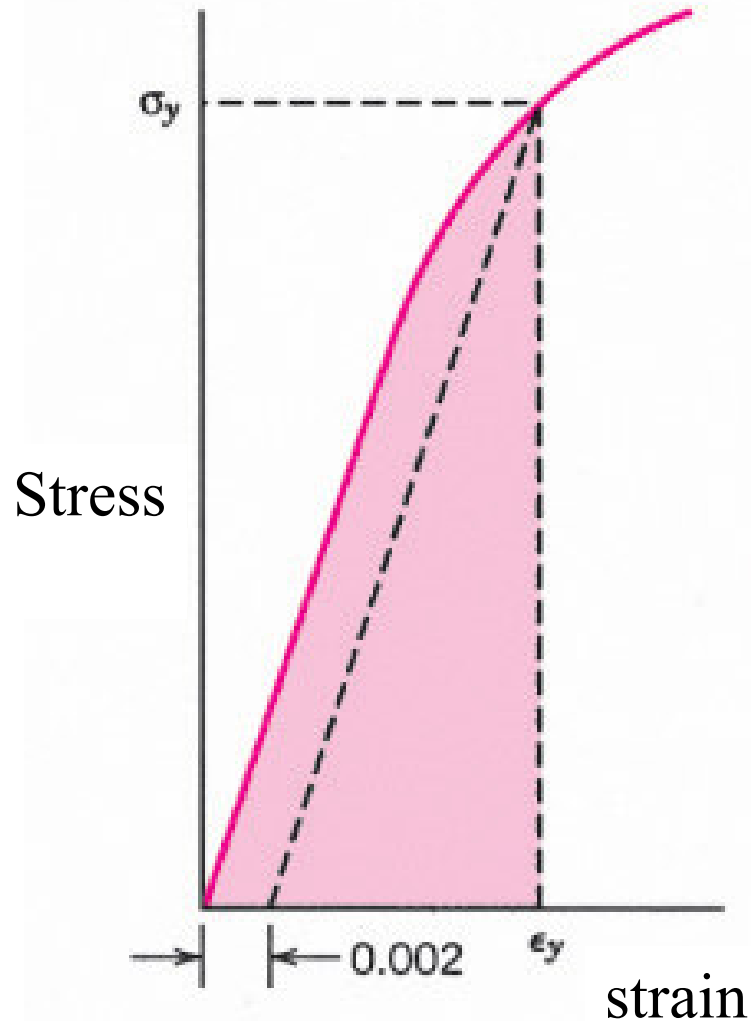


Toughness and resilience

- **Toughness = energy absorbed up to fracture = area of σ - ϵ curve up to fracture) (J / m³)**
- **Resilience = energy absorbed under mech. impact (notched samples)**



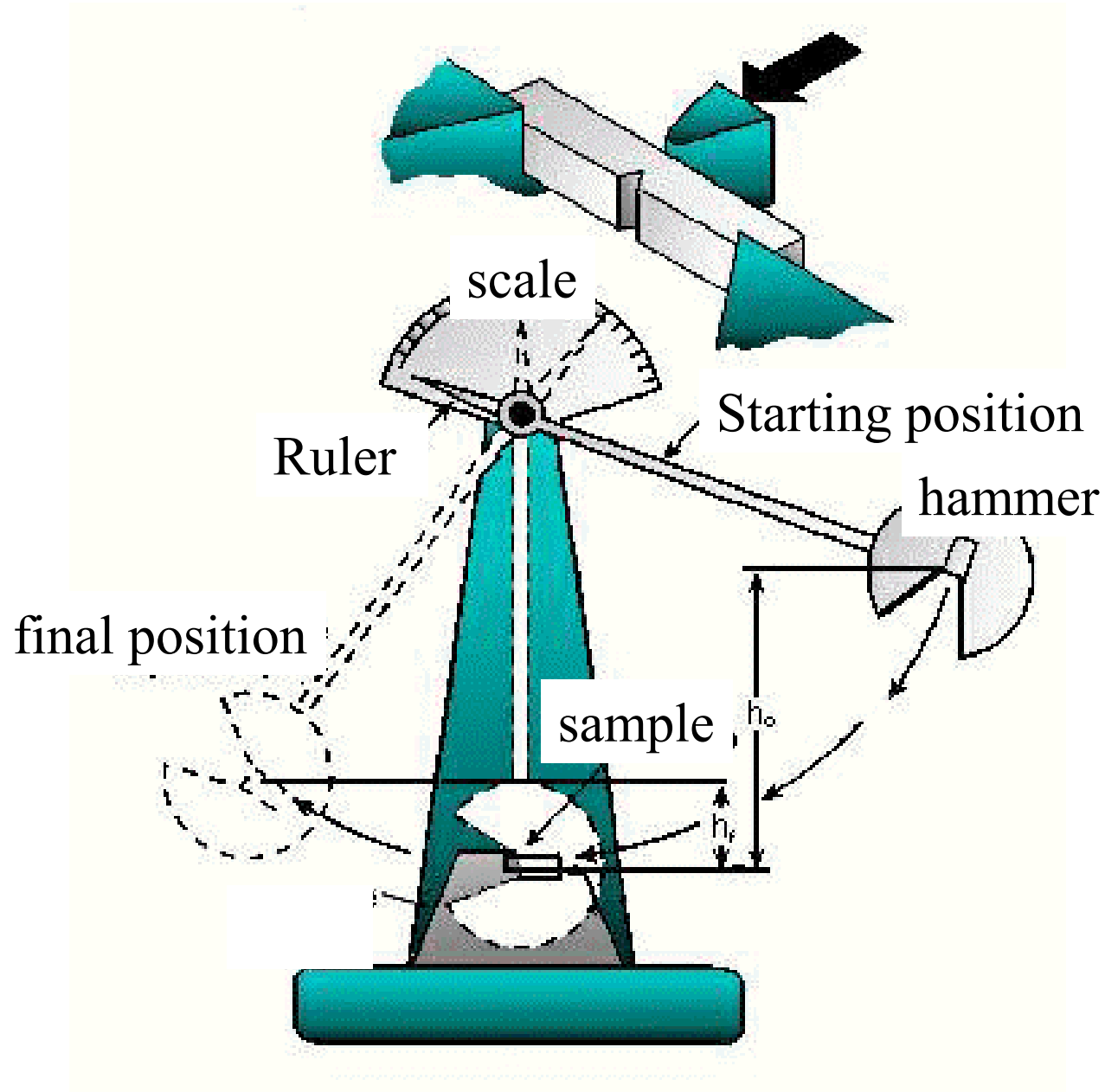
Fracture elastic energy or resilience modulus (U_r)



$$U_r = \int \sigma \, d\epsilon \quad (\text{between } \epsilon_0 \text{ and } \epsilon_y) \quad (\text{ELASTIC FIELD})$$

$$\sigma = E \, \epsilon \quad \epsilon_y = \sigma_y / E$$

$$U_r = \frac{1}{2} \sigma_y \epsilon_y = \frac{1}{2} \sigma_y^2 / E$$

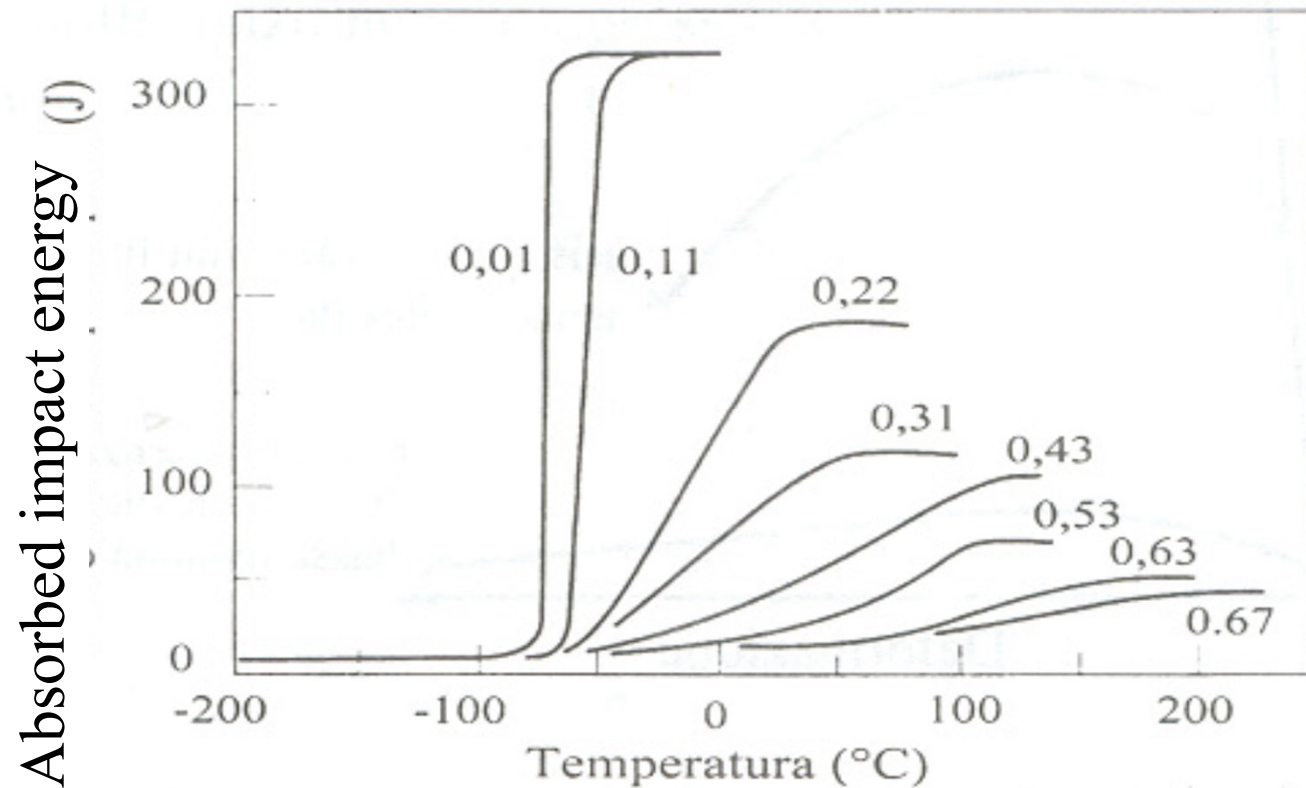


Charpy test:

Assessment of the resiliency → energy necessary to fracture a notched sample (impact of a hammer)



Absorbed impact energy vs. temperature for several steels: ductile to brittle transition



Role of C and Fe_3C on dislocation motion and ductile to brittle transition

Effect of ductile to brittle transition...

- 4 °C
- steel



Importance of Brittle to Ductile Transition Temperature

The Titanic

P/S too High !
Mn too low !

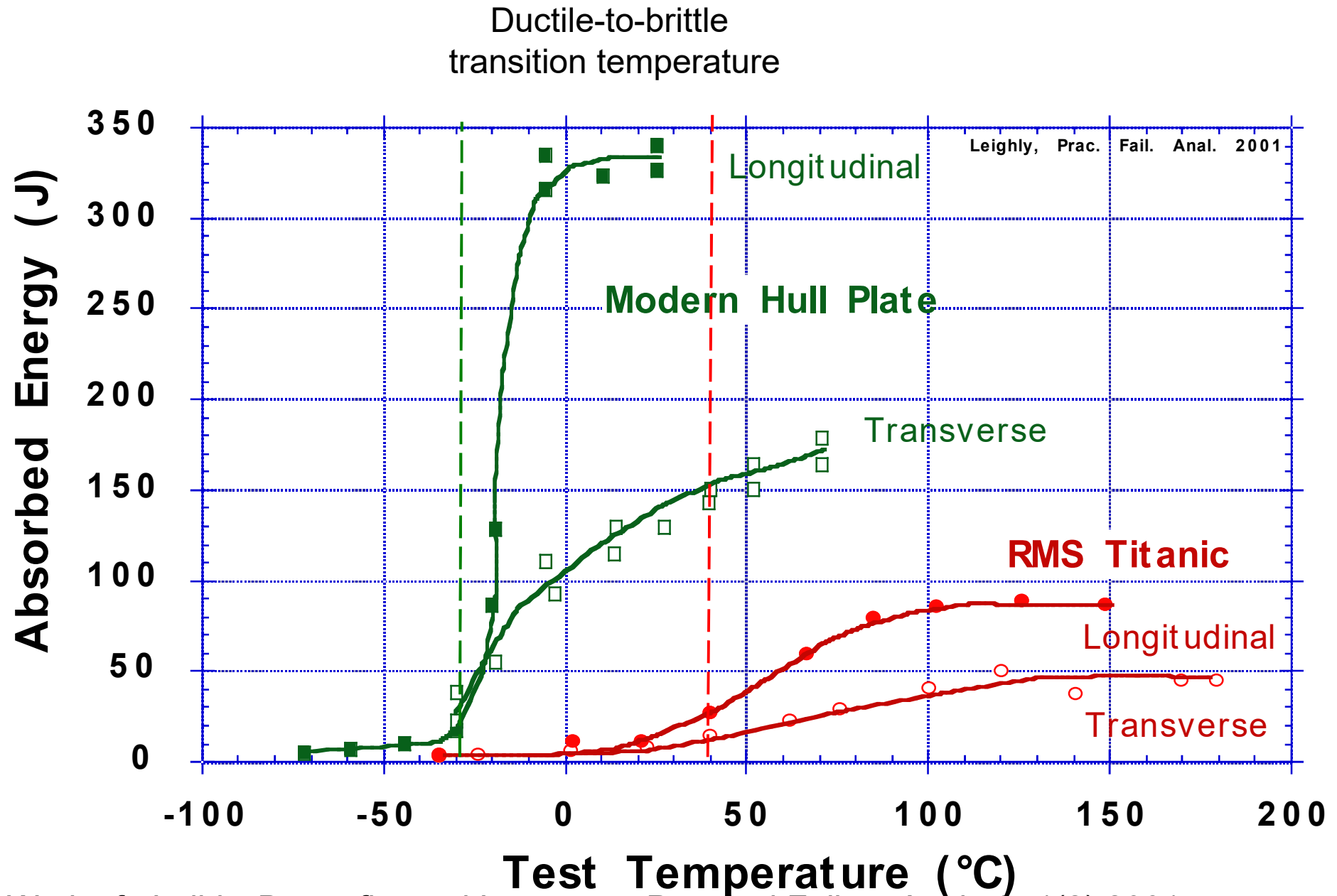
	Titanic hull plate	A36 modern structural steel
Mn:S Ratio	7:1	15:1 (typical)

Composition

	Titanic hull plate	A36 modern structural steel
C	0.21	0.20
Mn	0.47	0.55
P	0.045	0.012
S	0.069	0.01 to 0.04
Si	0.017	0.007
Cu	0.024	0.01
O	0.013	-
N	0.0035	0.0032
Mn:S Ratio	7:1	15:1 (typical)

Courtesy of L.L.Snead, ORNL, USA

Importance of “Ductility” for Engineering Materials

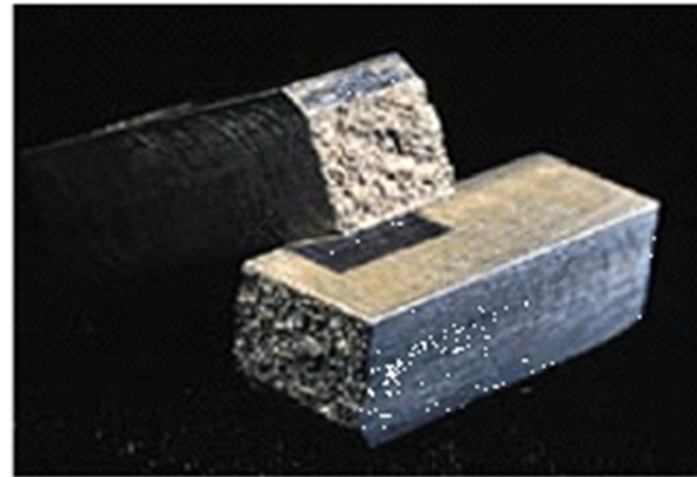
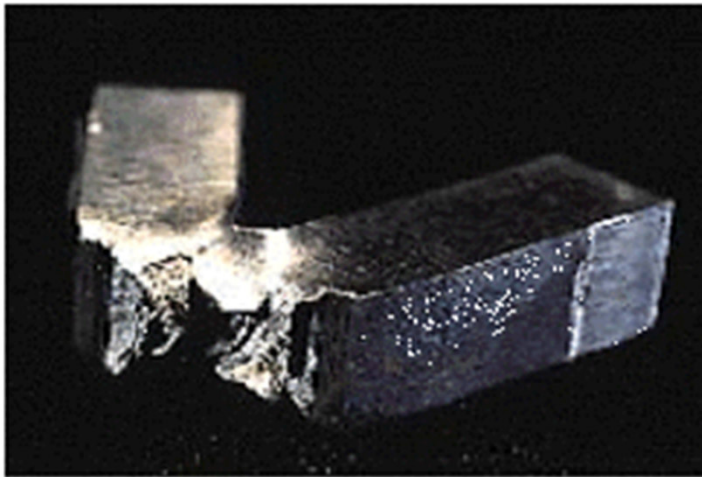
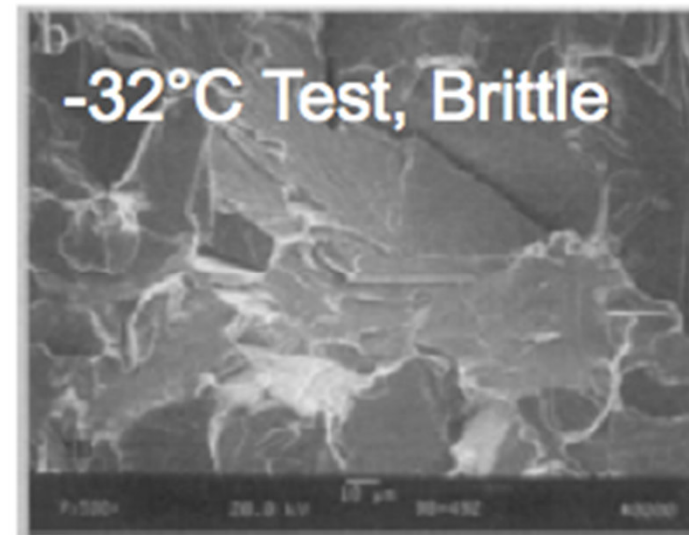
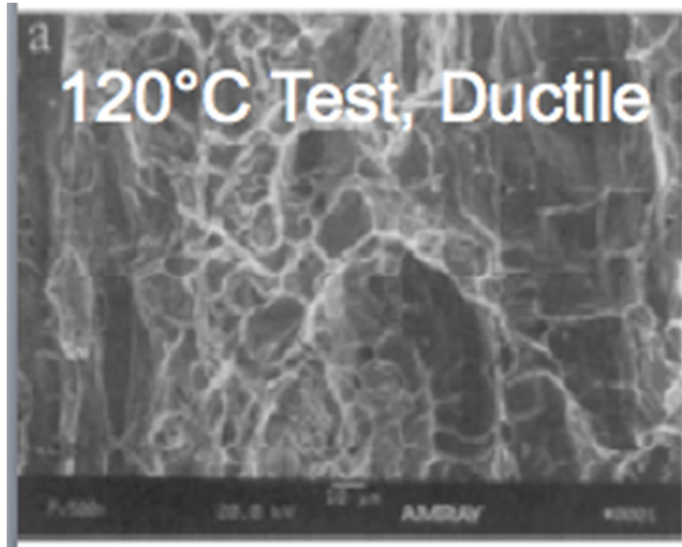


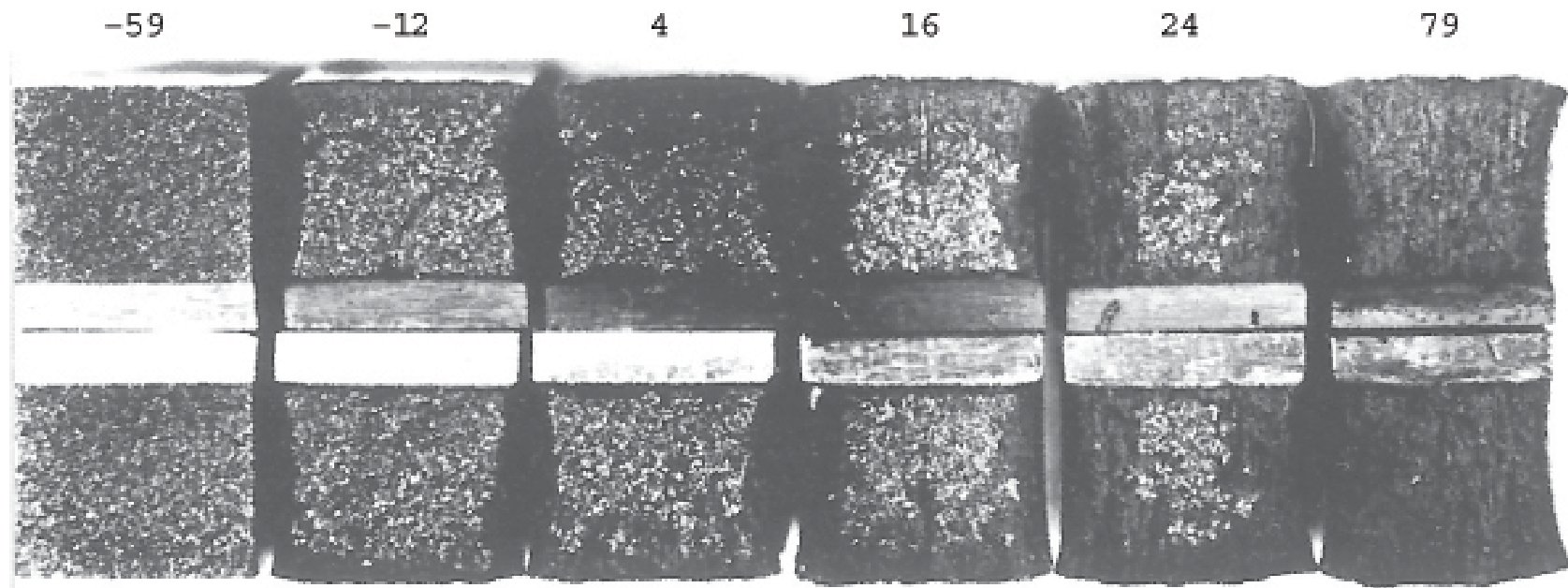
Work of : Leighly, Braamfitt, and Lawrence. Practical Failure Analysis 1(2) 2001.

Courtesy of L.L.Snead, ORNL, USA

Importance of Brittle to Ductile Transition Temperature

Titanic Hull Plate

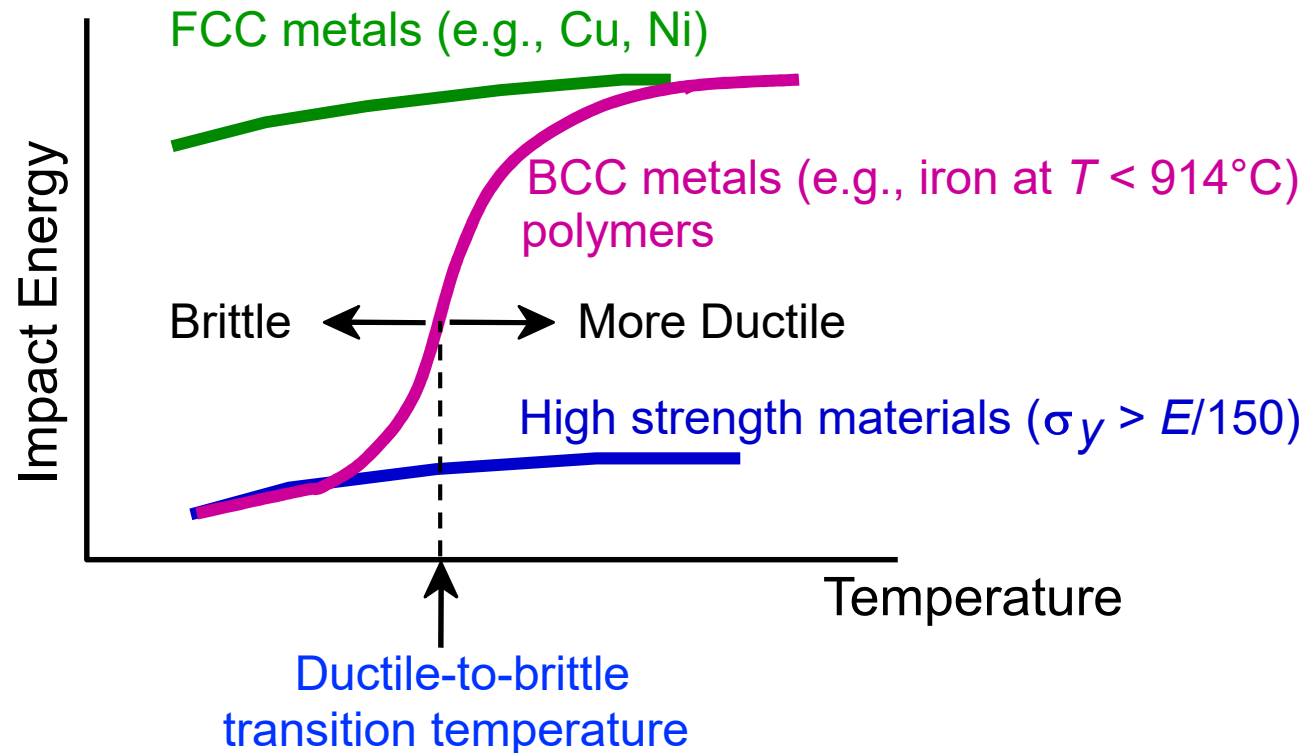




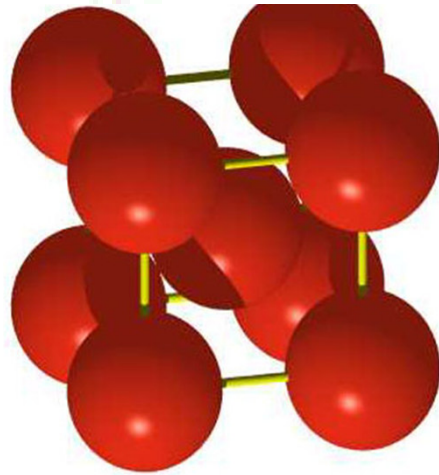
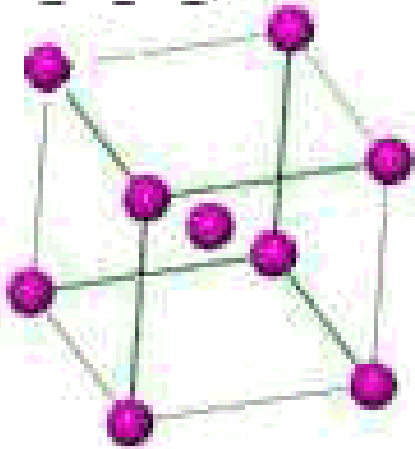
Fracture surfaces after Charpy test (V-notched) at given T
Ductile to brittle transition

D $\rightarrow$ B transition temperature

- Increasing temperature...
 - increases % EL and K_{IC}
- **Ductile-to-Brittle Transition Temperature (DBTT)...**

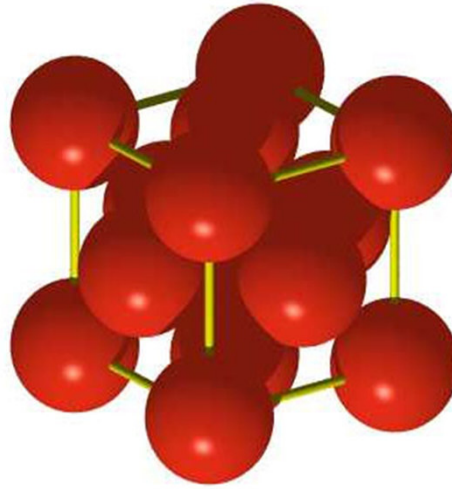
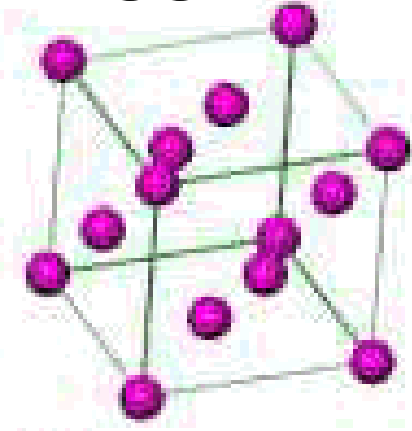


BCC



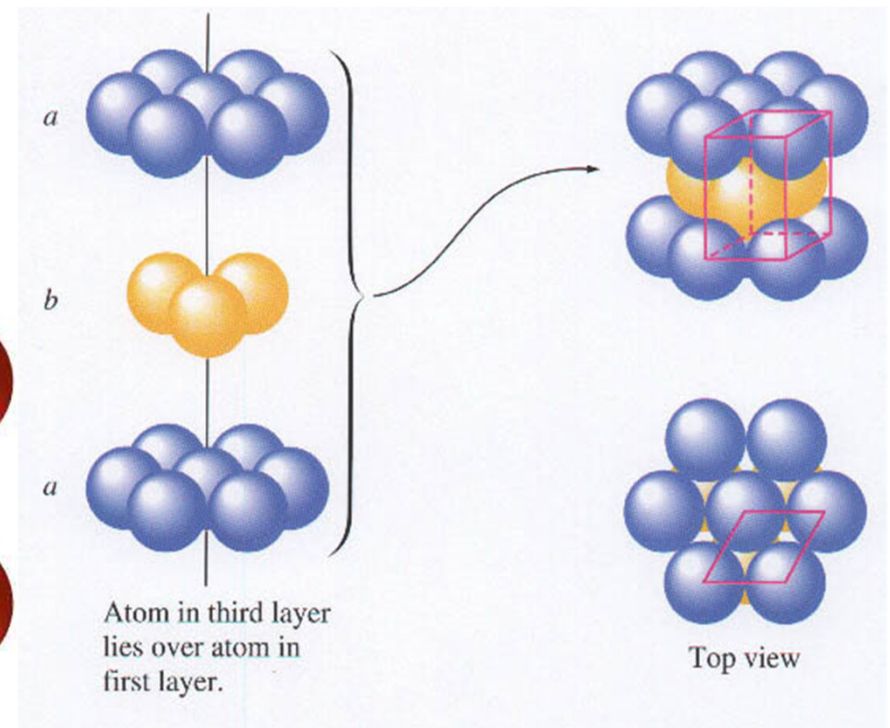
Ex: Fe, W, Cr

FCC



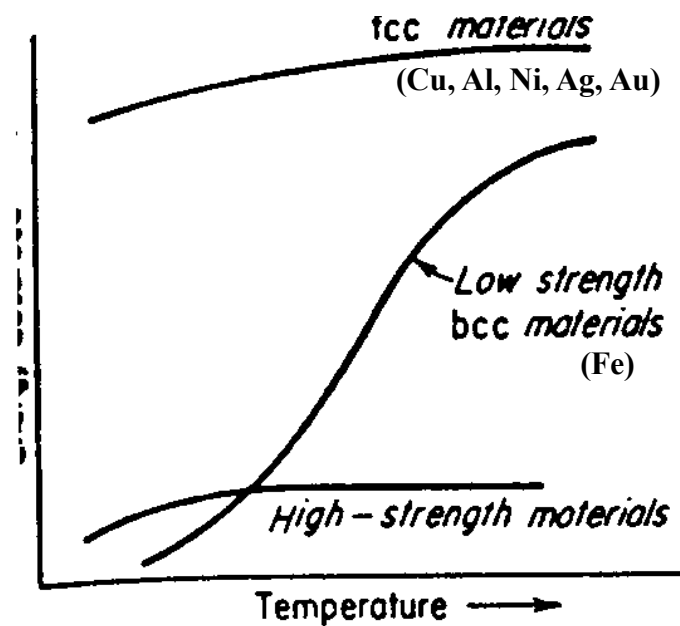
Ex: Cu, Al, Ag, Au

HEXAGONAL



Ex: Mg, Ti, Zn

- <https://www.tec-science.com/material-science/ductility-of-metals/influence-of-the-lattice-structure-on-the-ductility/>



	face-centered cubic (fcc)	body-centered cubic (bcc)	hexagonal closest packed (hcp)
lattice structure			
number of preferred slip planes	4	6	1
number of slip directions	3	2	3
number of preferred slip systems	12	12	3
packing density	74 % (densest packed)	68 % (loosely packed)	74 % (densest packed)
ductility	high	medium	low
metals	γ-iron (<1392°C), aluminium, β-cobalt (>417°C), lead, copper, nickel	α-iron (<911°C), chrome, molybdenum, β-titanium (>882°C), vanadium, tungsten	α-cobalt (<417°C), α-titanium (<882°C), zinc, magnesium

For good deformability, a lattice structure should have at least 5 slip systems to be activated during a deformation process, so that the workpiece can deform in any direction as desired.

p
1.]

For good deformability, a grid structure should have at least 5 slip systems!

CHARPY tests on
steel at different T:

at room T
2.22 J/mm²

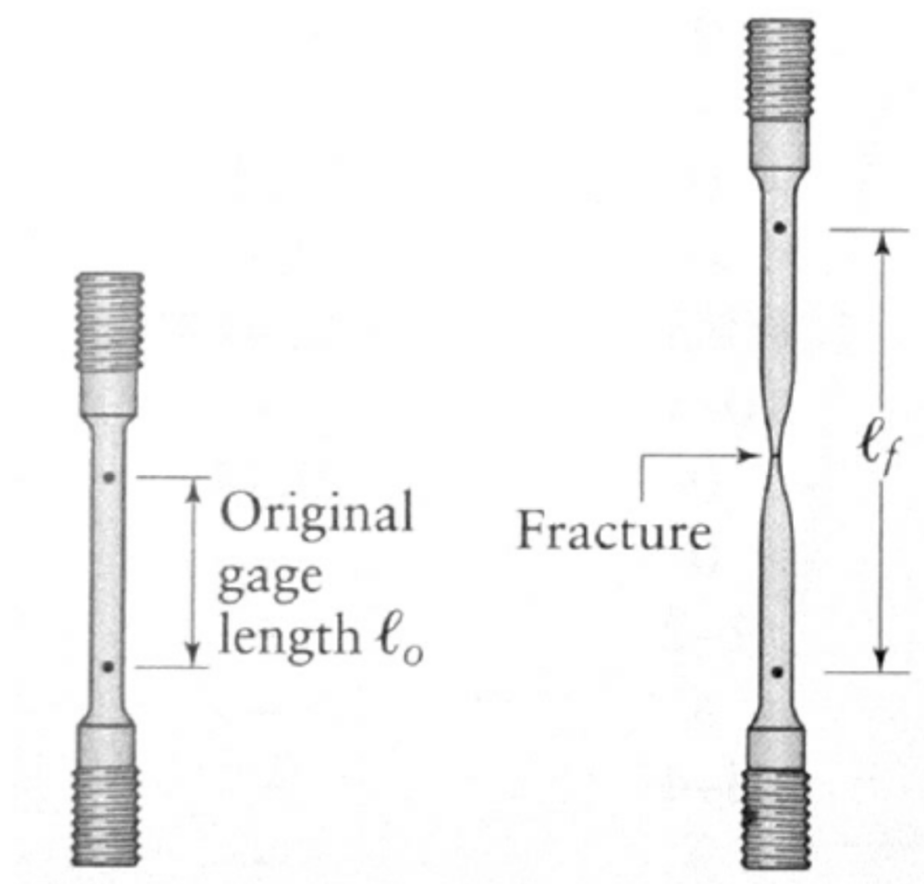
at -200 °C
0.04 J/mm²

CHARPY test on steels
with same composition
but different thermal
treatments:

C40 steel annealed
0.89 J/mm²

C40 steel quenched
0.07 J/mm²

DUCTILITY TEST



$$\%_{STRAIN} = \frac{100 \cdot (l_f - l_o)}{l_o}$$

$$\%_{necking} = \frac{100 \cdot (A_o - A_f)}{A_o}$$

95

Table 7.2 Room-Temperature Mechanical Properties (in Tension) for Various Materials

Material	Yield Strength		Tensile Strength		Ductility, %EL [in 50 mm (2 in.)] ^a
	MPa	ksi	MPa	ksi	
Metal Alloys ^b					
Molybdenum	565	82	655	95	35
Titanium	450	65	520	75	25
Steel (1020)	180	26	380	55	25
Nickel	138	20	480	70	40
Iron	130	19	262	38	45
Brass (70 Cu–30 Zn)	75	11	300	44	68
Copper	69	10	200	29	45
Aluminum	35	5	90	13	40
Ceramic Materials ^c					
Zirconia (ZrO ₂) ^d	—	—	800–1500	115–215	—
Silicon nitride (Si ₃ N ₄)	—	—	250–1000	35–145	—
Aluminum oxide (Al ₂ O ₃)	—	—	275–700	40–100	—
Silicon carbide (SiC)	—	—	100–820	15–120	—
Glass–ceramic (Pyroceram)	—	—	247	36	—
Mullite (3Al ₂ O ₃ –2SiO ₂)	—	—	185	27	—
Spinel (MgAl ₂ O ₄)	—	—	110–245	16–36	—
Fused silica (SiO ₂)	—	—	110	16	—
Magnesium oxide (MgO) ^e	—	—	105	15	—
Soda–lime glass	—	—	69	10	—
Polymers					
Nylon 6,6	44.8–82.8	6.5–12	75.9–94.5	11.0–13.7	15–300
Polycarbonate (PC)	62.1	9.0	62.8–72.4	9.1–10.5	110–150
Polyester (PET)	59.3	8.6	48.3–72.4	7.0–10.5	30–300
Polymethyl methacrylate (PMMA)	53.8–73.1	7.8–10.6	48.3–72.4	7.0–10.5	2.0–5.5
Polyvinyl chloride (PVC)	40.7–44.8	5.9–6.5	40.7–51.7	5.9–7.5	40–80
Phenol-formaldehyde	—	—	34.5–62.1	5.0–9.0	1.5–2.0
Polystyrene (PS)	—	—	35.9–51.7	5.2–7.5	1.2–2.5
Polypropylene (PP)	31.0–37.2	4.5–5.4	31.0–41.4	4.5–6.0	100–600
Polyethylene—high density (HDPE)	26.2–33.1	3.8–4.8	22.1–31.0	3.2–4.5	10–1200
Polytetrafluoroethylene (PTFE)	—	—	20.7–34.5	3.0–5.0	200–400
Polyethylene—low density (LDPE)	9.0–14.5	1.3–2.1	8.3–31.4	1.2–4.55	100–650

^a For polymers, percent elongation at break.

^b Property values are for metal alloys in an annealed state.

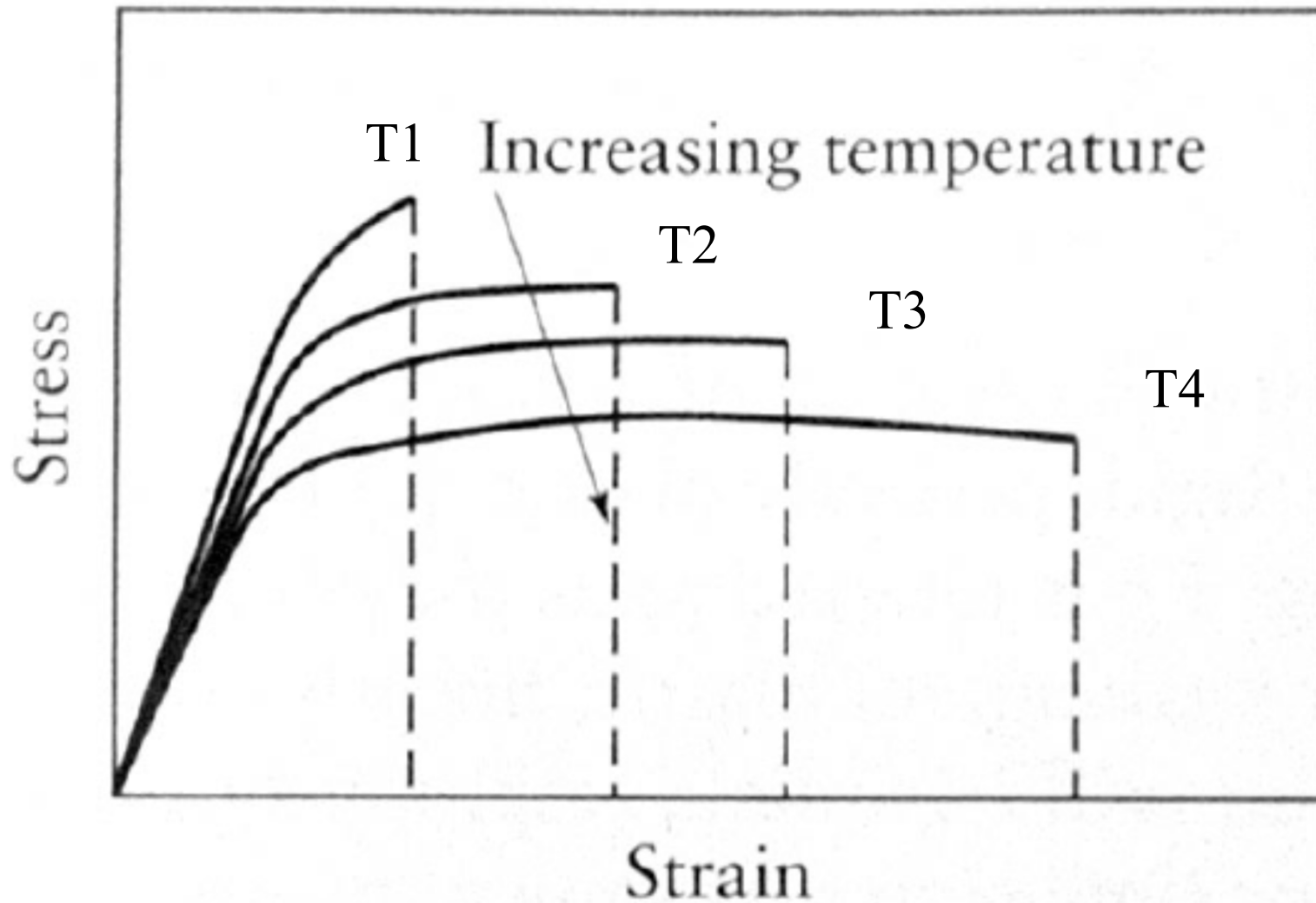
^c The tensile strength of ceramic materials is taken as flexural strength (Section 7.10).

^d Partially stabilized with 3 mol% Y₂O₃.

^e Sintered and containing approximately 5% porosity.

Effect of T

$$T_1 < T_2 < T_3 < T_4$$



Stress/strain curves vs Temperature

